

Xolvon Backend — Handover Documentation

Official handover entry point for the Xolvon backend (NestJS 12, strict ESM, Node.js 22+). This folder is the canonical, code-verified reference. The root `README.md` and `ARCHITECTURE.md` predate some hardening work; where they disagree with this folder, the documents here (and the code) win.

Document index

Document	Purpose
<code>README.md</code> (this file)	From-zero setup, environment variables, commands, migrations, PDF build
<code>architecture.md</code>	Module map, data layer (SQLite/D1), storage port, auth model, full data model
<code>security.md</code>	Security headers, input sanitization, consolidated controls, audit results, limitations
<code>deployment-runbook.md</code>	Staging → production deployment, migration apply/rollback, admin bootstrap, monitoring status
<code>api-contract.md</code>	Canonical endpoint contract (auth/role per route)
<code>decision-log.md</code>	Locked decisions (DL-nnn), conflicts, open items
<code>production-operations.md</code>	Environment separation, admin bootstrap, rollback, observability boundary
<code>qa-results.md</code>	Live-deployment verification status (currently BLOCKED — recorded honestly)

1. Prerequisites

- **Node.js** v22.0.0+ (NestJS 12 runs native ESM)
- **npm** v10+
- No Cloudflare account, database, or bucket is needed for local development.

2. From-zero local setup

2.1 Clone and install

```
git clone <repository-url>
cd Project-SaaS-Xolvon
npm install
```

`.npmrc` sets `legacy-peer-deps=true` (required by the `@nestjs/throttler` peer range, see DL-016); a clean `npm install` on a fresh clone must exit 0.

2.2 Create `.env`

```
cp .env.example .env
```

`.env` is gitignored — never commit it. The local defaults in `.env.example` run fully offline: `APP_ENV=local`, `DB_DRIVER=sqlite`, `STORAGE_DRIVER=local-test`. You must supply only two values locally (names below; put real values in `.env`, never in docs or Git):

- `JWT_SECRET` — any high-entropy string of at least 32 characters (e.g. generate one with `openssl rand -hex 32`).
- `ADMIN_BOOTSTRAP_EMAIL` / `ADMIN_BOOTSTRAP_PASSWORD` — only needed when you run `npm run seed:admin` (section 5); not read by the HTTP server.

All other blank variables (`CLOUDFLARE_*`, `R2_*`) are validated only when the matching driver is selected (`DB_DRIVER=d1` / `STORAGE_DRIVER=r2`) — see `src/config/env.validation.ts`. The validator fails fast at bootstrap and never echoes values.

2.3 Build

```
npm run build
```

2.4 Apply database migrations (local SQLite)

> **Important:** the HTTP server does **not** auto-apply migrations at startup. `src/main.ts` never runs them; `local.db` starts empty. You must apply migrations once before the first run.

```
# Applies all pending files in src/database/migrations/ to ./local.db
npx ts-node src/database/migrate.ts
```

(`npm run seed:admin` also applies pending migrations first, so seeding implicitly migrates.)

2.5 Run locally

```
npm run start:dev # watch mode
# or
npm run start:prod # node dist/main (after npm run build)
```

The API listens on `http://localhost:PORT/api` (default `PORT=3000`). Smoke

`test` is the health route is `GET /api/` (it returns `Hello World!` and is exempt from all throttling groups):

```
curl -i http://localhost:3000/api
curl -s http://localhost:3000/api/courses # public catalog, paginated
```

3. Environment variable reference

Names and purposes only — ****no secret values belong in this repo, in Git, in logs, or in any document.**** The three templates are `.env.example` (local, 17 names), `.env.staging.example` and `.env.production.example` (15 names each); `CORS_ALLOWED_ORIGINS` is present but blank in all three. One further name is read at runtime but is not in the example files (see the row marked ***runtime-optional***).

Variable	Purpose	Required		
APP_ENV	Runtime mode: <code>local</code> \	<code>staging</code> \	<code>production</code> . <code>production</code> enables HSTS in security headers.	Optional, defaults <code>local</code>
PORT	HTTP listen port (integer 1–65535).	Optional, defaults <code>3000</code>		
DB_DRIVER	Persistence driver: <code>sqlite</code> (better- <code>sqlite3</code> on <code>local,db</code>) \	<code>d1</code> (Cloudflare D1 REST adapter).	Optional, defaults <code>sqlite</code>	
CLOUDFLARE_ACCOUNT_ID	Cloudflare account id for the D1 REST endpoint.	Required when <code>DB_DRIVER=d1</code>		
CLOUDFLARE_D1_DATABASE_ID	D1 database id the REST adapter queries.	Required when <code>DB_DRIVER=d1</code>		
CLOUDFLARE_API_TOKEN	Bearer token used per-query against the D1 REST API (never stored on the service instance).	Required when <code>DB_DRIVER=d1</code>		
STORAGE_DRIVER	Object storage driver: <code>local-test</code> (in-process stub, <code>http://local-storage.test</code> URLs) \	<code>r2</code> (Cloudflare R2 via S3 presigning).	Optional, defaults <code>local-test</code>	
R2_ENDPOINT	S3-compatible R2 endpoint (must be absolute <code>https:</code> URL). Its origin also feeds CSP source lists.	Required when <code>STORAGE_DRIVER=r2</code>		
R2_BUCKET	R2 bucket name.	Required when <code>STORAGE_DRIVER=r2</code>		
R2_ACCESS_KEY_ID	R2 access key id.	Required when <code>STORAGE_DRIVER=r2</code>		
R2_SECRET_ACCESS_KEY	R2 secret access key.	Required when <code>STORAGE_DRIVER=r2</code>		
JWT_SECRET	HS secret for signing/verifying access tokens (<code>AuthGuard</code>).	Always required (min 32 chars)		
FRONTEND_URL	Primary allowed browser origin; CORS and request-integrity fallback when <code>CORS_ALLOWED_ORIGINS</code> is unset.	Always required (valid <code>http(s)</code> URL)		
CORS_ALLOWED_ORIGINS	Comma-separated explicit whitelist of allowed origins (credentials mode; wildcards rejected). Also the request-integrity guard's allowed-origin set. Falls back to <code>FRONTEND_URL</code> , then <code>http://localhost:3001</code> .	Optional (blank in all three example files; set real origins per environment)		
CSP_CONNECT_SRC	Comma/space-separated origins added to CSP <code>connect-src</code> / <code>style-src</code> / <code>img-src</code> / <code>font-src</code> . Malformed entries are dropped. Defaults to <code>'self'</code> .	Runtime-optional (not in the example files)		
TRUST_PROXY_HOPS	Express <code>trust proxy</code> hop count for client-IP resolution behind the edge proxy. Unset/blank → <code>false</code> (fail closed: <code>X-Forwarded-For</code> cannot mint fresh throttle buckets); non-numeric → bootstrap aborts. Staging/production templates set <code>1</code> .	Optional (validated at bootstrap, not by <code>env.validation.ts</code>)		
ADMIN_BOOTSTRAP_EMAIL	Email of the first admin account, consumed only by <code>npm run seed:admin</code> .	Required for <code>seed:admin</code>		
ADMIN_BOOTSTRAP_PASSWORD	Password for the first admin account, consumed only by <code>npm run seed:admin</code> .	Required for <code>seed:admin</code>		

Cross-field validation lives in `src/config/env.validation.ts` and keys off `DB_DRIVER` / `STORAGE_DRIVER` — not off `APP_ENV` .

4. Commands

Command	What it does
<code>npm run test:esm</code>	Unit/behavior suite (Jest with <code>--experimental-vm-modules</code> , via cross-env). Current baseline on this branch: 44 suites / 411 tests.
<code>npm run test:e2e</code>	End-to-end suite (<code>test/jest-e2e.json</code> harness with overridden providers). Baseline: 107 tests. Runs fully against in-memory/local SQLite — zero cloud credentials.
<code>npm run build</code>	<code>nest build</code> → <code>dist/</code> .
<code>npm run lint</code>	<code>oxlint src/ test/</code> .
<code>npm run format</code>	Prettier over <code>src/</code> and <code>test/</code> .
<code>npm run test:cov</code>	Coverage report.
<code>npm run seed:admin</code>	Controlled first-admin bootstrap (section 5).
<code>npm run docs:pdf</code>	Regenerate committed PDFs (section 6).

Plain `npm test` (`jest` without the ESM flag) is **not** usable with this ESM-only codebase — always use `test:esm` / `test:e2e` (DL-008).

5. Adding and running a migration

Migrations are plain SQL files in `src/database/migrations/` , applied by `src/database/migrate.ts` in **filename sort order**, tracked in the `_migrations` table (filename + SHA-256 content hash). Rules:

- Add** a new file `NNNN_short_description.sql` continuing the numeric prefix (next free number after `0007_...`). Name only — content of applied files is immutable.
- Never edit an applied migration**. The runner compares the recorded hash and throws on drift (`"changed after it was applied"`), fail-closed. Fix forward with a new file.

3. Statements are split with a string/comment/trigger-aware splitter, so plain

`separated.DDL` is safe. `xolvon-backend-full-documentation.pdf`

4. **Run locally:** `npx ts-node src/database/migrate.ts` (applies to `./local.db` under the current working directory; prints `applied/skipped` counts). The HTTP server itself never runs it.

5. **Run against D1 (staging/production):**

`npx wrangler d1 migrations apply xolvon-staging --remote` — the database name and `migrations_dir` are configured in `wrangler.jsonc`. See `deployment-runbook.md` for the full procedure and rollback policy (forward-only; compensating migration + deployment revert).

6. The `src/database/migrations/_reference/` folder is a reference copy and is **not** applied (the runner only enumerates top-level `*.sql` files).

7. Schema changes must be preceded by a `decision-log.md` entry (RULES §84); `SCHEMA.md` remains the schema source of truth.

6. Generating documentation PDFs

Per DL-019, official PDFs are produced with Playwright/Chromium and committed:

```
npm run docs:pdf
```

- Runs `scripts/docs-pdf.mjs`: renders the **four official handover docs**

(`README.md`, `architecture.md`, `security.md`, `deployment-runbook.md`) to

`docs/pdf/<name>.pdf` (A4, Chromium print pipeline), plus a combined `docs/pdf/xolvon-backend-full-documentation.pdf` in that order. It does **not** render the other `docs/*.md` (`api-contract`, `decision-log`, `qa-results`, `production-operations`) — the deliverable set is pinned in the script.

- Requires the Chromium browser binary once per machine:

```
npx playwright install chromium.
```

- Output is deterministic for the same Markdown input; regenerate and

secret-scan whenever these docs change.

- `docs/Xolvon-API-Documentation.pdf` (and its generator

`scripts/generate-api-pdf.py`) were a non-conforming, out-of-band pair

(per DL-019) and were **deleted at plan T9**; they are not part of this

workflow and no longer exist in the tree. See DL-026 in

`decision-log.md`.

7. House rules for contributors

- No secret or credential values in docs, tests, logs, Postman files, PDFs, or

Git history — variable **names** only (DL-024, cleanup policy DL-025).

- No production deployment claims without real target + HTTPS domain +

verification evidence (current status: blocked, see

`qa-results.md`).

- Endpoint or auth-model changes require updating

`api-contract.md` and a `decision-log.md` entry in the

same change.

API Contract — Backend xolvon.com

Xolvon Backend Documentation · xolvon-backend-full-documentation.pdf

Daftar endpoint + kebutuhan auth. Dokumen ini bertambah setiap modul baru dan

disempurnakan penuh di T16 (`.omo/plans/xolvon-backend-build.md`).

- **Status dokumen:** skeleton per T4. Kolom status menandai `aktif` (sudah ada di repo) atau `planned` (didefinisikan plan, belum diimplementasikan).
- **Global prefix:** semua route dilayani di bawah `/api/*` (`src/main.ts`).
- **Auth:** Authorization: Bearer <accessToken> (JWT) — Bearer-only, tidak ada cookie session (DL-016/DL-018). Role: `user` (default) dan `admin` (`RolesGuard`). Mutation browser request (`POST/PATCH/DELETE`) additionally lolos request-integrity guard `src/common/guards/request-integrity.guard.ts` (T3): `Origin/Referer` harus ada dalam set asal yang diizinkan, `Sec-Fetch-Site: cross-site` ditolak; request non-browser tanpa header asal lolos (tetap butuh Bearer valid).
- **Validasi:** global `ValidationPipe` (`whitelist`, `forbidNonWhitelisted`, `transform`) — field di luar DTO ditolak.
- **Error:** global `AllExceptionsFilter` — respons aman, tanpa stack trace/internal.
- **Pagination:** kontrak terkunci di decision log DL-011 — `?page=` 1-based, `?limit=` default 20 maks 100, respons `{items, page, limit, total}`.
- **Rate limit** (per IP, jendela 60 detik, via `@nestjs/throttler` named groups — DL-012 + implementasi T6 di `src/config/throttle.config.ts`): `/auth/login` + `/auth/register` **5/min**; katalog publik list (`GET /courses`, `/projects`, `/marketplace`, `/collective`, `/home`) **30/min** (group `catalog`); `GET /search` **30/min** (group `search`); signed-URL issuance (`GET /lessons/:id/video-url`, `POST /admin/media/upload-url`, `POST /admin/media/read-url`) **10/min** (group `signed`); `POST /orders` **10/min** (group `order`); rute tak berhias lain **100/min** (group `default`). Health `GET /api` dikecualikan dari SEMUA group (`@SkipThrottle` eksplisit per nama tracker).
- **Audit:** mutation sensitif (verify/activate/revoke/publish/cancel) menulis `admin_audit_logs` (plan T11/T15).

Endpoint aktif (sudah ada di repo)

Auth (`src/auth/auth.controller.ts`)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
POST	<code>/api/auth/register</code>	Public	-	aktif	Registrasi user baru; <code>role</code> tidak pernah diterima dari input (DL-005)	<code>RegisterDto</code>	<code>SafeUserDto</code> (201)
POST	<code>/api/auth/login</code>	Public	-	aktif	Login; mencatat IP + user agent ke sesi; rate limit menyusul (DL-012)	<code>LoginDto</code>	<code>AuthResponseDto</code> { <code>accessToken</code> , <code>refreshToken</code> , <code>user</code> : <code>SafeUserDto</code> } (200)
POST	<code>/api/auth/refresh</code>	Public	-	aktif	Terbit access token baru via refresh token	<code>RefreshTokenDto</code>	{ <code>accessToken</code> } (200)
POST	<code>/api/auth/logout</code>	Public	-	aktif	Revoke sesi refresh token	<code>LogoutDto</code>	{ <code>message</code> } (200)
GET	<code>/api/auth/me</code>	Bearer JWT	user	aktif	Profil user dari payload JWT sesi	-	<code>JwtPayload</code> (200)
GET	<code>/api/auth/admin</code>	Bearer JWT	admin	aktif	Cek akses admin	-	{ <code>message</code> , <code>user</code> } (200)

Orders (`src/orders/orders.controller.ts`)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
POST	<code>/api/orders</code>	Bearer JWT	user	aktif	Checkout; total dihitung server-side; status awal <code>pending</code> (DL-002)	<code>CreateOrderDto</code>	<code>OrderResponseDto</code> (201)
POST	<code>/api/orders/:id/payment-proof</code>	Bearer JWT	owner order	aktif	Submit object key bukti bayar; cek kepemilikan anti-IDOR	<code>SubmitPaymentProofDto</code>	{ <code>message</code> , <code>proofId</code> } (201)
PATCH	<code>/api/orders/:id/verify</code>	Bearer JWT	admin	aktif	Verifikasi pembayaran → <code>paid</code> ; setelah rekonsiliasi T3 hanya dari <code>pending</code> (DL-002) + audit	-	<code>OrderResponseDto</code> (200)
POST	<code>/api/orders/:id/activate</code>	Bearer JWT	admin	aktif	Aktivasi enrollment dari order <code>paid</code> ; idempoten via upsert	-	<code>OrderActivationResponseDto</code> (200)

Enrollments (`src/enrollments/enrollments.controller.ts`)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
GET	<code>/api/enrollments/me</code>	Bearer JWT	user	aktif	Daftar enrollment milik sesi (anti-IDOR, dari <code>user.sub</code>)	-	<code>EnrollmentResponseDto[]</code> (200)
PATCH	<code>/api/enrollments/:id/revoke</code>	Bearer JWT	admin	aktif	Revoke enrollment → <code>revoked</code> + audit (plan T15)	-	<code>EnrollmentResponseDto</code> (200)

App (`src/app.controller.ts`)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
GET	<code>/api</code>	Public	-	aktif	Health check / hello	-	<code>string</code> (200)

Search (`src/search/search.controller.ts`)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
GET	<code>/api/search</code>	Public	-	aktif	Pencarian lintas course, project, dan marketplace; hanya <code>published</code> ; FTSS; <code>?q=</code> opsional, <code>?type=course</code>	<code>SearchQueryDto</code>	{ <code>items</code> : <code>SearchResultDto[]</code> , <code>query</code> , <code>page</code> , <code>limit</code> , <code>total</code> } (200)

`SearchResultDto` hanya memuat {`type`, `id`, `title`, `slug`, `description`, `status`}. Search tidak pernah mengembalikan draft, private, admin-only, unpublished, atau revoked content; seluruh search term dan pagination memakai DTO validation serta bind parameters.

Home (`src/home/home.controller.ts`)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
GET	<code>/api/home</code>	Public	-	aktif	Agregasi homepage published-only (featured courses, projects, marketplace, collective) tanpa N+1; throttled group <code>catalog</code>	-	<code>HomeResponseDT0</code> { <code>courses[]</code> , <code>projects[]</code> , <code>marketplace[]</code> , <code>collective[]</code> } (200)

Courses (`src/courses/courses.controller.ts`)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
GET	<code>/api/courses</code>	Public	-	aktif	Katalog published-only, <code>?q=</code> contains pada title (LIKE, wildcard di-escape), <code>?sort=latest</code>	<code>oldest</code> (default latest), pagination DL-011	- { <code>items</code> : <code>CourseCardDto[]</code> , <code>page</code> , <code>limit</code> , <code>total</code> , <code>query</code> } (200)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
GET	/api/courses/:slug	Public	-	aktif	Detail published-only (draft → 404) + lessons terurut order_index ASC — ringkasan aman tanpa content / video_object_key /signed URL (SCHEMA.md §17)	-	CourseDetailDto (200) / 404
POST	/api/courses	Bearer JWT	admin	aktif	Buat course (status awal draft, UUID server-generated) + audit create ; slug unik	CreateCourseDto	CourseCardDto (201)
PATCH	/api/courses/:id	Bearer JWT	admin	aktif	Update parsial + audit update ; slug baru dicek unik	UpdateCourseDto	CourseCardDto (200)
POST	/api/courses/:id/publish	Bearer JWT	admin	aktif	Transisi ketat draft → published, gate field wajib (description, price) + audit publish	-	CourseCardDto (200)
POST	/api/courses/:id/unpublish	Bearer JWT	admin	aktif	Transisi ketat published → draft + audit unpublish	-	CourseCardDto (200)
GET	/api/courses/:slug/lessons	Public	-	aktif	Daftar lesson published milik course published (src/lessons/course-lessons.controller.ts), urut order_index ASC; ringkasan aman tanpa content / video_object_key	-	LessonSummaryDto[] (200)

CourseCardDto (SCHEMA.md §16): {id, title, slug, description, price,

thumbnailUrl, status} . CourseDetailDto menambahkan lessons:

LessonSummaryDto[] dengan LessonSummaryDto = {id, title, orderIndex, status}.

Collective (src/collective/collective.controller.ts)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
GET	/api/collective	Public	-	aktif	Member published-only, ?q= (LIKE name/role/skills), pagination DL-011, urut display_order ASC lalu created_at DESC	-	PaginatedCollectiveResponseDto {items, page, limit, total} (200)
GET	/api/collective/:slug	Public	-	aktif	Detail member + relatedProjects via project_members (published); TIDAK pernah memuat email/telepon (SCHEMA.md §57)	-	CollectiveMemberDetailDto (200) / 404
POST	/api/collective	Bearer JWT	admin	aktif	Buat member (status awal draft) + audit create	CreateCollectiveMemberDto	CollectiveMemberResponseDto (201)
PATCH	/api/collective/:id	Bearer JWT	admin	aktif	Ubah member + audit update	UpdateCollectiveMemberDto	CollectiveMemberResponseDto (200)
POST	/api/collective/:id/publish	Bearer JWT	admin	aktif	Set status published + audit publish	-	CollectiveMemberResponseDto (200)
POST	/api/collective/:id/unpublish	Bearer JWT	admin	aktif	Set status draft + audit unpublish	-	CollectiveMemberResponseDto (200)

Projects (src/projects/projects.controller.ts, src/project-media/, src/project-members/)

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
GET	/api/projects	Public	-	aktif	Portfolio published-only, ?q= contains pada title/summary, ?sort=latest	oldest (default latest), pagination DL-011	- {items: ProjectCardDto[], page, limit, total} (200)
GET	/api/projects/:slug	Public	-	aktif	Detail published-only (draft → 404, sama seperti slug tak dikenal); urutan cerita Problem→Solution→Tech Stack→Result→Media (sort_order ASC, video dan object key dieksklusi) → members[].role dari JOIN collective_members (SCHEMA.md §47-48)	-	ProjectDetailDto (200) / 404
POST	/api/projects	Bearer JWT	admin	aktif	Buat project (status awal draft) + audit create ; slug unik; type string terkontrol (bukan enum final)	CreateProjectDto	ProjectResponseDto (201)
PATCH	/api/projects/:id	Bearer JWT	admin	aktif	Update parsial + audit update ; slug baru dicek unik	UpdateProjectDto	ProjectResponseDto (200)
POST	/api/projects/:id/publish	Bearer JWT	admin	aktif	Transisi ketat draft → published, gate field wajib (title, slug, type, summary, problem, solution) + audit publish	-	ProjectResponseDto (200)
POST	/api/projects/:id/unpublish	Bearer JWT	admin	aktif	Transisi ketat published → draft + audit unpublish	-	ProjectResponseDto (200)
POST	/api/projects/:id/media	Bearer JWT	admin	aktif	Attach media (objectKey, mediaType string terkontrol — enum final open decision DL-014, sortOrder) + audit create	AttachProjectMediaDto	ProjectMediaResponseDto (201)
DELETE	/api/projects/:id/media/:mediaId	Bearer JWT	admin	aktif	Detach media ter-scope ke project (lintas project → 404) + audit delete	-	{message} (200)
POST	/api/projects/:id/members	Bearer JWT	admin	aktif	Assign collective member dengan role eksplisit (upsert ON CONFLICT(project_id, member_id) ; role string terkontrol (bukan enum final) + audit create	AssignProjectMemberDto	{projectId, memberId, role} (201)
DELETE	/api/projects/:id/members/:memberId	Bearer JWT	admin	aktif	Remove assignment member ter-scope ke project + audit delete	-	{message} (200)

ProjectCardDto (SCHEMA.md §46): {id, title, slug, type, summary, status}.

ProjectDetailDto (§47, urutan cerita §48): {id, title, slug, type, summary,

problem, solution, techStack: string[], result, media: {id, mediaType, sortOrder[], members: {memberId, name, role}[], status} — media publik

tanpa objectKey, video dieksklusi (butuh signed private access, T13/T14).

Admin (src/admin/admin.controller.ts)

Method	Path	Auth	Role	Status	Deskripsi
GET	/api/admin/users	Bearer JWT	admin	aktif	Paginated user list.
GET	/api/admin/users/:id	Bearer JWT	admin	aktif	User detail.

Method	Path	Auth	Role	Status	Deskripsi
GET	/api/admin/orders	Bearer JWT	admin	aktif	Paginated order list.
GET	/api/admin/overview	Bearer JWT	admin	aktif	User/order/enrollment summary.

Learning, media, and order operations

All routes in this table require Bearer JWT. Admin routes additionally require the `admin` role; user routes derive ownership from the JWT subject.

Method	Path	Role	Status	Deskripsi
POST	/api/courses/:courseId/lessons	admin	aktif	Create lesson.
PATCH	/api/lessons/:id	admin	aktif	Update lesson.
DELETE	/api/lessons/:id	admin	aktif	Delete lesson.
PATCH	/api/lessons/:id/reorder	admin	aktif	Reorder lesson.
POST	/api/lessons/:id/publish	admin	aktif	Publish lesson.
POST	/api/lessons/:id/unpublish	admin	aktif	Unpublish lesson.
POST	/api/lessons/:lessonId/resources	admin	aktif	Create resource.
DELETE	/api/resources/:id	admin	aktif	Delete resource.
GET	/api/lessons/:id/video-url	user	aktif	Signed lesson video URL for active enrollment.
POST	/api/progress	user	aktif	Write owned progress.
GET	/api/progress	user	aktif	Read owned progress.
PATCH	/api/enrollments/:id/revoke	admin	aktif	Revoke enrollment and audit.
POST	/api/orders/:id/cancel	admin	aktif	Cancel pending order and audit.
POST	/api/admin/media/upload-url	admin	aktif	Create validated media upload URL.
POST	/api/admin/media/confirm	admin	aktif	Confirm media upload.
POST	/api/admin/media/read-url	admin	aktif	Create private media read URL.

Historical implementation notes

The endpoint tables above are the active contract. The notes below preserve module-specific response and validation details from the implementation plan.

Courses — aktif (T5)

Endpoint sudah pindah ke tabel aktif di atas. `status` tidak pernah diterima dari input (DTO tidak memilikinya; global `ValidationPipe` `forbidNonWhitelisted` menolak field di luar DTO) — transisi status hanya via endpoint `publish/unpublish`.

Lessons & Course Resources — aktif (T6)

CRUD admin untuk lesson (`order_index`, draft/published) dan resource

'pdf	resource	assignment'; summary publik tanpa field privat.
------	----------	---

Projects / Project Media / Project Members — aktif (T7)

Endpoint sudah pindah ke tabel aktif di atas. `type` (project), `mediaType` (media), dan `role` (member) divalidasi sebagai string terkontrol dengan batas panjang — BUKAN enum final (SCHEMA.md §45/§50/§52; open decision DL-014). Object key media privat tidak pernah muncul di respons publik.

Collective — aktif (T8)

Endpoint sudah pindah ke tabel aktif di atas. `skills` disimpan comma-separated TEXT dan `social_links` sebagai JSON array TEXT (SCHEMA.md §56); keduanya diparse di mapper menjadi `string[]` / `SocialLink[]` yang tidak pernah null.

Marketplace & Marketplace Media (T9, `src/marketplace/`)

Showcase murni — tidak ada logika transaksi/checkout (SCHEMA.md §60); satu-satunya permukaan komersial adalah `externalUrl` (validasi minimal https, DL-013; governance tetap OPEN — PRD §91.11).

Method	Path	Auth	Role	Status	Deskripsi	Request	Response
GET	/api/marketplace	Public	-	aktif	Listing published-only, <code>?q=</code> (LIKE title/description, wildcard di-escape), <code>?sort=latest</code>	oldest' (default latest), pagination DL-011	- <code>{items: MarketplaceItemDto[], page, limit, total}</code> (200)
GET	/api/marketplace/:slug	Public	-	aktif	Detail published-only (draft → 404); mediaurut <code>sort_order</code> ASC; capabilities diparse dari comma-separated TEXT	-	<code>MarketplaceItemDetailDto</code> (200) / 404

Method	Path	Auth	Role	Status	Deskripsi	Request	Response		
POST	/api/marketplace	Bearer JWT	admin	aktif	Buat listing (status awal draft) + audit create : externalUrl wajib https valid	CreateMarketplaceItemDto	MarketplaceItemDto (201)		
PATCH	/api/marketplace/:id	Bearer JWT	admin	aktif	Update parsial + audit update : slug baru dicek unik; externalUrl divalidasi ulang	UpdateMarketplaceItemDto	MarketplaceItemDto (200)		
POST	/api/marketplace/:id/publish	Bearer JWT	admin	aktif	Transisi ketat draft → published + audit publish	-	MarketplaceItemDto (200)		
POST	/api/marketplace/:id/unpublish	Bearer JWT	admin	aktif	Transisi ketat published → draft + audit unpublish	-	MarketplaceItemDto (200)		
POST	/api/marketplace/:id/media	Bearer JWT	admin	aktif	Attach media (objectKey R2, mediaType image)	video\	deck per SCHEMA.md §63) + audit create	AttachMarketplaceMediaDto	MarketplaceMediaDto (201)
DELETE	/api/marketplace/:id/media/:mediaId	Bearer JWT	admin	aktif	Detach media ter-scope ke item (lintas item → 404) + audit delete	-	{message} (200)		

MarketplaceItemDto (SCHEMA.md §59): {id, title, slug, description, capabilities: string[], externalUrl, status}`.

Home — aktif (T10)

Agregasi published-only (featured courses, projects, marketplace, collective) tanpa N+1.

Audit — aktif internal (T11)

Belum ada endpoint — dan tidak akan punya endpoint publik. record() internal untuk modul lain; metadata tidak pernah keluar ke API publik/user.

Admin — aktif (T12)

Method	Path	Auth	Role	Deskripsi
GET	/api/admin/users	Bearer JWT	admin	Tabel users
GET	/api/admin/orders	Bearer JWT	admin	Tabel orders (kolom per PRD §45)
GET	/api/admin/overview	Bearer JWT	admin	userCount / pendingOrders / activeEnrollments

Media (R2) — aktif (T13)

Method	Path	Auth	Role	Deskripsi
POST	/api/admin/media/upload-url	Bearer JWT	admin	Presigned PUT 3600s; MIME/size/prefix divalidasi; key server-generated
POST	/api/admin/media/confirm	Bearer JWT	admin	Konfirmasi upload

Lessons video & Progress — aktif (T14)

Method	Path	Auth	Role	Deskripsi
GET	/api/lessons/:id/video-url	Bearer JWT	user aktif	Signed URL 300s; wajib enrollment aktif + lesson milik course (anti-IDOR)
POST	/api/progress	Bearer JWT	user aktif	Tulis progress; ownership dari sesi, bukan body

Orders (tambahan) — aktif (T15)

Endpoint sudah pindah ke tabel aktif di atas pada path kanonik

```
POST /api/orders/:id/cancel (controller src/orders/orders.controller.ts, @Controller('orders') + @Post('/:id/cancel') ). Catatan rekonsiliasi: dokumen awal pernah menuliskannya sebagai POST /api/admin/orders/:id/cancel ; path tersebut TIDAK pernah ada di controller dan TIDAK ada di docs/openapi.json maupun Postman — kontrak kanonik adalah POST /api/orders/:id/cancel . Semantik tetap: hanya transisi pending → cancelled yang diizinkan, dan cancel menulis audit (plan T15).
```

Search — aktif (T16)

Endpoint lintas entitas sudah aktif pada tabel di atas. Search bersifat public, published-only, memakai FTS5, dan mengikuti kontrak pagination DL-011.

Catatan kontrak lintas-endpoint

- Status order: pending | paid | cancelled (DL-002); status enrollment: active | revoked (DL-004).
- Tidak ada endpoint delete / role-change di V1 (DL-005, PRD §44 view-only).
- Tidak ada endpoint yang menerima role dari input user; tidak ada registrasi admin publik.

- Respons tidak pernah memuat `password_hash` , `refresh_token` , object key privat, kredensial R2, atau stack trace.

• Semua `endpoint list` (aktif maupun planned) mengikuti kontrak pagination DL-011.

Architecture — Xolvon Backend

Xolvon Backend Documentation · xolvon-backend-full-documentation.pdf

Code-verified architecture reference for the NestJS 12 (strict ESM, Express platform) backend. Every claim below cites the file that implements it.

Companion documents: `api-contract.md` (endpoint contract),

`security.md` (controls), `deployment-runbook.md`

(operations), `decision-log.md` (why).

1. Request lifecycle

Bootstrap order in `src/main.ts` (all guards/pipes below are process-wide):

```
Request → Express (trust proxy: TRUST_PROXY_HOPS, src/main.ts:25–28)
→ Helmet security headers (enableSecurityHeaders, src/main.ts:35–39)
→ CORS (explicit origin whitelist + credentials, src/main.ts:43–50)
→ Global prefix /api (src/main.ts:17)
→ APP_GUARD #1 ThrottlerGuard (src/app.module.ts:65)
→ APP_GUARD #2 RequestIntegrityGuard (src/app.module.ts:66)
→ @UseGuards(AuthGuard, RolesGuard) where declared per route/controller
→ ValidationPipe { whitelist, forbidNonWhitelisted, transform } (src/main.ts:53–59)
→ Controller → Service → DatabaseService token / StoragePort token
→ AllExceptionsFilter (safe JSON error envelope, src/main.ts:62)
```

There is no middleware layer and no `NestModule` configuration anywhere — the

only `app.use()` is Helmet inside `src/security/security-headers.ts`.

`AuthGuard` is deliberately **not** global: routes that don't attach it are

public (catalog/search/home/collective/marketplace GETs); `@Public()` is

metadata consumed by `AuthGuard` only where it is attached

(`src/auth/guards/auth.guard.ts:23–30`).

2. NestJS module map

`src/app.module.ts:30–61` (import order):

Module	Directory	Responsibility		
<code>ConfigModule</code> (global)	<code>src/config/</code>	<code>.env</code> load + fail-fast <code>validateEnvironment</code> (<code>env.validation.ts</code>); CORS, CSP, trust-proxy, throttle resolvers		
<code>ThrottlerModule</code> (global, via <code>APP_GUARD</code>)	<code>src/config/throttle.config.ts</code>	5 named limit groups (§5.3)		
<code>DatabaseModule</code>	<code>src/database/</code>	<code>DatabaseService</code> DI token; driver factory <code>sqlite</code> ↔ <code>D1</code> REST adapter; migration runner		
<code>AuthModule</code>	<code>src/auth/</code>	Register/login/refresh/logout, Argon2id hashing, JWT issue/verify, hashed refresh sessions, <code>AuthGuard</code> / <code>RolesGuard</code>		
<code>EnrollmentsModule</code>	<code>src/enrollments/</code>	Entitlement checks, idempotent activation, admin revoke		
<code>OrdersModule</code>	<code>src/orders/</code>	Server-computed order totals, payment-proof submission (anti-IDOR), verify/activate/cancel transitions		
<code>CoursesModule</code>	<code>src/courses/</code>	Published-only catalog, draft-gated admin CRUD, publish gates, slug uniqueness		
<code>HomeModule</code>	<code>src/home/</code>	Published-only aggregation (featured courses/projects/marketplace/collective), no N+1		
<code>MarketplaceModule</code>	<code>src/marketplace/</code>	Showcase-only SaaS catalog; <code>externalUrl</code> minimal https validation (DL-013)		
<code>AuditModule</code>	<code>src/audit/</code>	Internal <code>record()</code> writer for <code>admin_audit_logs</code> ; no public endpoint, metadata never exits the API		
<code>CollectiveModule</code>	<code>src/collective/</code>	Talent roster (draft/published), never exposes email/phone on public reads		
<code>ProjectsModule</code> + <code>ProjectMediaModule</code> + <code>ProjectMembersModule</code>	<code>src/projects/</code> , <code>src/project-media/</code> , <code>src/project-members/</code>	Portfolio CRUD, media attach/detach (private object keys never public), member assignment (<code>type</code> / <code>mediaType</code> / <code>role</code> are controlled strings, DL-014)		
<code>LessonsModule</code>	<code>src/lessons/</code>	Lesson CRUD/reorder/publish; <code>GET /api/lessons/:id/video-url</code> mints 300s signed URLs gated by active enrollment		
<code>CourseResourcesModule</code>	<code>src/course-resources/</code>	Resource attach ('pdf')	resource\	assignment') under lessons
<code>AdminModule</code>	<code>src/admin/</code>	Read-only admin surfaces: users, orders, overview counters		
<code>MediaModule</code>	<code>src/media/</code>	<code>STORAGE_PORT</code> binding + admin upload-url/confirm/read-url endpoints		
<code>ProgressModule</code>	<code>src/progress/</code>	Lesson completion state, ownership from JWT subject		
<code>SearchModule</code>	<code>src/search/</code>	Cross-entity FTS5 search (courses/projects/marketplace), published-only		

Health endpoint: `GET /api` (`src/app.controller.ts`) — returns a static

string, exempt from every throttle group via

`@SkipThrottle(HEALTH_SKIP_THROTTLE)`.

3. Data layer — one token, two drivers

All services inject the `DatabaseService` class as DI token; the factory in

`src/database/database.module.ts:25–44` selects the implementation from

`DB_DRIVER`:

3.1 `sqlite` (local development & e2e)

`DatabaseService` (`src/database/database.service.ts`) opens `better-sqlite3`

on `./local.db` (relative to cwd) with `foreign_keys = ON`, exposing

`queryAll`, `queryOne`, `execute` — always positional ? bind parameters

(prepared statements are mandatory per HANDBOOK_BACKEND §4.1). Zero cloud

credentials needed.

3.2 d1 (staging/production)

D1RestAdapter (src/database/d1-rest.adapter.ts) wraps D1Service (src/database/d1.service.ts) behind the same three methods. D1Service POSTs {sql, params} to https://api.cloudflare.com/client/v4/accounts/<CLOUDFLARE_ACCOUNT_ID>/d1/database/<CLOUDFLARE_D1_DATABASE_ID>/query with Authorization: Bearer <CLOUDFLARE_API_TOKEN> **resolved on demand per query and never stored as an instance field** (hardening invariant, d1.service.ts:77-79). Timeout 10s; timeouts→504, network/non-OK→502 — the adapter never leaks the upstream body. onModuleInit fail-fasts on missing CLOUDFLARE_* values. (src/database/d1.module.ts exists but is imported nowhere — dead code kept for reference.) Migrations run through the same execute() contract, so the runner (src/database/migrate.ts) works on either driver; the operational path for D1 is wrangler d1 migrations apply (see runbook). Applied files are recorded in _migrations with a SHA-256 hash; hash drift throws (fail-closed, forward-only policy).

4. Storage layer — StoragePort interface-first (DL-007)

src/media/storage.port.ts defines the token STORAGE_PORT and the interface:

```
createUploadUrl(request: { key; contentType; contentLength }): Promise<string>
createReadUrl(key: string, expiresIn?: number): Promise<string>
confirmUpload(key: string): Promise<void>
```

src/media/media.module.ts:12 binds the implementation from

STORAGE_DRIVER:

Driver	Class	Behavior
local-test	src/media/local-test-storage.service.ts	Deterministic stub URLs (http://local-storage.test/...); confirmUpload records keys in memory. Used by all unit/e2e runs — tests never touch the network.
r2	src/media/r2-storage.service.ts	S3 SDK v3 against R2 (region: 'auto'); presigned PUT 3600s (signable headers: content-type/content-length) and presigned GET default 300s. Credentials only from env, never in responses.

Consumers (src/media/media.service.ts , src/lessons/lessons.service.ts)

inject the token and are unchanged when drivers are swapped. Private object keys never appear in public API responses; entitlement is checked before a read URL is minted (server owns trust).

5. Auth model — JWT Bearer + hashed refresh sessions

5.1 Tokens

- **Access:** JWT signed/verified with JWT_SECRET , TTL 15m (src/auth/auth.service.ts:132-133, 211-212). Sent as Authorization: Bearer <token>; extraction and verification in src/auth/guards/auth.guard.ts (generic 401 messages, no token echo).
- **Refresh:** cryptographically random string returned once at login; the database stores only its SHA-256 hash in sessions.refresh_token (auth.service.ts:31-33, 136-149) alongside user_id , ip_address , user_agent , expires_at . refresh looks the session up by hash and purges expired rows; logout deletes the session by hash (auth.service.ts:173-225). Plaintext refresh tokens can never be read back from a leaked database.

5.2 CSRF model (explicit non-decision to adopt cookies)

The cookie-session CSRF pattern (synchronizer token in a cookie) was **NOT adopted**. The API is Bearer-only — no authentication happens via cookies — so classic cross-site request forgery cannot carry credentials automatically. Instead, browser state-changing requests are constrained by the request-integrity guard: POST / PATCH / DELETE requests must not declare Sec-Fetch-Site: cross-site, and any Origin / Referer they carry must be in the explicit allowed-origin set (CORS_ALLOWED_ORIGINS , falling back to FRONTEND_URL). Non-mutating requests are unaffected, and headerless service-to-service clients remain compatible; authentication is still required independently on every protected route. This is the locked decision DL-018 (with DL-003 keeping the repo's JWT+refresh model over the handbook's getSession() sample); implementation src/common/guards/request-integrity.guard.ts , rationale security.md .

5.3 Rate limiting

`src/config/throttle.ts` with five named groups; all 60s Windows

(`src/config/throttle.config.ts`): default 100/min (undecorated routes),

search 30/min (GET /api/search), catalog 30/min (courses/projects/

marketplace/collective listings + GET /api/home), signed 10/min (lesson

video URL, media upload-url/read-url), order 10/min (POST /api/orders).

POST /api/auth/login and POST /api/auth/register are tightened inline to

5/min per IP (DL-012/DL-016). Client-IP accuracy behind an edge proxy is

controlled by TRUST_PROXY_HOPS ; locally it fails closed to false so

X-Forwarded-For cannot mint fresh buckets (`src/config/trust-proxy.ts`).

5.4 Authorization

RolesGuard enforces @Roles('admin') on all admin surfaces; roles come

exclusively from the users row / verified JWT — never from request input

(DL-005). First-admin creation is a controlled seed command

(npm run seed:admin), not a public route.

6. Data model — 19 tables + FTS5

Schema source of truth: SCHEMA.md ; applied set:

src/database/migrations/0000...0007 (19 regular tables + 3 FTS5 virtual

tables; plus the runtime bookkeeping table _migrations). Conventions

(HANDBOOK_BACKEND §3.1): TEXT UUID v4 primary keys, ISO 8601 TEXT timestamps,

INTEGER money (Rupiah) and booleans (0/1), TEXT + CHECK for enums,

prepared statements only.

6.1 Tables by domain

Domain	Tables	Notes		
Identity	users , sessions	users.role CHECK 'user'	admin ; users.status defaults 'active' (no CHECK constraint); sessions.refresh_token stores SHA-256 hash (§5.1)	
Course	courses , lessons , course_resources , course_tags	courses.slug UNIQUE; lessons.order_index sequencing, is_preview 0/1, video_object_key private; resource_type 'pdf'	resource\	assignment'
Transaction	orders , order_items , payment_proofs	orders.status final 'pending'	paid\	cancelled (DL-002); server-computed orders.amount (CHECK > 0); proofs store R2 object keys + uploaded_by', anti-IDOR at submit (DL-006)
Entitlement	enrollments	UNIQUE (user_id , course_id) prevents double enrollment; status final 'active'	revoked' (DL-004); idempotent upsert on activation	
Learning	progress	Per-user lesson completion; ownership from JWT subject		
Portfolio	projects , project_media , project_tags	Narrative fields Problem→Solution→Tech→Result; project_media.object_key private, media_type controlled string (DL-014)		
Collective	collective_members , project_members	project_members(project_id , member_id) is the Portfolio→Collective M:N bridge with explicit role ; display_order for listings		
Marketplace	marketplace_items , marketplace_media	Showcase only — no transactions (SCHEMA.md §60); external_url https-minimal (DL-013)		
Security	admin_audit_logs	Append-only trail for admin state mutations (verify/activate/cancel/ revoke/publish/create/update/delete); never exposed publicly		
Search	course_fts , project_fts , marketplace_fts	FTS5 virtual tables kept in sync by 9 insert/delete/update triggers (0007_fts.sql); power GET /api/search		

6.2 Relationships

users 1-N sessions	(user_id FK)
users 1-N orders	(user_id FK)
users 1-N enrollments	N-1 courses (both FKs; UNIQUE pair)
users 1-N progress	(user_id FK; lesson-scoped rows)
courses 1-N lessons	(course_id FK, ON DELETE CASCADE)
courses 1-N course_resources	(via lesson/course, course_resources FK)
courses N-M course_tags	
orders 1-N order_items	N-1 courses (order_id , course_id FKs)
orders 1-N payment_proofs	(order_id FK; object keys only)
projects 1-N project_media	(project_id FK)
projects N-M collective_members	(via project_members: composite PK (project_id-projects FK CASCADE, member_id – lookup enforced in code)
projects N-M project_tags	
marketplace_items 1-N marketplace_media	
users 1-N admin_audit_logs	(actor_user_id FK)

Critical invariants (HANDBOOK §3.3, DL-002/DL-004): entitlement is

enrollments.status='active' per (user, course); expires_at columns are

enforced in queries; no delete/role-change endpoints exist in V1, so

soft-state transitions (revoke/cancel/unpublish) are the only mutations, and

each writes an audit row.

7. Framework notes

- ESM-only: NestJS 12 with "type": "module"; hence

`NODE_OPTIONS=--experimental-vm-modules` in test scripts (DL-008) and

XoivgSP extension imports in source backend-full-documentation.pdf

- **Deviation from HANDBOOK_BACKEND §8** (Langkah 6/8): the handbook's

`vinext` /Workers binding path targets the frontend/Workers deployment. The backend is Node/Express and reaches D1 through the REST adapter behind the

`DatabaseService` token — a documented deviation, conflict K-06 in

`decision-log.md`.

- **Error surface:** `AllExceptionsFilter` returns

`{statusCode, message, error, timestamp, path}` with a generic 500 message —

no stack traces, upstream bodies, or secrets.

Format entri mengikuti PRD.md §92 (Open Decision Protocol) dan aturan RULES.md §84: setiap keputusan memiliki **Owner, Options, Decision, Date, Impact** — tidak ada keputusan yang diselesaikan diam-diam.

- Tanggal dokumen dibuat: 2026-09-12 (plan T4, .omo/plans/xolvon-backend-build.md).
- Entri DL-001–DL-010 adalah keputusan owner yang terkunci (draft .omo/drafts/xolvon-backend-build.md bagian "Keputusan owner (LOCKED oleh user)").
- Entri DL-011–DL-012 adalah default yang diadopsi backend dan menunggu tanda tangan owner.
- Konflik antar-dokumen dicatat di bagian [Konflik dokumen](#).
- Item yang masih terbuka diindeks di bagian [Keputusan masih terbuka](#) tanpa entri final.

Keputusan terkunci owner (LOCKED)

DL-001 — Lapisan database

Problem:
Repo memakai D1Service.query() langsung (D1 REST) sehingga pengembangan lokal butuh kredensial Cloudflare; handbook menetapkan DatabaseService + better-sqlite3 untuk lokal dan jalur migrasi ke D1 untuk production.

Existing Requirement:
HANDBOOK_BACKEND.md §2:27–63 dan §8:484–518; draft keputusan owner #1.

Options:
– Tetap memakai D1Service langsung di semua environment.
– DatabaseService better-sqlite3 (local.db) untuk lokal + adapter D1 di balik kontrak yang sama.

Owner:
Founder/BE Lead

Decision:
Ikuti handbook – DatabaseService berbasis better-sqlite3 (local.db) untuk development lokal; D1 REST dipertahankan hanya sebagai adapter di balik token DatabaseService saat DB_DRIVER=d1. Diimplementasikan di T1/T3.

Date:
2026-09-12

Impact:
Semua modul meng-inject token DatabaseService (bukan D1Service langsung); bootstrap lokal sukses tanpa kredensial Cloudflare; src/database/d1.service.ts tidak diubah sehingga invarian hardening (commit 0c83956) tetap terjaga.

DL-002 — Status order final

Problem:
Repo memiliki status keempat awaiting_verification, sementara handbook §3.4 FINAL dan SCHEMA.md §33 menetapkan hanya tiga status.

Existing Requirement:
HANDBOOK_BACKEND.md §3.4:112–120; SCHEMA.md §33; draft keputusan owner #2.

Options:
– Pertahankan awaiting_verification sebagai status terpisah.
– Collapse ke 3 status handbook (pending | paid | cancelled).

Owner:
Founder/BE Lead

Decision:
Status order FINAL: pending | paid | cancelled. awaiting_verification dihapus; submit payment proof tetap diizinkan dan mempertahankan status pending. Rekonsiliasi service, interface, migrasi, tes, dan tabel admin dikerjakan di T3.

Date:
2026-09-12

Impact:
verifyOrder hanya menerima order pending lalu menyetel paid; tes lama yang memakai awaiting_verification diperbarui eksplisit; SCHEMA.md §33 tidak berubah karena sudah final.

DL-003 — Model sesi

Problem:
Handbook §5.1/§5.5 memakai model session-token getSession(); repo memakai JWT access + refresh token berbasis tabel sessions. Perlu satu model yang dikunci.

Existing Requirement:
HANDBOOK_BACKEND.md §5.1:160–233, §5.5; draft keputusan owner #3.

Options:
– Session-token model handbook (getSession()).
– Cookie httpOnly.
– Pertahankan JWT + refresh token milik repo.

Owner:
Founder/BE Lead

Decision:
Pertahankan model JWT + refresh token milik repo. Deviasi dari handbook getSession() didokumentasikan lewat entri ini.

Date:
2026-09-12

Impact:
AuthGuard tetap membaca Authorization: Bearer; kolom sessions.refresh_token dan sessions.expires_at wajib ada di skema (DL-004); hardening expiry/refresh eksplisit dan pesan generik dikerjakan di T17.

DL-004 — Adopsi Schema V2

Xolvor Backend Documentation - xolvor-backend-full-documentation.pdf

Problem:
Kode repo meng-query kolom yang tidak ada di SCHEMA.md maupun migrasi repo (users.status/email_verified/phone_verified, sessions.refresh_token/expires_at, courses.price); handbook §3 menambah tabel dan kolom V2 yang belum diadopsi.

Existing Requirement:
HANDBOOK_BACKEND.md §3:67–154; SCHEMA.md §28, §125–128; draft keputusan owner #4.

Options:
– Kembalikan kode ke skema lama.
– Adopsi Schema V2 penuh + sinkronisasi SCHEMA.md.

Owner:
Founder/BE Lead

Decision:
Adopsi Schema V2: tabel course_tags, project_tags, dan FTSS (course_fts, project_fts, marketplace_fts); kolom sessions.refresh_token/expires_at/ip_address/user_agent, users.status/email_verified/phone_verified, courses.price, collective_members.display_order. Status enrollment FINAL active|revoked – expired dihapus, kolom expires_at dipertahankan dan ditegakkan di query. SCHEMA.md disinkronkan melalui proses perubahan skema eksplisit (T2).

Date:
2026-09-12

Impact:
Migrasi V2 19 tabel + index + FTSS idempoten; SCHEMA.md menjadi sumber kebenaran tunggal; entity/field/enum baru di luar kontrak wajib memperbaiki SCHEMA.md + entri decision log sebelum implementasi.

DL-005 — Admin bootstrap

Problem:
PRD §91.8 – mekanisme membuat admin pertama belum final; route registrasi admin publik dilarang.

Existing Requirement:
PRD §91.8, §44; HANDBOOK_BACKEND.md §7; ARCHITECTURE.md §165; draft keputusan owner #5.

Options:
– Route registrasi admin publik.
– Seed script server-side terkontrol.

Owner:
Founder/BE Lead

Decision:
Bootstrap admin via seed script server-side (dijalankan terkontrol di T19). Tidak ada route publik yang membuat admin; role tidak pernah diterima dari input user.

Date:
2026-09-12

Impact:
Delete dan role-change tidak memiliki endpoint di V1 (PRD §44 view-only); keadaan ini dicatat resmi di sini, bukan diselesaikan diam-diam. Endpoint admin (T12) murni read + aksi transisi status.

DL-006 — Ratifikasi payment_proofs

Problem:
PRD §91.6 menyatakan penyimpanan payment proof belum diputuskan, tetapi repo sudah mengimplementasikan tabel payment_proofs + alur submit proof. Konflik implementasi vs dokumen.

Existing Requirement:
PRD §91.6; SCHEMA.md §1023–1033 (open decision); draft keputusan owner #6.

Options:
– Hapus tabel dan alur proof.
– Ratifikasi implementasi sebagai keputusan resmi.

Owner:
Founder/BE Lead

Decision:
Tabel payment_proofs dan alur POST /orders/:id/payment-proof diratifikasi sebagai bagian kontrak V1.

Date:
2026-09-12

Impact:
PRD §91.6 terjawab; yang disimpan adalah object key (bukan berkas); antrean verifikasi admin memakai status order pending (DL-002).

DL-007 — R2 StorageService interface-first

Problem:
Modul media/R2 belum ada dan kredensial R2 production belum tentu tersedia saat pembangunan; implementasi tidak boleh menunggu atau menebak.

Existing Requirement:
HANDBOOK_BACKEND.md §6:393–414, §8:521–559; ARCHITECTURE.md §45–48; draft keputusan owner #7.

Options:
– Tunggu kredensial sebelum menulis kode media.
– Bangun StorageService ke interface, verifikasi menyusul.

Owner:
Founder/BE Lead

Decision:

Bangun StorageService interface-first sebagai provider yang dapat ditukar:
presigned PUT 3600s, private read URL 200s, object key digenerate server,
prefix sesuai ARCHITECTURE.md §48. Verifikasi terhadap R2 nyata menyusul saat
kredensial tersedia (T13/T19).

Date:
2026-09-12

Impact:
Konsumen (media, lesson video, proof) tidak berubah saat provider ditukar;
kredensial tidak pernah masuk kode, tes, atau respons client.

DL-008 — Test runner

Problem:
NestJS 12 memakai modul ESM murni sehingga runner test default tidak aktif
tanpa konfigurasi khusus; perlu keputusan runner yang stabil untuk semua todo.

Existing Requirement:
README bagian Run tests; draft keputusan owner #8.

Options:
- Migrasi ke runner/framework test lain.
- Pertahankan test:esm dan test:e2e.

Owner:
Founder/BE Lead

Decision:
Pertahankan npm run test:esm (unit/behavior) dan npm run test:e2e (harness
deterministik yang meng-override provider DatabaseService + PasswordService).

Date:
2026-09-12

Impact:
Seluruh todo plan memakai kedua runner tersebut; tidak menambah framework
test baru.

DL-009 — Scope build

Problem:
Cakupan build perlu dikunci agar tidak terjadi scope creep fitur V2.

Existing Requirement:
Draft keputusan owner #9; PRD §89-90; SCHEMA.md §169.

Options:
- Partial scope (modul inti saja).
- Scope penuh sampai production terverifikasi.

Owner:
Founder/BE Lead

Decision:
Scope penuh sampai deployment production terverifikasi (setara Hari 1-14
tracker BE), termasuk env/secrets/monitoring/rollback.

Date:
2026-09-12

Impact:
Fitur V2 (payment gateway otomatis, community/chat real-time, sertifikat,
review/rating, kupon/diskon, affiliate) tetap out of scope.

DL-010 — Deliverable docs

Problem:
Keputusan teknis tersebar di plan/chat; RULES.md §84 mewajibkan decision log;
SCHEMA.md §95 mewajibkan kontrak pagination dikunci sebelum implementasi.

Existing Requirement:
RULES.md §84:2340-2366; SCHEMA.md §95; draft keputusan owner #10.

Options:
- Catat keputusan hanya di plan/chat.
- Deliverable docs resmi di dalam repo.

Owner:
Founder/BE Lead

Decision:
Wajib ada docs/decision-log.md (berkas ini) dan docs/api-contract.md;
api-contract.md diperbarui setiap modul baru dan disempurnakan penuh di T16.

Date:
2026-09-12

Impact:
Tidak ada keputusan yang diselesaikan diam-diam; setiap [OPEN DECISION]
memiliki entri Owner/Options/Decision/Date/Impact di berkas ini.

Default yang diadopsi backend

Entri berikut adalah default teknis yang diadopsi backend agar implementasi
tidak terblokir. Keduanya menunggu tanda tangan owner; perubahan nilai setelah
implementasi wajib lewat entri decision log baru dengan owner manusia.

DL-011 — Kontrak pagination

Problem:
SCHEMA.md §95 – pagination behavior belum ditentukan sumber dokumen dan

DO NOT ASSUME berlaku; volume data mengharuskan pagination sehingga contract
wajib dikunci sebelum implementasi.

Xolvo Backend Documentation - xolvo-backend-full-documentation.pdf

Existing Requirement:

SCHEMA.md §95:2287-2298; plan T4 (memblokir T5/T7/T9/T16).

Options:

- Offset-based: ?offset= & ?count=.
- Page-based: ?page= & ?limit=.

Owner:

Backend (default adopted, track owner sign-off)

Decision:

Kontrak pagination: ?page= 1-based, ?limit= default 20 maksimum 100; respons list berbentuk {items, page, limit, total}. Berlaku untuk semua endpoint list (courses, projects, marketplace, search, admin).

Date:

2026-09-12

Impact:

T5/T7/T9/T16 mengikuti kontrak ini; owner dapat mengganti kontrak hanya dengan entri decision log baru sebelum T5 mulai.

DL-012 — Ambang rate limit auth

Problem:

PRD §104 mewajibkan pembatasan percobaan login; auth.controller.ts masih TODO; nilai ambang belum dikunci siapa pun.

Existing Requirement:

PRD §104:2718-2753; HANDBOOK_BACKEND.md §5.1; plan T17.

Options:

- 3 percobaan per menit per IP.
- 5 percobaan per menit per IP.
- 10 percobaan per menit per IP.

Owner:

Backend (default adopted, track owner sign-off)

Decision:

Rate limit /auth/login dan /auth/register: 5 percobaan per menit per IP via @nestjs/throttler (T17). Melebihi ambang mengembalikan HTTP 429.

Date:

2026-09-12

Impact:

Implementasi T17 memakai nilai ini; owner dapat mengganti ambang dengan entri decision log baru sebelum T17.

DL-013 — Validasi minimal external_url marketplace (governance tetap OPEN)

Problem:

Marketplace item menampilkan CTA ke website SaaS eksternal (PRD §55), tetapi governance `external\_url` belum final – PRD §91.11 dan SCHEMA.md §61 menegaskan aturan validasi/pengelolaan URL (valid + approved + managed) masih open decision dan melarang membangun business logic tambahan berdasarkan asumsi. Implementasi T9 tetap butuh aturan minimal agar endpoint tidak menerima nilai berbahaya.

Existing Requirement:

PRD §91.11 (External SaaS URL Governance); SCHEMA.md §61; plan T9.

Options:

- Tanpa validasi (menunggu governance final).
- Validasi minimal: URL absolut valid (URL parse) + https-only; http ditolak.
- Governance penuh: allowlist domain / approval workflow admin.

Owner:

Backend (default adopted, track owner sign-off)

Decision:

Validasi minimal diadopsi untuk V1 di MarketplaceService (create + update): `external\_url` harus parse sebagai URL absolut valid DAN berprotokol https; http ditolak dengan 400 (CTA me-redirect visitor keluar situs, target insecure adalah celah redirection security), string non-URL ditolak dengan 400. Governance final (approval/allowlist/management workflow) TETAP OPEN – tidak diimplementasikan dan tidak digantikan asumsi.

Date:

2026-09-12

Impact:

Endpoint marketplace menerima hanya https URL valid sejak T9; owner dapat mengganti/memperketat aturan dengan entri decision log baru tanpa mengubah kontrak respons. Kepatuhan MarketplaceModule sebagai showcase murni (tanpa pembayaran/transaksi, SCHEMA.md §60) tetap berlaku.

DL-014 — type / mediaType / role portfolio sebagai string terkontrol (enum final tetap OPEN)

Problem:

Plan T7 membangun modul Portfolio, tetapi SCHEMA.md §45 (project.type), §50 (project_media.media_type), dan §52 (project_members.role) menegaskan enumerasi final belum diputuskan dan melarang mengarang enum sendiri (RULES.md §84:2340-2366). Implementasi tetap butuh aturan validasi minimal.

Existing Requirement:

SCHEMA.md §45/§50/§52; RULES.md §84; plan T7; PRD §26-30.

Options:

- Mengarang enum final (dilarang SCHEMA.md – DO NOT INVENT).
- Menunda modul sampai enum final (memblokir T7 tanpa dasar owner).
- String terkontrol: wajib string non-kosong dengan batas panjang; nilai bebas sampai enum final dikunci owner.

Owner:

Backend (default adopted, track owner sign-off)

Decision:
kolom type, mediaType, dan role tidak ada di database sebagai string terkontrol (isRequired + @IsNotEmpty + @MaxLength: type 100, mediaType 50, role 100) di CreateProjectDto/AttachProjectMediaDto/AssignProjectMemberDto – TIDAK ada @IsEnum. Enumerasi final TETAP OPEN (indeks "Keputusan masih terbuka"); menguncinya nanti hanya menambah validator di DTO tanpa mengubah kontrak respons maupun query.
Date:
2026-09-12
Impact:
T7 selesai tanpa asumsi enum; media publik tetap dieksklusi video/object key; perubahan nilai setelah enum final dikunci owner wajib lewat entri decision log baru.

DL-015 — Kolom "WhatsApp / Proof Reference" tabel admin orders

Problem:
PRD §45 mendaftarkan kolom "WhatsApp / Proof Reference if applicable" untuk tabel admin orders/activation, tetapi menandainya belum final. Tabel skema tidak memiliki kolom whatsapp/proof terdedikasi, dan menambahkan field baru di luar kontrak tanpa keputusan owner dilarang (SCHEMA.md §127-128). Implementasi T12 butuh keputusan eksplisit untuk kolom ini.
Existing Requirement:
PRD §45:1266 (kolom belum final); SCHEMA.md §127-128; DL-006 (ratifikasi payment_proofs); plan T12.
Options:
- Menyimpan referensi WhatsApp/proof sebagai kolom dedikasi baru (pada orders atau tabel terpisah) dan mengeksposnya di respons admin. - Tidak menyimpan kolom dedikasi; baris payment_proofs adalah referensi proof yang sah, dan kontak WhatsApp admin tercantum terpisah di halaman publik.
Owner:
Founder/Product
Decision:
TIDAK ada kolom dedikasi untuk WhatsApp/proof reference di database maupun di respons endpoint admin. Referensi proof of payment tetap merujuk pada baris payment_proofs (object key per order) yang telah diratifikasi di DL-006; kebutuhan tampilan "Proof" di FE membaca dari sana. Kontak WhatsApp admin untuk recovery/reset sudah tersedia sebagai teks statis di halaman publik (PRD §42) dan bukan data transaksi.
Date:
2026-09-12
Impact:
GET /admin/orders (T12) tidak mengembalikan field whatsapp/proof; tidak ada perubahan skema untuk kolom ini. Bila owner kelak menghendaki kolom dedikasi, wajib lewat entri decision log baru + perubahan skema eksplisit (SCHEMA.md §127-128).

DL-015 implementation note — T12 admin proof surface

T12 implements the locked DL-015 decision without adding a WhatsApp/proof field to the admin response. GET /api/admin/orders exposes derived activation state and safe order/user fields only; payment proof references remain owned by the existing payment_proofs flow, while WhatsApp remains static public contact copy. Any dedicated transaction reference requires a new owner decision and schema change.

DL-016 — Implementasi rate limit auth dengan @nestjs/throttler

Problem:
DL-012 mengunci ambang (5 percobaan/menit/IP untuk /auth/login dan /auth/register) tetapi belum menetapkan mekanisme implementasinya; auth.controller.ts:37 masih TODO saat T17 dimulai.
Existing Requirement:
DL-012; PRD §104:2718-2753; HANDBOOK_BACKEND.md §5.1:160-233; plan T17.
Options:
- Implementasi manual (counter in-memory + middleware custom). - @nestjs/throttler (ThrottlerModule.forRoot + APP_GUARD + @Throttle per-route).
Owner:
Backend (default adopted per DL-012, track owner sign-off)
Decision:
Mekanisme = @nestjs/throttler ^6.5.0 (versi tercatat di package.json; peer range paket masih sampai ^11 sehingga .npmrc legacy-peer-deps=true ditambahkan). Konfigurasi produksi: global default 100 req/menit/IP per route (ThrottlerModule.forRoot, name 'default', ttl 60000), login DAN register diperketat ke 5 req/menit/IP via decorator @Throttle di AuthController (route metadata meng-override module default di @nestjs/throttler). Semua route lain memakai default 100/menit – /api/health dan konten publik GET tidak dibatasi agresif. Melebihi ambang – HTTP 429 dengan pesan standar throttler ("ThrottlerException: Too Many Requests") tanpa detail internal. Pesan kegagalan login tetap generik "Invalid email or password." untuk unknown-email maupun wrong-password (anti-enumerasi, dipertahankan + diuji). Expiry/refresh eksplisit: sesi kedaluwarsa dihapus (DELETE by id) saat refresh() dan dikembalikan 401 generik – diuji unit.
Date:
2026-09-12
Impact:
Ambang DL-012 kini aktif secara produksi via APP_GUARD ThrottlerGuard. Owner dapat mengganti ambang dengan entri decision log baru; e2e memakai

override module-options ketat (3/menit) untuk membuktikan pipeline guard myata, dan menguji limit login 5/menit sesuai angka produksi.

Xolvo Backend Documentation - xolvo-backend-full-documentation.pdf

Konflik dokumen (tercatat + resolusinya)

ID	Konflik	Resolusi		
K-01	Order status: repo memakai <code>awaiting_verification</code> (<code>0001_orders_enrollments.sql:16</code> , <code>orders.service.ts</code>) vs <code>HANDBOOK_BACKEND.md §3.4 FINAL</code> dan <code>SCHEMA.md §33</code> <code>pending</code>	paid	cancelled	DL-002 — repo direkonsiliasi ke 3 status (T3)
K-02	Enrollment status: migrasi repo mengizinkan <code>expired</code> vs <code>SCHEMA.md §28</code> hanya <code>active</code>	revoked	DL-004 — <code>expired</code> dihapus; <code>expires_at</code> dipertahankan dan ditegakkan di query	
K-03	Kolom ekstra: kode meng-query <code>users.status/email_verified/phone_verified</code> (<code>auth.service.ts:58-61</code>) dan <code>sessions.refresh_token/expires_at/ip_address/user_agent</code> (<code>auth.service.ts:150-152</code>) tanpa migrasi maupun entri <code>SCHEMA.md</code>	DL-004 — diadopsi ke Schema V2, <code>SCHEMA.md</code> disinkronkan (T2)		
K-04	Model sesi: <code>handbook getSession()</code> (<code>session-token</code>) vs repo <code>JWT + refresh bearer</code>	DL-003 — repo dipertahankan; deviasi didokumentasikan		
K-05	<code>payment_proofs</code> sudah terimplementasi di repo padahal <code>PRD §91.6</code> dan <code>SCHEMA.md</code> menandainya <code>open decision</code>	DL-006 — diratifikasi		
K-06	Deployment: <code>handbook §8 Langkah 6/8</code> memakai jalur <code>vinext /Workers</code> binding; repo <code>BE</code> memakai <code>Node/Express</code> (<code>@nestjs/platform-express</code>) + <code>adapter D1 REST</code>	Akan dicatat resmi sebagai deviasi terdokumentasi saat <code>production (T20)</code> ; <code>BE</code> tetap <code>Node/Express</code>		

Keputusan masih terbuka

- Indeks item `PRD §91` dan plan yang **belum** diputuskan — entri final dengan `Owner/Options/Decision/Date/Impact` akan ditambahkan saat masing-masing diputuskan (tidak ada entri placeholder di log ini):
- PRD §91.1 — penempatan `Solve On (/solve-on)` vs `Home Section/Anchor`.
 - PRD §91.2 — `auth library` final (model sesi sudah dikunci di `DL-003`).
 - PRD §91.3 — `password hashing` final disesuaikan runtime (repo saat ini: `argon2`).
 - PRD §91.4 — `runtime/adapter deployment` final (dikunci di `T20`, lihat `K-06`).
 - PRD §91.5 — mekanisme/provider `QR` atau `payment`.
 - PRD §91.7 — `assignment: resource` saja vs `submission workflow`.
 - PRD §91.9 — `legal copy (Terms/Privacy/Disclaimer)`.
 - PRD §91.10 — jumlah konten awal.
 - PRD §91.12 — `analytics provider` pihak ketiga.
 - Plan T7 — `enum type media project` (string terkontrol diadopsi di `DL-014`; enumerasi final masih terbuka).
 - Plan T9 — `governance external_url marketplace` (validasi minimal `https` diadopsi di `DL-013`; `approval/allowlist/management` masih terbuka).
 - Plan T12 — kolom `WhatsApp/proof` pada tabel `admin` (diputuskan di `DL-015`: tidak ada kolom dedikasi; `payment_proofs` adalah referensinya).
 - T19 — `analytics/monitoring provider` pihak ketiga (interface `MonitoringPort` tersedia; provider dan credential tetap `OPEN` sampai keputusan owner).

DL-017 — Environment, admin bootstrap, and monitoring boundary

Problem:
T19 membutuhkan pemisahan `local/staging/production`, bootstrap admin pertama, dan structured logging/error monitoring tanpa memasukkan credential provider yang belum dipilih.

Existing Requirement:
`ARCHITECTURE.md §§79-90/165`, `RULES.md §52`, `PRD.md §105`, T19 plan.

Options:
- Public admin registration or ad-hoc production seed data.
- Controlled server-side command using `ADMIN_BOOTSTRAP_EMAIL/PASSWORD`.

Owner:
Founder/BE Lead

Decision:
Use `APP_ENV` values `local/staging/production` with separate secret stores. The server-side ``npm run seed:admin`` command hashes through `PasswordService`, inserts `role=admin`, and is idempotent without a public route or committed credentials. Structured monitoring is represented by `MonitoringPort` only; provider selection and credentials remain `OPEN DECISION`. Rollback is deployment revert plus backup restore or a reviewed compensating migration; applied migrations are never edited.

Date:
2026-09-12

Impact:
Local dry-run is supported without Cloudflare credentials. Staging/production live deployment and provider selection remain T20 work and are not claimed by T19.

Addendum hardening decisions (T1)

DL-018 — Bearer-only request integrity

Owner:
Founder/BE Lead

Options:
- Add a cookie-backed CSRF token flow.
- Keep the API Bearer-only and enforce browser request-integrity checks using allowed `Origin/Referer` and `Fetch-Metadata` policy for mutation requests.

Decision:
Use Bearer-only request integrity. Do not add a CSRF cookie token because the

current API authentication contract uses Authorization Bearer tokens rather than cookie authentication. Browser mutation requests will be checked against the explicit request integrity policy in the Xdovvo-Request-Integrity-Policy-Header.pdf Date: 2026-09-12 Impact: No CSRF cookie or token secret is added. No-cookie Bearer API clients remain supported; GET requests are unaffected. Non-browser service-to-service access requires the explicit policy defined during implementation.
--

DL-019 — Playwright PDF artifacts

Owner: Founder/BE Lead Options: - Keep documentation as Markdown only or generate PDFs outside the repository. - Generate Markdown-to-PDF artifacts with Playwright/Chromium and commit them. Decision: Use Playwright/Chromium for the approved Markdown-to-PDF workflow. Generated documentation PDFs are committed, including the combined document in the order README, Architecture, Security, and Deployment Runbook. Date: 2026-09-12 Impact: PDF output becomes a reviewed repository artifact and must be regenerated and secret-scanned when its source documentation changes. No credentials or private environment values may appear in the source or generated PDFs.
--

DL-020 — Environment CORS whitelist and origin placeholders

Owner: Founder/BE Lead Options: - Continue using one FRONTEND_URL origin. - Parse CORS_ALLOWED_ORIGINS as an environment-specific explicit whitelist. Decision: Use CORS_ALLOWED_ORIGINS as the explicit whitelist, with a localhost fallback for local development. Staging and production origins remain OPEN placeholders until the owner supplies final domains; wildcard origins are not approved when credentials are enabled. Date: 2026-09-12 Impact: T5 must preserve local startup without a final deployment domain and must reject unknown credentialed origins. Staging and production deployment evidence remains blocked until real target origins are provided; no domain is fabricated here.
--

DL-021 — CSP connect and media origins

Owner: Founder/BE Lead Options: - Allow broad CSP connect-src and media-src values. - Use explicit environment origin lists and leave unavailable deployment origins open until owner confirmation. Decision: Use explicit CSP connect-src and media-src origin lists. Final frontend, storage, and other deployment-specific origins remain OPEN placeholders until the owner confirms them. Do not enable unsafe-inline or unsafe-eval as a default. Date: 2026-09-12 Impact: T4 must document the approved local baseline and make staging/production CSP origins configurable without embedding unverified domains or secrets.

DL-022 — Monitoring provider and recipients

Owner: Founder/BE Lead Options: - Select a monitoring vendor and alert recipients now. - Keep MonitoringPort as the boundary until provider and recipient decisions are approved. Decision: Monitoring provider and alert recipients remain OPEN. The existing MonitoringPort interface is the only approved integration boundary for this stage; no vendor SDK, endpoint, recipient address, or credential is selected. Date: 2026-09-12 Impact: Later monitoring work must record the provider, recipients, data boundary, and secret-store location in a new decision entry before production wiring.

DL-023 — Security scanner choice

Owner: Founder/BE Lead

Options:
- Adopt a named third-party or paid security scanner.
- Use repository-native scans and npm audit while scanner selection is open.

Decision:
Security scanner choice remains OPEN. T1 records no vendor commitment; current verification uses the repository-native secret-pattern review and the existing quality commands. A future scanner requires explicit owner approval and a new decision entry.

Date:
2026-09-12

Impact:
No scanner credentials, paid service, or CI integration is added. Residual scan coverage and any selected tool must be documented before the final security gate.

DL-024 — Postman production placeholders

Owner:
Founder/BE Lead

Options:
- Commit populated production credentials or target-specific values.
- Commit production environment structure with blank placeholders only.

Decision:
Postman production environments may contain names, URLs marked as placeholders, and blank secret variables only. No access token, password, API key, cookie, or other production credential may be committed.

Date:
2026-09-12

Impact:
T11 can provide importable production structure without claiming production readiness. Operators must supply values through an external secret-safe workflow.

DL-025 — Conservative cleanup policy

Owner:
Founder/BE Lead

Options:
- Remove ambiguous files and generated artifacts during hardening.
- Preserve source, migrations, evidence, and user changes; remove only proven temporary artifacts with a deletion ledger entry.

Decision:
Use conservative, reversible cleanup. Preserve source, migrations, evidence, approved PDFs, and unrelated user changes. Delete only an artifact proven to be temporary, and record its path and rationale in the cleanup ledger. Do not edit or delete secrets as a substitute for rotation or incident handling.

Date:
2026-09-12

Impact:
T12 must produce an explainable file-by-file cleanup diff and must leave required files intact. Any tracked environment secret requires separate incident handling, not silent cleanup.

DL-026 — DL-019 disposition: official PDF pipeline vs stray Python artifact

Problem:
DL-019 locked the official PDF workflow as Playwright/Chromium Markdown-PDF ('npm run docs:pdf' → 'scripts/docs-pdf.mjs', output 'docs/pdf/*.pdf', committed). Commit '9a6d1d7' ("feat: add throttle and trust proxy configurations with API documentation and Postman collection") arrived out-of-band and introduced two non-conforming artifacts: 'scripts/generate-api-pdf.py' (Python/reportlab generator reading the root Postman collection) and 'docs/Xolvon-API-Documentation.pdf' (its output, sitting loose in 'docs/').

Existing Requirement:
DL-019; plan T8/T9/T12 ('.omo/plans/xolvon-addendum-hardening-docs.md'); conservative cleanup policy DL-025 (nothing deleted without a ledger entry).

Options:
- Accept the Python-generated PDF as an additional official artifact.
- Delete the stray files immediately during T8.
- Record them as non-conforming and schedule removal/replacement at T9, where the committed Playwright PDFs (individual + combined, order README → Architecture → Security → Deployment Runbook) supersede them and the T12 deletion ledger documents the removal.

Owner:
Founder/BE Lead

Decision:
The only approved PDF generation path is Playwright: official handover docs are Markdown under 'docs/' and PDFs are generated exclusively via 'npm run docs:pdf' ('scripts/docs-pdf.mjs'). 'scripts/generate-api-pdf.py' and 'docs/Xolvon-API-Documentation.pdf' are formally declared NON-CONFORMING to DL-019 and scheduled for removal/replacement at T9 (Playwright committed PDFs; combined output covers the Postman-derived API content once T10/T11 reconcile the contract). This entry records the disposition only – per DL-025 the files are NOT deleted at T8; deletion happens in the T9/T12 pass with a deletion ledger line.

Date:
2026-09-12

Impact:
T8 documentation ships as Markdown + the sanctioned Playwright pipeline only; no second PDF generator becomes canonical. Reviewers must not treat the stray Python/PDF pair as authoritative. T9 regenerates all committed PDFs; T12's cleanup ledger carries the eventual deletion of both stray paths.

Disposition (executed at T9, 2026-09-12 — append-only ledger line):

~~Deleted as scheduled. Files removed: scripts/generate-api-pdf.py and docs/Xolvon-API-Documentation.pdf~~. Reason: non-conforming to the locked Playwright-only PDF pipeline (DL-019/DL-026); superseded by the committed Playwright outputs docs/pdf/README.pdf, docs/pdf/architecture.pdf, docs/pdf/security.pdf, docs/pdf/deployment-runbook.pdf and docs/pdf/xolvon-backend-full-documentation.pdf (combined order README → Architecture → Security → Deployment Runbook), all generated by `npm run docs:pdf (scripts/docs-pdf.mjs)`. No source, test, or script referenced the deleted paths (workspace grep: zero hits); docs/README.md \$6 mentions them only as scheduled-for-removal notes, which this entry closes. The commit 9a6d1d7 strays 'Xolvon-API.postman_collection.json' and root README.md remain untouched pending T11/T12 disposition.

DL-027 — Final staging/production origins and CSP origins remain OPEN

Problem:
T8 documentation must describe staging/production deployment without inventing domains. DL-020 locked the CORS whitelist mechanism and DL-021 locked the explicit CSP origin-list mechanism, but the concrete values – final staging and production frontend origins and the CSP 'connect-src'/media origins that depend on them – have never been supplied by the owner, and 'docs/qa-results.md' confirms no real domains exist in this workspace.

Existing Requirement:
DL-020, DL-021; plan T8 acceptance ("all current route/auth decisions are consistent"); addendum open-decision list ("Final staging and production frontend origins", "Final CSP connect-src/media origins").

Options:
– Populate example values in the docs/templates and treat them as final.
– Leave the decision open with a named owner, documenting the placeholder fallback behavior until final domains exist.

Owner:
Founder/Backend

Decision:
FINAL staging and production frontend origins, and the CSP 'CSP_CONNECT_SRC' connect/media origin lists derived from them, REMAIN OPEN DECISIONS with owner = Founder/Backend. Until they are supplied: 'CORS_ALLOWED_ORIGINS' and 'CSP_CONNECT_SRC' stay unset (CORS falls back to 'FRONTEND_URL', CSP defaults to 'self' + 'R2_ENDPOINT' origin), no domain values are fabricated in docs, templates, Postman, or PDFs, and deployment evidence for those environments remains blocked. Closing this decision requires a new entry here with the exact approved origins per environment.

Date:
2026-09-12

Impact:
'docs/README.md', 'docs/security.md', and 'docs/deployment-runbook.md' describe these as open with the safe fallback behavior; the final gate (F3/F4) cannot claim CORS/CSP production readiness until the owner records the origins here.

DL-028 — OpenAPI contract is exported as JSON only; no Swagger UI is served

Problem:
T10 must publish the HTTP contract as 'docs/openapi.json' without weakening the Bearer-only security posture. Serving the generated Swagger UI as an unauthenticated route would expose a full API map (and an interactive "try it" console) to anonymous visitors on every environment, including staging and production, at zero product benefit for API clients that already have the committed artifact.

Existing Requirement:
DL-016 (JWT Bearer-only auth); addendum Must-NOT-have "no paid/extra public attack surface without owner decision"; plan T10 acceptance ("decide Swagger UI exposure (local/internal only unless auth protected) in decision log").

Options:
– Serve 'SwaggerModule.setup()' at '/api/docs' in all environments.
– Serve the UI only when 'APP_ENV=local'.
– Export the document as a committed JSON artifact and never serve a UI route.

Owner:
Founder/Backend

Decision:
The application serves NO Swagger UI in ANY environment (local included). The contract ships as the committed 'docs/openapi.json', regenerated only by 'npm run docs:openapi' ('src/openapi/export-openapi.ts'), which bootstraps the real app offline, forces local drivers, and fails if any live route is missing from the document. Reviewers and clients render the JSON with their own tools. Reopening UI exposure requires a new entry here plus an auth protection design.

Date:
2026-09-12

Impact:
'package.json' gains 'docs:openapi'; 'src/openapi/*' adds the document factory, export script, and contract tests; 'docs/openapi.json' becomes a generated artifact checked for secret-free content by 'findSecretLeaks' and the export spec. No new HTTP route is added to the running application.

DL-029 — Stray root Postman collection relocated into postman/ and removed

Problem:
Plan T11 requires the Postman artifacts at 'postman/xolvon-backend.postman_collection.json'

plus `postman/environments/\*.postman\_environment.json`. Commit `9a6d1d7` (already flagged by DL-026 for its stray PDF-psir) additionally placed `Xolvon-API.postman_collection.json` in the repo ROOT. That collection covers all 65 canonical operations (66 requests) but cannot serve the T11 acceptance criteria: zero test scripts (so `npx newman run` asserts nothing and cannot be a smoke gate), zero saved success/error examples, its second login silently OVERRIDES `accessToken` (the admin/user token chain breaks mid-folder), no `postman/environments/` files exist (staging/production placeholders per DL-024 were never created), and the root location contradicts the planned layout and the single-source-of-truth rule of DL-026/T14 reconciliation.

Existing Requirement:

DL-024 (production Postman environments = names + placeholder URLs + blank secrets only); DL-025 (conservative cleanup, every deletion logged); DL-026 (this is the Postman half of the 9a6d1d7 stray disposition); plan T11 (`.omo/plans/xolvon-addendum-hardening-docs.md`).

Options:

- Keep both collections and hope consumers pick the right one.
- Move the root file as-is into postman/ (keeps the coverage gap and broken layout).
- Rebuild the canonical collection at postman/ from docs/openapi.json, port every useful request from the stray file (its token-capture login flow, folder taxonomy, and placeholder bodies are all carried over), then delete the root copy under this ledger entry.

Owner:

Founder/Backend

Decision:

Option 3 executed at T11. `postman/xolvon-backend.postman\_collection.json` (11 folders / 75 requests) now covers ALL 65 openapi operations (checker-verified against docs/openapi.json) with class-validator-legal placeholder bodies, per-request Newman test scripts, saved success + 401/403/404/409/429 examples whose error envelopes match the real guard/service throw strings, Bearer token capture into collection variables (cookie jar explicitly unused – the API is Bearer-only), and a self-contained `10. Newman Smoke` folder. Environments ship under `postman/environments/{local,staging,production}.postman\_environment.json`; staging/production contain placeholder URLs and blank secret variables per DL-024, and final staging/production origins remain OPEN per DL-027. `npm run test:newman` is the documented local smoke path. The root `Xolvon-API.postman\_collection.json` is DELETED in the same commit as the replacement (this entry is its ledger line). The stale root README \$ "Postman Collection Integration" link is left for the T14 docs-reconciliation pass rather than expanded into this test commit.

Date:

2026-09-12

Impact:

Single canonical Postman artifact; the stray root file no longer misleads consumers; T14 reconciles README prose. No environment secret is committed (secret scan recorded in .omo notepad evidence t11). Deletion is reversible: the old file remains recoverable from `git show 9a6d1d7:Xolvon-API.postman\_collection.json`.

Cleanup ledger (T12)

Executed 2026-09-12 under the DL-025 conservative cleanup policy. Audit covered

`git status`, `git ls-files`, `git ls-files -o --exclude-standard`, and a

root/subfolder scan for `*.tmp`, `*.bak`, `*.log`, `Thumbs.db`, `.DS_Store`

(none tracked, none present). Tracked total before and after: 266 files — no

tracked file was deleted by this task; every deletion below is an untracked,

gitignored working-copy remnant of a scratch file that was previously committed

and already purged from tracking by commit `e91dc97` ("chore: cleanup gitignore,

purge scratch artifacts"). Each deleted file's full content remains recoverable

from history via `git show e91dc97^:<path>`.

Deleted (proven stale scratch, zero tracked references — verified by `git grep`)

Path	Reason	Recovery
t15-verification.md	Stale T15 verification scratch at repo root; no reference in any tracked doc/script/test.	<code>git show e91dc97^:t15-verification.md</code>
t3-happy.txt	Stale T3 evidence scratch at repo root; only cross-referenced by the untracked root <code>DoneClaim</code> scratch file itself, not by tracked material.	<code>git show e91dc97^:t3-happy.txt</code>
t3-failure.txt	Same as above.	<code>git show e91dc97^:t3-failure.txt</code>
learning-t3.md	Stale T3 learning scratch at repo root; only referenced by the untracked <code>DoneClaim</code> scratch, not by tracked material.	<code>git show e91dc97^:learning-t3.md</code>

Kept — needs owner decision (referenced by tracked `docs/qa-results.md`)

Path	Why kept
DoneClaim	Untracked scratch, but <code>docs/qa-results.md</code> §T19 cites it by name as the historical record that production credentials were never exercised. Deleting would strand a live documentation reference. Owner may delete once T14 reconciles the qa-results prose.
evidence-t19-happy.txt	Same: cited by name in <code>docs/qa-results.md</code> :34-35.
evidence-t19-failure.txt	Same: cited by name in <code>docs/qa-results.md</code> :34-35.
t20-happy.txt	Cited by name in <code>docs/qa-results.md</code> (BLOCKED-gate evidence for T20).
t20-failure.txt	Cited by name in <code>docs/qa-results.md</code> (BLOCKED-gate evidence for T20).

Disposition notes (NOT deletions; guarded per T12 scope)

- No tracked file under `src/`, `migrations/`, `test/`, or this

`docs/decision-log.md` was removed or altered beyond this ledger append.

- The out-of-band root `README.md` rewrite (commit `65ecc4d` era / post-T work)

and its stale `Xolvon-API.postman_collection.json` link are left untouched —

owned by the T14 docs-reconciliation pass (already anticipated by DL-029).

~~Xolvon Bomo/ and docs/pdf/ were not touched (committed artifacts by owner~~

decision, DL-019/DL-026).

- No git history was rewritten; nothing was reset, rebased, or reverted.

Environment / secret hygiene findings

- `git ls-files | grep -E '^(^/)\.env$' → empty`: no real `.env` is tracked.
- `git log --all --oneline -- .env '**/.env' → empty`: no real `.env` was ever

tracked in history. No security-incident/rotation action required.

- `.gitignore` verified: ignores `.env`, `.env.*` (with `!.env.example / !.env.*.example` exceptions), `local.db / local.db-journal / local.db-wal / local.db-shm / *.db*`, `DoneClaim`, `evidence-*.txt`, `learning-*.md`, `t[0-9]*-*.txt`, `t[0-9]*-*.md`, `/.omo`, `.codegraph`, `dist`, `buildinfo`,

OS junk (`.DS_Store`, `Thumbs.db`), and `*.tmp / tmp/` patterns.

- `.env.example`, `.env.local.example`, `.env.staging.example`, `.env.production.example` all remain present and unmodified.
- Variable-name reconciliation vs `src/config/env.validation.ts`: MATCH — every

validated variable (`JWT_SECRET`, `FRONTEND_URL`, `PORT`, `APP_ENV`,

`DB_DRIVER`, `STORAGE_DRIVER`, `CLOUDFLARE_ACCOUNT_ID`,

`CLOUDFLARE_D1_DATABASE_ID`, `CLOUDFLARE_API_TOKEN`, `R2_ENDPOINT`,

`R2_BUCKET`, `R2_ACCESS_KEY_ID`, `R2_SECRET_ACCESS_KEY`) appears in the

example files. No silent source edits were made.

- FLAG (informational, no action by T12): `CORS_ALLOWED_ORIGINS` is consumed by

`src/config/cors.ts` (falls back to `FRONTEND_URL`, then the local dev

origin, so bootstrapping is unaffected) but is documented in none of the

four example files. Owner/docs task should add it to `.env.staging.example`

and `.env.production.example` alongside the DL-027 final-origin decision.

Dependency vulnerability management (T13)

DL-030 — Dependabot + npm audit as the scanning approach; 4 residual HIGHs are owner decisions

Problem:
T2/T13 `npm audit --json` (2026-09-12, evidence
`.omo/notepads/xolvon-addendum-hardening-docs/evidence/t13-audit-before.json`)
reports 0 critical, 4 high, 1 moderate, 2 low. `npm audit fix` (non-breaking,
no `--force`) is a verified no-op: it exits 1 with the identical 7 findings and
produces an empty git diff on package.json/package-lock.json — every fix is
either unavailable or semver-major/breaking, so no safe automated remediation
exists today. Per the addendum rule "no paid scanner without owner decision"
(DL-023), the recurring-scan mechanism must be chosen and the unfixable HIGHs
must be recorded, not silently left open.

Existing Requirement:
DL-023 (scanner choice was OPEN, paid vendor requires owner approval); plan T13
(`.omo/plans/xolvon-addendum-hardening-docs.md`); addendum §1.5.

Options:
(a) Paid scanner vendor — rejected, requires owner approval per DL-023.
(b) Dependabot (free, repository-native, GitHub-hosted) — selected.
(c) Manual periodic `npm audit` only — no continuous coverage.

Decision:
1. Scanning approach: GitHub Dependabot via `.github/dependabot.yml` — weekly
npm version updates (mon 07:00 Asia/Jakarta, minor+patch grouped,
open-pull-requests-limit: 5, commit-message prefix `chore` to match repo
style) plus the github-actions ecosystem. npm audit remains the local gate
(`npm audit --json` in final verification). No paid scanner/vendor was added.
Enabling Dependabot security updates in repository settings is an owner/repo
admin action (GitHub UI), noted here as the activation dependency.
2. Residual HIGH advisories — owner decisions, deliberately NOT auto-fixed
(all fixes are breaking or nonexistent; forcing them would violate the
no-break-the-build rule and this repo does not touch app source for audit):

- multer <=2.2.0 [HIGH] (4 advisories incl. GHSA-wc9g-mqfw-jrwm DoS via
crafted multipart field names; GHSA-qfvm-cv95-jqjf fd-leak DoS;
GHSA-535w-7cp7-47q4 oversized-array-index DoS; GHSA-qvfw-j98x-7q72 size
-limit bypass [low-severity sub-advisory]) pulled in by
@nestjsjs/platform-express@12.0.1. fixAvailable=false: the newest
@nestjsjs/platform-express (12.0.1) still pins multer@2.2.0 and multer@2.3.0
(the fixed line) is not consumed by any platform-express release. Why not
auto-fixed: would require either waiting for an upstream NestJS patch or
adding a package.json "overrides" entry for multer@^2.3.0 — an unaudited
manifest change outside this task's permitted remediation set.
Owner options: add "overrides: {\"multer\": \"^2.3.0\"}" after validating the
upload paths (admin media upload, payment-proof), or wait for
@nestjsjs/platform-express to ship the bump. Mitigating facts: multer is
only reached through Nest platform-express multipart routes; core app
uploads flow through presigned R2 URLs, not multipart bodies.
- @nestjsjs/platform-express * [HIGH] — same finding surfaced on the direct
dependency (via: multer). Same owner decision as above.
- undici <=6.27.0 [HIGH] (15 advisories incl. GHSA-f269-vfmq-vjvj WebSocket
length overflow crash; GHSA-vm6-8vpv-qv8q unbounded memory;
GHSA-v9p9-hfj2-hcw8 unhandled exception; GHSA-vxpw-j846-p89q fragment DoS)
and tmp <=0.2.5 [HIGH] (GHSA-ph9p-34f9-6g65 path traversal;
GHSA-52f5-9888-hmc6 symlink write), plus @nestjsjs/mau * [MODERATE] and the
two linked LOWs (inquirer 3.0.0-8.2.6||9.0.0-9.3.7 via external-editor;
external-editor >=1.1.1 via tmp): ALL reachable only through the
devDependency @nestjsjs/mau@0.2.6 (chain: @nestjsjs/mau -> ... -> undici;
@nestjsjs/mau -> inquirer -> external-editor -> tmp). The only fix npm

offers is @nestjs/mau@0.0.6 marked isSemVerMajor:true – a downgrade to a much older artifact, breaking, and 0.0.6 does not even appear in the package's published version list. Why not auto-fixed: major/breaking, explicitly out of safe scope. Owner options (recommended): verify @nestjs/mau is genuinely unused (grep of src/test/scripts/nest-cli.json finds ZERO references – it appears to be leftover Nest CLI tooling) and `npm uninstall @nestjs/mau`, which closes 1 moderate + 2 high + 2 low in one step with no runtime impact; or pin and accept, since all four dev-only packages never ship in `dist/` and are not reachable in the production runtime (npm-start path is `node dist/main` with no inquirer/tmp/undici imports from app code).

Date:
2026-09-12

Impact:
Continuous dependency scanning lands without any paid vendor (DL-023 closed: Dependabot selected). Gates stay green untouched by dependency churn (package.json/package-lock.json unchanged this task). The 4 HIGH + 1 MODERATE + 2 LOW residuals become explicit, dated owner decisions with concrete paths instead of audit noise; production runtime is dev-only-surface for 6 of the 7 findings, and the multer pair depends on an upstream NestJS release or a reviewed overrides entry.

ID: DL-033
Title: Video ObjectKey Attachment to Lesson (B.1)
Scope: Module (lessons, admin)
Status: Decided – default engineering, menunggu konfirmasi bisnis

Problem:
Tidak ada jalur API untuk mengaitkan `video\_object\_key` yang sudah di-upload dan dikonfirmasi via storage presigned URL ke record `lessons` yang bersangkutan.

Existing Requirement:
Admin harus dapat mengaitkan video lesson yang tersimpan di private storage bucket ke record database pelajaran terkait (PRD §21-23).

Options Considered:
1. Menambahkan parameter `videoObjectKey` ke `PATCH /api/lessons/:id`. (Ditolak: memungkinkan perubahan metadata video tanpa validasi kepemilikan path/prefix dan riwayat konfirmasi media).
2. Endpoint terpisah `PATCH /admin/lessons/:id/video` (admin only) dengan body `{ objectKey: string }`, validasi prefix ketat `private/courses/{course\_id}/lessons/{lesson\_id}/video/`, verifikasi status `confirmed` pada tabel `media\_objects`, dan pencatatan audit log `attach\_lesson\_video`. (Dipilih).

Decision:
Implementasikan `PATCH /admin/lessons/:id/video` (dan alias `PATCH /lessons/:id/video`) dengan:
1. Validasi lesson existence (404).
2. Validasi traversal `..`, `\\` (400).
3. Validasi course-lesson prefix matching: penolakan cross-lesson key dengan 403 Forbidden.
4. Validasi rekaman konfirmasi pada `media\_objects` (400 bila belum dikonfirmasi).
5. Audit log mutasi `attach\_lesson\_video` ke `admin\_audit\_logs`.

Date:
2026-09-14

Impact:
Admin panel memiliki alur deterministik aman untuk memasang video pelajaran tanpa celah salah taut atau akses sembarang file di private storage.

ID: DL-034
Title: Checkout Policy for Active Enrollment & Draft Course Checkout (B.2)
Scope: Module (orders)
Status: Decided – default engineering, menunggu konfirmasi bisnis

Problem:
User dapat membuat order baru untuk course yang sudah aktif dimilikinya, dan user dapat mengecek ID course berstatus draft melalui `POST /orders` jika mengetahui UUID-nya.

Existing Requirement:
E1b order ganda saat enrollment aktif dan FIND-T9-06 checkout course draft belum memiliki definisi status code yang konsisten.

Options Considered:
1. Mengizinkan order ganda (misal untuk perpanjangan masa aktif). (Ditolak: sistem belum memiliki konsep subscription/renewal di V1).
2. Menolak order untuk course yang sudah berstatus `active` dengan 409 Conflict, dan menolak course yang bukan `published` (draft) dengan 404 Not Found agar konsisten dengan public catalog. (Dipilih).

Decision:
Di `OrdersService.checkout()`:
1. Cek status course di database: jika ada course yang statusnya bukan `published`, tolak dengan 404 Not Found ("Course tidak ditemukan atau belum dipublikasikan").
2. Cek tabel `enrollments`: jika user sudah memiliki enrollment berstatus `active` untuk course yang dipesan, tolak dengan 409 Conflict ("Anda sudah memiliki akses aktif ke course ini").

Date:
2026-09-14

Impact:
Mencegah user membayar ganda untuk materi yang sudah diakses, dan mencegah kebocoran status materi draft ke pengguna biasa.

ID: DL-035
Title: Server-Minted User-Scoped Payment Proof Keys (B.3)
Scope: Module (orders, media)
Status: Decided – default engineering, menunggu konfirmasi bisnis

Problem:
User berpotensi menyerempet atau menggunakan objectKey milik user lain jika object key generation bukti bayar tidak memiliki kepemilikan yang terisolasi per user di path storage.

Existing Requirement:

BUG-T9-01 residual: pencegahan IDOR dan substitusi file pada bukti transfer pembayaran.

Xolvo Backend Documentation - xolvo-backend-full-documentation.pdf

Options Considered:

1. Mengizinkan client menentukan object key acak di bawah prefix umum `private/proofs/`. (Ditolak: security by obscurity).
2. Endpoint server-minted `POST /orders/:id/payment-proof-url` yang menghasilkan presigned PUT dengan path spesifik `private/users/{userId}/proofs/{uuid}.{ext}`, menyimpan rekaman status `issued` di tabel `media\_objects`, dan memvalidasi kepemilikan strictly via `assertOwnProofKey` (403 Forbidden bila berbeda user). (Dipilih).

Decision:

1. Buat migration `0009\_media\_objects.sql` untuk pelacakan metadata objek penyimpanan.
2. Endpoint `POST /orders/:id/payment-proof-url` menghasilkan URL PUT dengan path terisolasi per user ID JWT.
3. `POST /orders/:id/payment-proof` memverifikasi kesesuaian path dengan JWT caller.

Date:

2026-09-14

Impact:

Tutup tuntas potensi kebocoran dan substitusi bukti pembayaran lintas pengguna dengan isolasi path kriptografis dan verifikasi token.

ID: DL-036

Title: Stateless JWT Logout Window ≤ 15 Minutes as Known Limitation (B.4)

Scope: Architecture (auth, security)

Status: Decided - default engineering, menunggu konfirmasi bisnis

Problem:

Setelah logout, JWT access token yang sudah diterbitkan tetap valid secara kriptografis hingga masa berlaku 15 menit habis karena sistem menggunakan stateless JWT tanpa central blacklist/Redis.

Existing Requirement:

BUG-T6-window: audit menemukan token masih dapat mengakses route ber-guard sebelum TTL 15 menit berakhir.

Options Considered:

1. Membangun distributed cache / Redis denylist di edge Cloudflare. (Ditolak: perubahan arsitektur besar yang menambah latensi jaringan dan dependensi infrastruktur baru di luar lingkup V1).
2. Menerima window 15 menit sebagai known limitation arsitektur stateless V1, dengan dokumentasi eksplisit di `docs/security.md` dan pembatalan segera refresh token di database. (Dipilih).

Decision:

Diterima sebagai known limitation V1. Sesi refresh token dihapus seketika di database saat logout, sehingga perpanjangan token tidak dimungkinkan. Token access habis secara alami dalam ≤ 15 menit. Rencana denylist terdistribusi dijadwalkan untuk evaluasi arsitektur V2 jika bisnis membutuhkan.

Date:

2026-09-14

Impact:

Mempertahankan kesederhanaan infrastruktur edge Cloudflare Workers dan D1 tanpa dependensi pihak ketiga.

ID: DL-037

Title: HTTP 201 vs 200 REST Contract on Confirm-Upload and Read-URL (B.5)

Scope: Contract (media, api)

Status: Decided - default engineering, menunggu konfirmasi bisnis

Problem:

Inkonsistensi status code antara pembuatan resource / metadata baru (201 Created) versus pembuatan URL baca sementara tanpa mutasi resource (200 OK).

Existing Requirement:

BUG-T4-14/15: keselarasan kontrak REST standar pada media operations.

Options Considered:

1. Menyeragamkan semua POST menjadi 200 OK. (Ditolak: melanggar semantik REST pada operasi yang membuat entitas).
2. `POST /admin/media/confirm` mengembalikan 201 Created karena memutasi dan membuat rekaman metadata; seluruh endpoint penyerahan read URL sementara (`POST /admin/media/read-url`, `POST /orders/:id/payment-proof-url`, `GET /lessons/:id/video-url`) mengembalikan 200 OK. (Dipilih).

Decision:

1. `POST /admin/media/confirm` eksplisit `@HttpCode(HttpStatus.CREATED)` (201).
2. `POST /admin/media/read-url` eksplisit `@HttpCode(HttpStatus.OK)` (200).
3. Update OpenAPI spec dan dokumentasi kontrak `docs/api-contract.md`.

Date:

2026-09-14

Impact:

Kontrak status code REST standar dan konsisten di seluruh modul backend.

Deployment Runbook — Xolvon Backend

Xolvon Backend Documentation · xolvon-backend-full-documentation.pdf

Step-by-step operating procedure for promoting this backend to staging and production, including migrations, rollback, first-admin bootstrap, and the (currently open) monitoring decision. Read `production-operations.md` for policy and `qa-results.md` for the verified state.

0. Current honest status

- **Local:** fully working — SQLite (`DB_DRIVER=sqlite`) + local-test storage (`STORAGE_DRIVER=local-test`), no cloud credentials.
- **Staging/production:** **NOT deployed and NOT verified**. The D1/R2 live path is blocked for lack of a deployment target, secret stores, wrangler access, and final HTTPS domains (recorded in `qa-results.md`). Nothing in this runbook may be marked done without real evidence; do not claim success from a local dry run.
- **Unblock requirements** (from `qa-results.md`): separate staging/production secret stores with valid D1/R2/JWT values, an authenticated deployment provider/target for the Node service, pre-migration backups, log access, and real HTTPS domains for both environments.

1. Deployment architecture (documented deviation)

The backend runs as a **Node/Express NestJS service** (`@nestjs/platform-express` , `node dist/main`). Cloudflare D1 is reached **over its REST API through the `D1RestAdapter` behind the `DatabaseService` token** (`DB_DRIVER=d1`), and R2 through the S3-compatible SDK behind `STORAGE_PORT` (`STORAGE_DRIVER=r2`). HANDBOOK_BACKEND.md §8 Langkah 6/8 shows swapping `DatabaseService` to a Workers `env.DB` binding and deploying via `vinext / @vinext/cloudflare` — that path belongs to the **frontend/Workers deployment**. For this backend it is a **documented deviation** (conflict K-06 in `decision-log.md`): the service is not a Workers runtime target; do not attempt to deploy `dist/main` with vinext.

2. Configuration per environment

Copy names only from the committed templates and populate values in the target's secret store — never in Git, docs, logs, or shell history:

Template	Fixed shape (non-secret)
<code>.env.staging.example</code>	<code>APP_ENV=staging</code> , <code>DB_DRIVER=d1</code> , <code>STORAGE_DRIVER=r2</code> , <code>TRUST_PROXY_HOPS=1</code>
<code>.env.production.example</code>	<code>APP_ENV=production</code> , <code>DB_DRIVER=d1</code> , <code>STORAGE_DRIVER=r2</code> , <code>TRUST_PROXY_HOPS=1</code>

Both templates still need real values for `CLOUDFLARE_ACCOUNT_ID` , `CLOUDFLARE_D1_DATABASE_ID` , `CLOUDFLARE_API_TOKEN` , `R2_ENDPOINT` , `R2_BUCKET` , `R2_ACCESS_KEY_ID` , `R2_SECRET_ACCESS_KEY` , `JWT_SECRET` , `FRONTEND_URL` , and optionally `PORT` . In addition, set the two runtime-optional names per environment once domains are decided (DL-020/DL-021/DL-027):

- `CORS_ALLOWED_ORIGINS` — final staging/production frontend origins (comma separated). **OPEN — owner has not supplied final domains**. Wildcards are rejected when credentials mode is on.
- `CSP_CONNECT_SRC` — frontend origins allowed in CSP fetch/source lists.

OPEN until confirmed; unset is safe (defaults to `'self'` + `R2_ENDPOINT` origin).

Validation is fail-fast at bootstrap (`src/config/env.validation.ts`): `d1` requires the three `CLOUDFLARE_*` names; `r2` requires the four `R2_*` names; `JWT_SECRET` ≥ 32 chars and a valid `FRONTEND_URL` are always required. `TRUST_PROXY_HOPS` must be a non-negative integer string or the process aborts before binding a port.

3. Provisioning D1 and R2 (once, per environment)

1. Auth wrangler on the operator machine: `npm wrangler login` .
2. D1 database — staging already exists and is the committed binding target in `wrangler.jsonc` (`database_name: "xolvon-staging"` , binding `DB` , `migrations_dir: "src/database/migrations"`). Production requires a new `npm wrangler d1 create xolvon-production` (name TBD by owner); record the resulting id **only** in the secret store as `CLOUDFLARE_D1_DATABASE_ID` — never in documents.

3. R2 bucket: `npx wrangler r2 bucket create <bucket>`; mint an API token

~~X~~scoped to read/write on that bucket; store endpoint/keys as the four `R2_*`

env names.

4. Deployment target for the Node process: **not yet provisioned** — any

Node-capable host/PaaS that can hold a secret store and sit behind HTTPS

with a known proxy hop count. Decide and record it in

`decision-log.md` before first deploy (relates to PRD

§91.4, open).

4. Migration apply

- **Staging/production (D1):**

`npx wrangler d1 migrations apply xolvon-staging --remote` (repeat with the

production database name for production). Take a D1 backup before the first

apply of any release.

- The in-repo runner (`src/database/migrate.ts`) applies files in

`src/database/migrations/` in filename order, records each with a SHA-256

hash in `_migrations`, skips applied files, and **throws on hash drift** —

applied migrations are immutable. The HTTP server does not auto-migrate.

- Never edit an applied migration; add a new numbered file (see

`README.md` §5).

5. Staging → production deployment procedure

Do staging first; production is the same sequence with production values.

5.1 Staging

1. Prereqs from §0 satisfied; backup of the staging D1 taken.

2. Apply pending migrations (§4) against the staging database.

3. Deploy the built artifact (`npm ci && npm run build → ship dist/`, or

the provider's equivalent) with the staging env populated (§2).

`package.json` exposes only `deploy: "nest deploy"` as a placeholder — it

targets no configured staging/production platform, so the actual

deployment mechanism is the provider target from §3.4 (still unprovisioned).

4. Health check: `curl -i https://<staging-host>/api → expect 200`. Verify

headers on the response: CSP present with the configured origins,

`X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`,

`Referrer-Policy: strict-origin-when-cross-origin`, no `X-Powered-By`;

no HSTS on staging (production-only, by design).

5. Functional smoke (clean accounts, throwaway data):

- `POST /api/auth/register` then `login`; repeat login 6× → 6th returns

429 (auth 5/min).

- `GET /api/courses → 200`; burst 31× → 429 (catalog 30/min).
- Browser-mutation integrity: `POST /api/orders` with a disallowed

`Origin` header → 403; without browser headers + valid Bearer → passes

auth path.

- CORS: allowed origin `OPTIONS` preflight returns

`Access-Control-Allow-Origin` + credentials; unknown origin does not.

6. First admin (§6).

7. Record evidence (status codes, header dumps) without copying tokens or

secret values anywhere.

5.2 Production

1. Same as staging with production secret store, plus:

2. `APP_ENV=production → HSTS max-age=31536000; includeSubDomains` must

appear; confirm only after HTTPS is terminating correctly, since HSTS is

sticky in browsers.

3. `CORS_ALLOWED_ORIGINS` / `CSP_CONNECT_SRC` set to the final owner-approved

domains (blocked on DL-027 — OPEN).

4. Repeat the §5.1 smoke matrix against the production host on a clean

account, then promote real content.

5. Any production success claim requires captured live evidence — the current

recorded state is **BLOCKED** (`qa-results.md`).

6. First-admin bootstrap (controlled)

There is no public admin registration route and no role input is ever accepted

from a client (DL-005/DL-017). Seed the first admin from a controlled operator

shell with the target environment's secret store attached, using the variable

names only (values from the secret store; avoid shell history — e.g. read them

from a file descriptor or interactive prompt) documentation.pdf

```
ADMIN_BOOTSTRAP_EMAIL='<admin email>' \
ADMIN_BOOTSTRAP_PASSWORD='<from secret store>' \
npm run seed:admin
```

Behavior (`src/database/seed-admin.ts` → `admin-bootstrap.ts`): applies

pending migrations, hashes the password with Argon2id via `PasswordService`,

inserts `role='admin', status='active'`, prints only

`created` or `already_exists`, is idempotent for the same admin email, and

refuses (conflict error) to touch an existing non-admin account.

7. Rollback

Authoritative procedure: `production-operations.md`

"Rollback procedure". Short form, in order:

1. Stop promotion; record deployment id, last-applied migration filename, and incident timestamp (no secret values).

2. **Revert the deploy** to the last known-good immutable build and confirm

it boots with the target secret store (`GET /api health`).

3. **Revert the last migration** only if the failed release applied one:

restore the pre-migration database backup/snapshot taken before that apply.

The runner is forward-only and creates no down-migrations.

4. If the platform cannot restore, ship a **compensating migration** as a new numbered file that reverses the affected change. **Never edit an applied migration** — `_migrations` hash drift must fail closed.

5. Re-run the migration set against a disposable local copy first; verify

`PRAGMA foreign_key_check` and the affected flows; redeploy known-good.

6. Keep the failed release isolated until root cause + forward fix are reviewed.

8. Monitoring & alerting — OPEN DECISION

`MonitoringPort` (`src/observability/monitoring.port.ts` :

`captureError`, `recordEvent`) is **interface-only** — no provider SDK,

endpoint, recipients, or credential is wired (DL-022). Log/monitor hygiene in

the meantime (enforced by `AllExceptionsFilter` and code review): event names,

route/status context, non-sensitive identifiers only; never passwords,

tokens, hashes, signed URLs, credentials, or echoed untrusted input.

Before any production monitoring wiring, a new decision entry must record

provider, alert recipients, data boundary, and secret-store location. Until

then, treat "monitoring" as: structured stdout logs + provider platform

metrics + the §5.1/§5.2 smoke checks per release.

9. Post-deploy checklist (per release)

- [] Migrations applied cleanly (count matches expected new files; no drift)
- [] Health `GET /api 200` through HTTPS
- [] Header matrix correct for the env (HSTS only in production)
- [] CORS allow/deny spot check; no wildcard-with-credentials
- [] Auth 5/min + one named group 429 observed
- [] Disallowed-Origin browser mutation → 403
- [] Admin seed/verify/activate/revoke happy path on clean accounts
- [] Evidence captured with zero secret values
- [] `qa-results.md` / runbook updated with real status (or an honest BLOCKED)

Environment separation

Use the same NestJS application architecture with separate values and secret stores:

Environment	Database	Storage	Required operational checks
local	sqlite / local.db	local-test	migrations, auth, course, lesson, admin flows
staging	d1	r2	Cloudflare bindings, auth, upload, private media, integration
production	d1	r2	HTTPS/domain, secrets, analytics, monitoring, rollback

Never copy production values into `.env`, tests, screenshots, logs, documentation, or a client bundle. `R2_ACCESS_KEY_ID`, `R2_SECRET_ACCESS_KEY`, `CLOUDFLARE_API_TOKEN`, `JWT_SECRET`, and admin bootstrap password are server-only.

Controlled admin bootstrap

There is no public admin registration route. Run this command only from a controlled operator shell with the target environment's secret store attached:

```
ADMIN_BOOTSTRAP_EMAIL='admin@example.invalid' \  
ADMIN_BOOTSTRAP_PASSWORD='set-in-secret-store-not-shell-history' \  
npm run seed:admin
```

The command runs pending migrations, hashes the password through `PasswordService`, inserts `role='admin'`, and prints only `created` or `already_exists`. It is idempotent for an existing admin. It refuses to change an existing non-admin account. Do not place either variable in a committed file or command history.

Rollback procedure

1. Stop promotion and record the deployment identifier, migration filename, and incident timestamp. Do not print or copy secret values.
2. Revert the application deployment to the last known-good immutable build using the hosting provider's deployment rollback command. T19 does not claim a live provider deployment; execute this only in the configured staging/production platform.
3. Confirm the reverted build starts with the target environment's secret store and run the health check. Check logs for event names and status only.
4. If the failed release applied the latest migration, restore a database backup/snapshot taken before that migration. The repository migration runner is forward-only; it does not invent down migrations.
5. If the platform requires an explicit compensating migration, create a new numbered migration that reverses the affected schema change. Never edit an applied migration because `_migrations` hash drift must fail closed.
6. Re-run the migration runner against a disposable local copy first, then verify `PRAGMA foreign_key_check` and the affected application flow.
7. Re-deploy the last known-good build, verify auth, admin authorization, storage access, and error responses, and record evidence without credentials.
8. Keep the failed deployment isolated until the root cause and a forward fix are reviewed. Do not claim production recovery from a local dry run.

Observability boundary

Application errors use the local `MonitoringPort` interface in `src/observability/monitoring.port.ts`; no provider SDK or credential is selected in T19. The provider decision remains open. Logs and monitoring events must contain event names, route/status context, and non-sensitive identifiers only. They must not contain passwords, tokens, hashes, signed URLs, R2 credentials, or exception messages copied from untrusted input.

T20 QA Results

Status

BLOCKED

T20 live deployment verification was not performed. No staging or production success is claimed, and no credential values, deployment identifiers, domains, metrics, or external service results were fabricated.

Prerequisite gate

The required deployment prerequisites are unavailable or unverifiable in this workspace:

- No separate staging secret store or populated staging target environment was found. `.env.staging.example` contains the required variable names but no usable target values.
- No separate production secret store or populated production target environment was found. `.env.production.example` contains the required variable names but no usable target values.
- A local `.env` exists and has populated entries, but its values were not read or printed and it does not establish which values belong to staging or production. It therefore cannot be treated as valid environment-specific deployment credentials.
- No process environment contains non-secret staging/production target or domain variables.
- No deployment target/provider configuration was found for the Node/Express

NestJS service. `package.json` has no staging/production deploy script; `wrangler.jsonc` contains only a D1 binding, with no application target, environment blocks, R2 binding, or custom-domain configuration. The `wrangler` executable is unavailable.

- No real staging or production domain/HTTPS endpoint was provided or found.
- T19 output (`DoneClaim`, `evidence-t19-happy.txt`, and `evidence-t19-failure.txt`) explicitly records that production credentials and infrastructure were unavailable and that live deployment remained T20.

Because `DB_DRIVER=d1` requires the three Cloudflare values and `STORAGE_DRIVER=r2` requires the four R2 values, the application cannot be started against the required staging or production services from this workspace. A deployment target, authenticated target identity, and real domains are also required for the staging-to-production and HTTPS checks.

T20 scenarios

Scenario	Result	Evidence
t20-happy : staging then production deployment and clean-account purchase/verify/activate/access/revoke/video/upload/login/logout/session-expiry checks	BLOCKED	No verifiable environment-specific D1/R2 credentials, deployment target identity, provider executable/configuration, or staging/production HTTPS domains
t20-failure : record failed checklist items before claiming completion	BLOCKED	Same prerequisite gate; no live endpoint was available to exercise

No live success is claimed. The blocked evidence records are `t20-happy.txt` and `t20-failure.txt`; both explicitly state that no live endpoint was available and contain no secrets.

Local verification only

The following checks are local repository checks and do not substitute for T20 deployment verification:

- `npm run build`
- `npm run lint`
- `npm run test:esm`
- `npm run test:e2e`

Observed on 2026-09-12:

- `npm run build` exited 0.
- `npm run lint` exited 0.
- `npm run test:esm` passed 36 suites and 314 tests.
- `npm run test:e2e` passed 1 suite and 83 tests.
- LSP diagnostics could not run because the TypeScript LSP is not installed;

installation had previously been declined. Markdown has no configured LSP.

These results cover local code only. They do not change the BLOCKED status of the live deployment gate.

Unblock requirements

Xolvon Backend Documentation - xolvon-backend-full-documentation.pdf

Provide separate staging and production secret stores containing valid D1, R2, JWT, and frontend URL values; provide the deployment provider/target, its authenticated access and executable/configuration; provide pre-migration backup/rollback and log access; and provide real HTTPS domains for both environments. The local .env must not be assumed to satisfy this gate.

After those prerequisites exist, T20 must deploy staging before production, run the complete checklist against clean accounts and real services, capture the resulting t20-happy and t20-failure evidence, monitor logs, and only then record a non-blocked result.

QA Addendum — Verifikasi Endpoint Lengkap Dua Target — 2026-09-13

Addendum ini adalah hasil eksekusi plan .omo/plans/xolvon-qa-endpoint-testing.md

(verifikasi endpoint penuh, dua target: LOKAL dan DEPLOYED). Bagian T20 lama di atas

TIDAK diubah — ini append-only sesuai aturan plan \$Must-NOT.

- Target LOKAL: http://localhost:3005/api (deviasi port — lihat §0.1)
- Target DEPLOYED: https://xolvon.canadev.my.id/api
- Seluruh nilai Actual pada tabel di bawah disalin VERBATIM dari artefak bukti

di .omo/evidence/xolvon-qa-endpoint-testing/ (direktori evidence yang sama untuk semua pointer). Tidak ada angka yang dikarang; sel Actual kosong = tidak ada.

- Format kolom milik addendum sumber: Endpoint | Skenario | Expected | Actual | Status | Catatan ;

Status ∈ {PASS, FAIL(→bug ID), BLOCKED-creds, N/A-adapted} (+label khusus yang tercatat

di artefak: N/A-DBAccess, N/A-noKeyAttachment, ENV-THROTTLE, DELEGATED-T13).

§0 Metode & deviasi (dicatat, bukan ditutupi)

1. Deviasi port :3000 → :3005 (LOKAL). :3000 dipegang proses MILIK USER —

hermes whatsapp-bridge (PID 1862; curl :3000/api menjawab 404 Express, bukan

Hello World!) — plus hazard ~/xolvon-local-supervisor.py (PID 12270→12635/12659)

yang crash-loop start:dev dan akan merebut :3000 begitu port lepas. QA TIDAK

membunuh proses milik user. Keputusan orchestrator (reversible): seluruh surface

lokal pindah ke PORT=3005 via shell-override. Bukti: task-1-preflight.txt

\$REDO Step 2 + \$FINISH 3005 (health Hello World! exact, lineage pid 24810→24834→24869).

2. Shell-override DB_DRIVER=sqlite STORAGE_DRIVER=local-test (WAJIB) + INSIDEN seed→D1.

.env mesin ini berisi DB_DRIVER=d1; npm run seed:admin TANPA override menulis

8 migrasi ke D1 REMOTE dan INSERT admin gagal 502 (task-1-preflight.txt §ROOT CAUSE).

Not perubahan data remote — INSERT gagal, hanya migration-bookkeeping remote tercatat.

Kontainmen: shell-override di SEMUA proses lokal (dotenv tidak menimpa process.env);

seed dengan override terbukti menulis LOKAL (local.db 319.488 B, admin count=1,

UNIQUE(user_id,course_id) ada). Bug produk terpetakan: BUG-seed-driver —

DIFIX di T16 (preflight refuse non-sqlite, commit 852f829→4c7d628).

3. Reset DB baseline. local.db pra-QA (0 byte) dipindah ke bukti

(local.db.pre-qa-1789270259), DB QA mulai dari nol → angka §3 bebas noise.

local.db dan .omo/ git-ignored (git check-ignore positif, task-1-preflight.txt Step 3/8).

4. Cookie → Bearer (N/A-adapted). Keputusan DL terkunci: login me-mint

{accessToken, refreshToken, user} di body, TANPA Set-Cookie. Baris "cookie flags"

di kedua target ditulis N/A — Bearer-only (keputusan DL terkunci), bukan PASS bohong.

5. Path-divergence /admin/courses → /api/courses. Matriks sumber menyebut

POST /admin/courses; rute aktual = POST /api/courses + @Roles('admin')

(courses.controller.ts:87-145). Baris PATH-DIVERGENCE N/A-adapted di §2.2 kedua target.

6. Referensi handbook diluruskan. Matriks sumber "Handbook §7/§12" → otoritatif

aktual: PRD §45 (kolom admin orders) dan PRD §52 (publish gate);

juga salah kutip awal pada teks-task X1 ("q hilang → 400 per SearchQueryDto")

yang terbukti salah kutip: q ber-@IsOptional() → aktual 200 listing published

(F-T12-01, ter-reproduksi deployed).

7. Metode HTTP. curl di shell mesin ini me-mask ! pada body JSON → semua

panggilan ber-password memakai node-fetch driver (t6-driver.mjs dst).

Semua request mutasi TANPA header Origin / Sec-Fetch-Site → lolos

RequestIntegrityGuard (konvensi global plan); sub-check guard menolak

Origin: https://evil.example dieksekusi di T6 KEDUA target (403 + kontras 409).

8. Pacing throttle. register/login 5/menit/IP/route; grup signed/order 10/menit;

catalog/search 30/menit;Assertion A3 (429@ke-6) dipindah ke window sepi Wave-4 (T6b).

Newman lokal: ---delay-request 12000 + ---env-var baseUrl=http://localhost:3005/api

(tanpa delay, 75 request membakar budget auth → false-failure). Hasil: 0 ENV-THROTTLE

di T4; 4×429 recover-by-retry di T6-deployed (bukan bug produk).

9. Fingerprint deployed (T2): MATCH + kaveat TARGET-DATA#REPO-SEED. 8 probe

field-by-field behavioral MATCH (envelope, pesan, validator verbatim, key set, docs-json 404, health string) → deployed menjalankan build kontrak repo. 1 MISMATCH DATA: `course-1` (non-UUID) + `course-2` + admin `admin-1` = seed out-of-band yang TIDAK ada di repo. Konsekuensi: semua fixture deployed dibuat sendiri via API (qa-*), id existing tidak pernah dipakai sebagai fixture.

10. ****Karantina qa-1.**** Semua mutasi deployed hanya mengenai entitas `qa-*` buatan run ini; exit karantina = unpublish/cancel terverifikasi response-surface (tidak ada API delete user/baris — residu tak-terekspose terdokumentasi per task).

11. **Cross-lane review.** Review ganda plan (momus + oracle) — sha256 transkrip IDENTIK `8b47f4f1c9b0c2dd58e941bdd1a5174def503f60abe8e3fe4fcd77fbd7bc7bca` (no retrieval drift). Koreksi dual-review yang folded ke plan & dieksekusi:

(B1) AD1 = 10-kunci + eksklusif DL-015 terkunci, bukan hard-fail 11 kolom;

(B2) seed admin = `ChangeMe-Admin-123!` (bukan `ChangeMe-Local-123!` yang student);

(B3) audit completeness TIDAK mid-wave → delegasi T13; (B4) assertion A3 429 → T6b window sepi Wave-4; (B5) newman `--delay-request`; non-blocking: race-A boleh 500 sanitised (invariant DB penentu), guard `grep -cF` var kosong, sumber limit 5/min = `@Throttle` auth.controller.ts:48,59 (bukan grup named throttle.config.ts).

§L1 — Automated tests (T3, T4, T5)

T3 — jest LOKAL (repo build, mocked):

Perintah	Suites	Tests	Exit	Status	Catatan
<code>npm run test:esm</code>	44 passed / 44 total	411 passed / 411 total	0	PASS	<code>task-3-esm.log</code> ; noise stderr <code>fetch failed</code> ×2 = assertion simulasi dalam test PASS
<code>npm run test:e2e</code>	1 passed / 1 total	107 passed / 107 total	0	PASS	<code>task-3-e2e.log</code> ; harness boot sendiri PORT 41234, mocked DB

Pasca-remediasi T16 (5 bug fix test-first, +14 test reproduksi/pinning):

`test:esm` **46/46 suites, 425/425 tests**; `test:e2e` **109/109 tests** — exit 0, build exit 0 (`task-16-final-suites.txt`). Kedua angka disitasi: 44/411+107/107 = baseline pra-fix (T3), 425/109 = pasca-fix (T16). Doc-drift badge README (38/335) → §10.

T4 — newman collection penuh vs LOKAL (server :3005 hidup):

Perintah literal (deviasi port + delay documented):

`npx newman run postman/xolvon-backend.postman_collection.json -e postman/environments/local.postman_environment.json --env-var baseUrl=http://localhost:3005/api --delay-request 12000`

Metrik	Nilai	Status	Catatan
iterations	1 executed / 0 failed	PASS	durasi 15m 23s
requests	75 executed / 0 failed	PASS	runtime request failures nol
assertions	150 executed / 27 failed spanning 15 requests	FAIL(→BUG-T4-01..15)	klasifikasi di bawah; 0 request-level fail
429 / ENV-THROTTLE	<code>grep -c 429</code> raw = 0	PASS	pacing 12s berhasil

Klasifikasi 27 assertion-fail (verdict T4: PARTIAL(15 product-bug candidates, 0 ENV-THROTTLE)):

- BUG-T4-01..13 = collection-asumsi-stale, produk BENAR: folder collection

mengasumsikan DB demo-seeded dengan content published (`course-contoh` dkk.) yang sengaja tidak ada di baseline QA-reset, plus ordering bug intra-collection (delete-lesson sebelum video-uri; `{{memberId}} / {{enrollmentId}}` placeholder zero-UUID belum di-capture; cancel order PAID → 400 = justru sesuai matriks E9). Semua actual (404/400/403) semantik benar per kontrak — lihat analisis RC per request di `task-4-newman-local.txt`.

- BUG-T4-14/15 = kandidat produk terkuat (2 request): `POST /admin/media/confirm`

dan `POST /admin/media/read-url` membalas 201 vs assertion 200 — kontrak `docs/api-contract.md` tidak mem-pin status sukses; Nest POST default 201.

Adjudikasi T17 (dokumentasikan 201 ATAU `@HttpCode(200)`) → §9 BUG-T4-14/15 OPEN-PARKED.

T5 — akses deployed + newman SMOKE vs DEPLOYED (hanya folder self-contained):

`npx newman run postman/xolvon-backend.postman_collection.json -e <mktemp env LUAR repo> --folder "10, Newman Smoke" --delay-request 12000`

Aspek	Actual	Status	Catatan
Login admin deployed	200, keys {accessToken, refreshToken, user}, user.role="admin", tanpa password_hash; refresh opaque 64-hex NON-JWT; user.id legacy admin-1	PASS	gate "deployed admin usable for T6d–T12d: YES"; creds hanya env-shell, tidak pernah ke file
Register+login student qa	201/200, id UUID v4	PASS	karantina dipatuhi
X3 pre-check /auth/admin	200 admin / 403 student / 401 no-token	PASS	envelope verbatim repo
Newman smoke deployed	requests 9/9 executed, 0 failed; assertions 18 executed, 0 failed; exit 0; 0×429	PASS	durasi 1m51.6s; <code>task-5-newman-smoke.txt</code>
Teardown	temp env di <code>mktemp -d</code> LUAR repo → <code>rm -rf</code> receipt tercap	PASS	secret scan evidence 0 hit

Verdict: T5: PASS.

§2 Matriks manual dua target (§2.1–§2.7)

Kolom seragam milik addendum. Rows disalin dari evidence per-task; Actual/Status verbatim.

§2.1 Auth (mapping A1–A5; bukti `task-6-auth-local.md` + `task-6b-throttle-local.md`)

Target: LOKAL (`http://localhost:3005/api`)

Endpoint	Skenario	Expected	Actual	Status	Catatan
POST /api/auth/register	A1: register email valid (qa student baru, phone sesuai regex, name)	201 user baru dibuat	201 body SafeUserDto {"id":"be455027-...", "email":"qa-t6-a1-1789275196@example.test", "role":"user", "status":"active", ...}	PASS	—
POST /api/auth/register	A1: role FORCED user bahkan saat body menyertakan "role":"admin"	201 dengan role:"user" — role client tidak pernah dipercaya	400 {"statusCode":400,"message":["property role should not exist"],"error":"Bad Request",...}; SQL: rolestr_count=0 (tidak ada baris sama sekali)	PASS	Naik level dari expected: whitelist+forbidNonWhitelisted MENOLAK total (bukan diam-diam memaksa user); INSERT service hardcoded 'user' sebagai defense-in-depth
POST /api/auth/register	A1: varian role-injection role: ["admin"] (array)	tetap ditolak/dipaksa	400 {"statusCode":400,"message":["property role should not exist"],...}; SQL: rolearr_count=0	PASS	whitelist bekerja struktural, bukan per-type
POST /api/auth/register	A1: email duplikat	409/400 DAN SELECT COUNT(*) users WHERE email=... = 1	409 {"statusCode":409,"message":"An account with these credentials is already registered.",...}; SQL: dup_count=1	PASS	Pesan anti-enumeration generik (email ATAU phone)
POST /api/auth/register	A1: format invalid email/phone + password pendek (+ name kosong)	400 dengan pesan per-field	400 message array persis: ["Name is required.", "A valid email address is required.", "Phone number format is invalid.", "Password must be at least 8 characters long."]	PASS	collect-all 1 request
POST /api/auth/login	A2: login benar (student)	200 {accessToken, refreshToken, user}	200 {"accessToken":"eyJhbGciOi...", "refreshToken":"5d2bc50f...", "user":{...}} — ketiga kunci ada: user.role="user"	PASS	token di-mask sesuai aturan bukti
(cookie flags session)	A2: baris cookie flags login	cookie HttpOnly/Secure/SameSite	Tidak ada Set-Cookie; token di body	N/A- adapted	N/A — Bearer-only (keputusan DL terkunci) — mapping A2
sqlite3 sessions	A2: sessions menyimpan SHA-256, BUKAN token mentah	hash tersimpan	proof_stored_is_sha256_of_raw=1 (sha256(rawRefresh) == stored, match penuh 64-hex); proof_raw_never_stored=0; stored_len=64; raw refresh opaque 64-hex NON-JWT	PASS	Bukti kriptografis via SQL read-only LOKAL
response shape (jq keys)	A2: response tidak pernah memuat password_hash	absen	keys user login/register → password_hash present? false (login student & admin)	PASS	mapper toSafeUser strip
POST /api/auth/login	A2: login password SALAH	401 pesan ambigu anti-enumeration	401 {"statusCode":401,"message":"Invalid email or password.",...}	PASS	identik user-tidak-ada vs pw-salah
POST /api/auth/login	A3: 6x login salah/menit → 429 (limit aktual 5/menit)	ke-6 = 429	401,401,401,401,401,429 — burst 6x node-fetch (<35ms), replikasi identik di window sepi kedua (+72s) → deterministik; #6 = 429 ThrottlerException: Too Many Requests TOO_MANY_REQUESTS	PASS	Dieksekusi T6b Wave-4 window sepi; t6b-attempt1.txt + t6b-attempt2-recovery.txt; TTL-60s reset terbukti
GET /api/auth/me	A4: tanpa token	401 envelope	401 {"statusCode":401,"message":"Missing or invalid authorization token.",...,"error":"Unauthorized",...,"path":"/api/auth/me"}	PASS	—
GET /api/auth/me	A4: token garbage	401	401 {"statusCode":401,"message":"Invalid or expired token.",...}	PASS	—
GET /api/auth/me	A4: token format-expired (JWT exp lampau, struktural valid)	401	401 {"statusCode":401,"message":"Invalid or expired token.",...}	PASS	—
GET /api/auth/me	A4: token valid	200 user, tanpa password_hash	200 {"sub":"be455027-...", "email":"qa-t6-a1-...", "role":"user", "iat":1789275235, "exp":1789276135}	PASS	shape aktual = JwtPayload; deviasi bentuk dari kata "user" di matriks dicatat
POST /api/auth/logout → /api/auth/refresh	A5: logout lalu refresh token lama ditolak	logout 200; refresh lama → 401	200 {"message":"Logged out successfully."} → 401 {"statusCode":401,"message":"Invalid or expired refresh token.",...}	PASS	Inti matriks TERBUKTI. FINDING: GET /auth/me dengan access-JWT ≤15m SETELAH logout → 200 — BUG-T6-window (Medium, Open-Parked — denylist = surface baru, dilarang plan)
GET /api/auth/admin	X3: no token → 401 / student → 403 / admin → 200	401/403/200	401 Missing or invalid authorization token. / 403 You do not have permission to access this resource. / 200 {"message":"Admin access granted.", "user":{"sub":"85cf38c6-...", "role":"admin", ...}}	PASS	3 kategori satu window
POST /api/auth/register	GUARD: Origin: https://evil.example + Sec-Fetch-Site: cross-site pada mutasi	WAJIB ditolak RequestIntegrityGuard (bukti status+body)	403 {"statusCode":403,"message":"Request metadata is not allowed.", "error":"Forbidden",...}; SQL setelahnya: dup_count_after_guard=1 (mati di guard, tak pernah kena service/DB)	PASS	order guard Throttler → RequestIntegrity → pipe
POST /api/auth/register	GUARD presisi: POST IDENTIK tanpa Origin harus lolos ke jalur validasi	lolos guard (bukan 403)	409 An account with these credentials is already registered. (body sama persis, header saja berbeda)	PASS	kontras 403-409 = bukti presisi guard

Verdict LOCAL: 18/19 PASS · 1 N/A-adapted (cookie flags — DL terkunci) · 0 FAIL · 0 DEFERRED tersisa (A3 ditutup PASS via T6b) · FINDING: BUG-T6-window (Medium, Open-Parked)

Target: DEPLOYED (https://xolvon.canadev.my.id/api)

Endpoint	Skenario	Expected	Actual	Status	Catatan
POST /api/auth/register	A1: register email valid → 201	201 user baru dibuat	201 {"id":"b25803bf-0de1-...", "email":"qa-t6d-1789276585@example.test", "role":"user", "status":"active", ...} (seq01)	PASS	bukti response-only; uuid v4 (bukan "course-1" legacy) = dibuat build ini
POST /api/auth/register	A1: role-injection "role":"admin" → forced user	role tak pernah admin	400 {"statusCode":400,"message":["property role should not exist"],...} (seq02)	PASS	deployed identik lokal: ditolak total
POST /api/auth/register	A1: role-injection array	ditolak/dipaksa	400 {"statusCode":400,"message":["property role should not exist"],...} (seq09)	PASS	ENV-THROTTLE: 4x429 window register dibagi worker se-IP → tembus 400 pada retry-5 (bukan FAIL produk)
POST /api/auth/register	A1: email duplikat → 409/400 tanpa baris kedua	409/400	409 {"statusCode":409,"message":"An account with these credentials is already registered.",...} (seq04)	PASS	single-account dibuktikan response-only: 409 + re-login resolve TEPAT SATU id
POST /api/auth/register	A1: format invalid → 400 per-field	400 pesan per-field	400 message array persis: ["Name is required.", "A valid email address is required.", "Phone number format is invalid.", "Password must be at least 8 characters long."] (seq03)	PASS	verbatim identik lokal
POST /api/auth/login	A2: login benar → 200 triple	200 {accessToken, refreshToken, user}	student 200 (seq10) & admin 200 (seq14) — {accessToken, refreshToken, user}; user.role = user / admin	PASS	login admin = satu-satunya pemakaian baris deployed yang ADA (legacy id admin-1). HANYA diautentikasi, tidak dimutasi
(cookie flags)	A2: cookie session flags	—	Tidak ada Set-Cookie; token di body	N/A- adapted	N/A — Bearer-only (keputusan DL terkunci) ; header login hanya access-control-allow-credentials:true
sessions hash	A2: SHA-256 di sessions	hash tersimpan	N/A-DBAccess — tanpa akses D1; bukti STRUKTURAL saja: raw refreshToken opaque 64-hex NON-JWT (student & admin)	N/A- DBAccess	hash-check TIDAK diklaim untuk deployed (mapping A2 — jujur)
response shape	A2: tidak ada password_hash	absen	keys user login → password_hash present? false (student & admin); body register likewise	PASS	diverifikasi dua peran deployed
POST /api/auth/login	A2: login salah → 401 ambigu	401 anti-enumeration	401 {"statusCode":401,"message":"Invalid email or password.",...} (seq12)	PASS	deployed = lokal
POST /api/auth/login	A3: ke-6 = 429	ke-6 = 429	401,401,401,401,401,429 — burst 6x node-fetch (span 2.1 s), #6 = 429 ThrottlerException: Too Many Requests — LAYER=APP/NetJS (amplop Nest verbatim, BUKAN html Cloudflare); header #1-#5 memuat x-ratelimit-limit: 5	PASS	T6b-DEPLOYED window IDLE attempt-1 bersih; edge CF tidak masking; t6bd-attempt1.txt

Endpoint	Skenario	Expected	Actual	Status	Catatan
GET /api/auth/me	A4: tanpa token → 401	401	401 {"statusCode":401,"message":"Missing or invalid authorization token.",_,,"path":"/api/auth/me"} (seq15)	PASS	—
GET /api/auth/me	A4: garbage → 401	401	401 {"statusCode":401,"message":"Invalid or expired token.",_,_} (seq16)	PASS	—
GET /api/auth/me	A4: expired → 401	401	401 {"statusCode":401,"message":"Invalid or expired token.",_,_} (seq17; JWT exp lampau + forged-sig)	PASS	—
GET /api/auth/me	A4: valid → 200 tanpa password_hash	200	200 {"sub":"b25803bf-...", "email":"qa-t6d-...", "role":"user", "iat":1789276716, "exp":1789277616} (seq18)	PASS	shape = JwtPayload; exp=iat+900 (15m)
POST /auth/logout → /auth/refresh	A5: invalidasi server-side + FINDING window	200→401; access JWT s15m 200	logout 200 (seq19) → refresh OLD 401 Invalid or expired refresh token. (seq20) → /auth/me access-JWT 200 (seq21)	PASS	BUG-T6-window PARITAS PERSIS dengan lokal — jendela s15m. no denylist — Open-Parked (owner); bukan FAIL matriks
GET /api/auth/admin	X3: 401/403/200	401/403/200	401 (seq22)/ 403 You do not have permission to access this resource. (seq23)/ 200 {"message":"Admin access granted.",_,_} (seq24)	PASS	3-kategori deployed lengkap
POST /api/auth/register	GUARD: Origin evil + cross-site → 403	WAJIB 403	403 {"statusCode":403,"message":"Request metadata is not allowed.",_,,"error":"Forbidden",_,,"path":"/api/auth/register"} (seq25)	PASS	guard menembak APP-layer deployed; request mati sebelum service (body sama dg baris dup → pasti 409 kalau lolos)
POST /api/auth/register	GUARD presisi: tanpa Origin lolos ke validasi	bukan 403	409 An account with these credentials is already registered. (seq04 — body IDENTIK, hanya header dihapus)	PASS	kontras 403→409 = presisi guard terbukti deployed

Verdict DEPLOYED: T6-DEPLOYED: 17/18 PASS, FAILS: [] – dalam 18 ada 1 N/A-adapted (cookie); di luar 18: 1 N/A-DBAccess (hash sessions); 0 DEFERRED tersisa (A3 PASS via T6b-DEPLOYED). Guard deployed PASS → tidak ada BUG-T6d-guard. BUG-T6-window terkonfirmasi PARITAS deployed. BUG-throttle-refresh-logout: bukti permukaan deployed x-ratelimit-limit: 100 pada /auth/refresh & /auth/logout (bukan 5/menit) → Medium, masuk T16.

\$2.2 Courses (mapping C1-C4; bukti task-7-courses-local.md + task-7-local/raw 33 file + task-7-deployed/raw)

Target: LOKAL — verdict baris: 15/16 PASS + 1 N/A-adapted (path-divergence), 0 FAIL.

Endpoint	Skenario	Expected	Actual	Status	Catatan			
GET /api/courses	C1-a katalog published-only: draft qa dibuat → absen; kontrol positif setelah publish	publik hanya published; q kosong → []; slug draft/absent → 404	Draft qa-t7-c1-draft-1789275270 (sqlite: draft) absen dari items. Saat cycle-course published (raw 12/27): statuses: ["published"] , draft c1 tetap absen; saat unpublished (raw 24): present 0	PASS	raw 01,02,12,24,27			
GET /api/courses?q=	C1-b q tanpa cocokan → 200 items:[] bukan error	sama	HTTP 200, {"items": [],"total":0,"query":"zzqa-nomatch-nothing-1789275270"} (query di-echo)	PASS	raw 03			
GET /api/courses	C1-c pagination keys	sama	keys top-level [items,limit,page,query,total] ; page:1,limit:100,total:0..2 konsisten DL-011	PASS	raw 02			
GET /api/courses/:slug	C2-a slug DRAFT qa → 404, BUKAN datanya	user biasa 403; field wajib 400 per-field (kelompok C2)	HTTP 404, body = envelope, 0 kemunculan judul/description draft (grep). Pesan: "Course tidak ditemukan."	PASS	raw 04; kesetaraan draft#missing (courses.service.ts:100-102)			
GET /api/courses/:slug	C2-b slug random → 404 envelope lengkap, tidak pernah 500	sama	HTTP 404, keys [error,message,path,statusCode,timestamp]	PASS	raw 05			
POST /api/courses	C2-c student (role user) → 403	sama	HTTP 403 "You do not have permission to access this resource." . error Forbidden , path /api/courses ; DB: slug forged TIDAK terbentuk	PASS	raw 06			
POST /api/courses	C2-c+ (adjacent) tanpa token → 401	sama	HTTP 401 "Missing or invalid authorization token."	PASS	raw 07			
POST /api/courses	C2-d field wajib hilang ({}) → 400 pesan per-field	sama	HTTP 400 message[] 12 pesan verbatim: ["title maksimal 200 karakter.", "title tidak boleh kosong.", "title harus berupa teks string.", "slug maksimal 120 karakter.", "slug hanya boleh berisi huruf kecil, angka, dan tanda hubung.", "slug harus berupa teks string.", "description maksimal 5000 karakter.", "description tidak boleh kosong.", "description harus berupa teks string.", "price terlalu besar.", "price minimal 0.", "price harus berupa integer."]	PASS	raw 08; perilaku aktual dicatat, bukan dinilai FAIL			
POST /api/courses	C2-e field asing → 400 forbidNonWhitelisted	sama	HTTP 400, verbatim: message: ["property evilField should not exist", "property status should not exist"] → injeksi status:"published" juga diblok whitelist	PASS	raw 09; DB tidak ada baris terbentuk			
POST /api/courses/:id/publish → GET /api/courses → .../unpublish	C3-a siklus penuh draft→publish→muncul→unpublish→hilang→republish	Publish gate data belum lengkap (§12 → aktual PRD §52)	201 draft → publish 200 status:"published" → muncul di items (present 1) → unpublish 200 draft → hilang (present 0) → republish 200. Setiap transisi punya row audit	PASS	raw 10–17,23–27,32.*			
POST /api/courses/:id/publish	C3-b gate incomplete → ditolak sesuai field wajib aktual	sama	Gate AKTUAL = assertPublishable: hanya description + price. Terbukti MENYALA: description:" " (whitespace, lolos create DTO tidak trim) → publish → HTTP 400: "Course belum lengkap untuk dipublish. Field wajib belum ada: description." → setelah PATCH description valid → publish 200. Publish yang ditolak gate TIDAK menulis row audit	PASS	raw 19–22: FINDING: cabang price tak terjangkau API (NOT NULL + @IsInt); 400 per-field 12 pesan			
POST /api/courses/:id/publish	C3-c publish oleh student → 403	sama	HTTP 403 "You do not have permission to access this resource."	PASS	raw 18			
POST /api/courses/:id/publish	C3-d (adjacent) publish ulang yang sudah published	sama	HTTP 400 "Course sudah berstatus published." (strict transition)	PASS	raw 14			
GET /api/courses/:slug	C4 detail publik tanpa video_object_key /signed URL — bahkan setelah lesson dibuat	raw video_object_key tak pernah muncul publik	POST lesson → 201 (DB video_object_key=NULL). Detail 200: scan forbidden [] ; scan leak text `video_object_key`	local-storage.test	X-Amz\ Signature\ content = 0 hit. Kontras: respons ADMIN 201 memang memuat videoObjectKey:null' → stripping di level DTO publik			
GET /api/home	Cross-check T7: draft qa tidak bocor ke home	— (tugas T7; logika penuh T12)	Home #1: 4/4 slug draft 0 hits, hanya published yang muncul; Home #2 pasca-teardown: qa_t7_slugs: []	PASS	raw 31,33			

Endpoint	Skenario	Expected	Actual	Status	Catatan
Xolvo Backend Documentation - xolvo-backend-full-documentation.pdf POST /api/admin/courses (matriks literal)	PATH-DIVERGENCE: matriks menyebut /admin/courses	POST /admin/courses → —	Rule literal tidak ada; aktual POST /api/courses + @UseGuards(AuthGuard, RolesGuard) @Roles('admin')	N/A-adapted	courses.controller.ts:87-145; seluruh C2/C3 dieksekusi di rute aktual

Target: **DEPLOYED** — verdict verbatim: T7-DEPLOYED: 13/13 PASS, FAILS: [] (baris PATH-DIVERGENCE N/A-adapted; C3-c & CROSS-home & QUARANTINE receipt tambahan, semua PASS).

Endpoint	Skenario	Expected	Actual	Status	Catatan			
GET /courses	C1-a published-only + draft qa absen	publik hanya published; q kosong → []; slug draft/absent → 404	HTTP 200; 1 item, statuses= ["published"] (asing course-1 tampil — tidak disentuh). Draft qa sendiri dibuat DRAFT dan tetap absen di semua snapshot; pasca-teardown. 0 slug qa-t7d	PASS	raw 04.17.19.30.31; kaveat T2 TARGET-DATA#REPO-SEED			
GET /courses?q=	C1-b q tanpa cocok → 200 []	sama	HTTP 200, {"items": [], "total": 0, "query": "zz-no-match-1789276600"} (query di-echo)	PASS	raw 05			
GET /courses	C1-c pagination keys	sama	keys top-level [items, limit, page, query, total]; page: 1 limit: 100 total: 1 (DL-011). Baris API-created = UUID v4 (381ffd46-...) → course-1 non-UUID = murni seed → kaveat T2 TERKONFIRMASI	PASS	raw 04.06.15			
GET /courses/:slug	C2-a slug draft → 404 bukan data	user biasa 403; field wajib 400 per-field	HTTP 404 "Course tidak ditemukan."; scan body: 0 kemunculan title/description draft sendiri	PASS	raw 07			
GET /courses/:slug	C2-b slug random → 404 envelope, bukan 500	sama	HTTP 404 (BUKAN 500), envelope keys tepat [error, message, path, statusCode, timestamp]	PASS	raw 08			
POST /api/courses	C2-c student 403	sama	HTTP 403 "You do not have permission to access this resource."; follow-up GET slug forged → 404	PASS	raw 09.10; student = akun sendiri di-register via API			
POST /api/courses	C2-c+ no-token 401	sama	HTTP 401 "Missing or invalid authorization token."	PASS	raw 11			
POST /api/courses	C2-d 400 per-field verbatim	sama	HTTP 400 message[] 12 pesan — IDENTIK lokal, verbatim (title/slug/description/price; lihat label LOKAL C2-d)	PASS	raw 12			
POST /api/courses	C2-e extra field verbatim	sama	HTTP 400, verbatim: message: ["property evilField should not exist", "property status should not exist"] → injeksi status diblok; follow-up GET → 404	PASS	raw 13,14			
POST /api/courses/:id/publish siklus	C3-a draft→publish→muncul→unpublish→hilang→republish	Publish gate (PRD §52)	create 201 draft → publish 200 → ?q= hits 1 → unpublish 200 draft → hits 0 → republish 200 → UNPUBLISH akhir 200 draft (karantina)	PASS	raw 15-20 + 29			
POST /api/courses/:id/publish	C3-b gate incomplete	sama	Create description: " " → 201 draft → publish → HTTP 400 verbatim: "Course belum lengkap untuk dipublish. Field wajib belum ada: description." — persis repo build, TIDAK ada delta build deployed	PASS	raw 26.27; course gate ditinggal DRAFT			
POST /api/courses/:id/publish	C3-c non-admin 403	sama	HTTP 403 (student)	PASS	raw 22			
POST /api/courses/:id/publish	C3-d double-publish 400	sama	HTTP 400 "Course sudah berstatus published." (identik lokal)	PASS	raw 21			
GET /courses/:slug	C4 tanpa video_object_key/signed URL pasca lesson	raw video_object_key tak pernah muncul publik	Detail PUBLIK 200: recursive scan forbidden hits [] ; leak: fragment scan ("video_object_key")	videoObjectKey	X-Amz	Signature=	r2.cloudflarestorage	sign cle /cou → 2 DR/
GET /home	Cross-check draft qa absen	—	HTTP 200; slug DRAFT qa-t7d ABSEN; receipt akhir: 0 substring qa-t7d di /home	PASS	raw 28,32			
POST /admin/courses (literal)	PATH-DIVERGENCE	POST /admin/courses → —	rule aktual POST /api/courses +admin guard; dikonfirmasi deployed: pesan verbatim identik repo	N/A-adapted	sama utk kedua target			

Karantina exit deployed: katalog kembali persis kondisi awal (total published=1 milik asing course-1), 3 course qa-t7d + lesson berakhir DRAFT, /home & /courses bersih dari qa-t7d.

§2.3 Lessons/Media/Upload (mapping M1-M3 + X2 + LESSONS; bukti task-8-media-local.md + task-8-local/raw 53 + task-8-deployed/raw 49)

Target: **LOKAL** — verdict verbatim: T8-LOCAL: 30/34 PASS, 4 N/A-adapted(stub-driver), FAILS: []

Endpoint	Skenario	Expected	Actual	Status	Catatan
POST /api/admin/media/upload-url	M1.1 contentType text/html → ditolak SEBELUM presign	MIME/size ditolak sebelum presign; non-admin 403	HTTP 400, verbatim: "contentType must be one of the following values: image/jpeg, image/png, image/webp, application/pdf, video/mp4" — body TIDAK memuat key / uploadUrl	PASS	raw 08
POST /api/admin/media/upload-url	M1.2 size 524288001 > MAX 524288000	sama	HTTP 400 "size must not be greater than 524288000". tanpa key/uploadUrl	PASS	raw 09
POST /api/admin/media/upload-url	M1.3 prefix private/secret (di luar MEDIA_PREFIXES)	sama	HTTP 400 "prefix must be one of the following values: public/site, public/projects, private/projects, private/courses"	PASS	raw 10
POST /api/admin/media/upload-url	M1.4 student token	non-admin 403	HTTP 403 "You do not have permission to access this resource."	PASS	raw 11
POST /api/admin/media/upload-url	M1.5 tanpa token	AD3: 401	HTTP 401 "Missing or invalid authorization token."	PASS	raw 12
POST /api/admin/media/upload-url	M2a happy: filename ".../EVIL-name.png" → key = \${prefix}/\${uuid}.ext BUKAN nama file	key server-generated	HTTP 201 {key: "private/projects/dcfbc54-...png", uploadUrl: "http://local-storage.test/upload/_", expiresIn: 3600}; key regex uuid v4 TRUE; EVIL TIDAK ada di key	PASS	raw 13; deviasi 201 vs teks-task 200 = F-T8-07 (→ BUG-T4-14/15 family)
PUT uploadUrl	M2b: upload nyata bytes 1x1 PNG ke presigned URL	upload PUT nyata	fetch failed — ConnectTimeoutError: attempted address local-storage.test:80 (network error, BUKAN respons HTTP) — driver local-test = stub, bytes tidak pernah mendarat	N/A-adapted	raw 14; F-T8-01; byte-level equivalence dibuktikan di DEPLOYED (T8d)

Endpoint	Skenario	Expected	Actual	Status	Catatan
<code>POST /api/admin/media/confirm</code>	M2c {key} → sukses	confirm alur penuh	HTTP 201 {key:"private/projects/dcfbc54-...png", confirmed:true} — TANPA validasi eksistensi objek: PUT barusan GAGAL di network, tetap "confirmed"	PASS (kontrak)	raw 15; integritas → F-T8-03 / BUG-T8-01 (→ §9)
DB: tabel metadata media	M2c-db proof: SELECT baris metadata untuk key	key terekam DB == key server-generated	Tabel media TIDAK ADA (.tables + DDL 0000-0007: nol); MediaService.confirm tidak menulis DB; substitusi persistensi yang ADA: course_resources.object_key	N/A-adapted (DOC-DRIFT PROVEN)	raw 18,40; README "register object metadata" = drift (F-T8-02 → §10)
<code>POST /api/admin/media/confirm</code>	M2c-edge traversal private/projects/././secret.png & key etc/passwd	negatif control	400 "Invalid media key." keduanya (gate isServerKey)	PASS	raw 16,17
<code>POST /api/admin/media/read-url</code>	M2d {key} valid → readUrl	alur M2	HTTP 201 {readUrl:"http://local-storage.test/read/...", expiresIn:300}	PASS	raw 19; 201-not-200 F-T8-07
<code>POST /api/admin/media/read-url</code>	M2d2 key invalid	sama	HTTP 400 "Invalid media key."	PASS	raw 20
GET readUrl + sha256	M2e: bytes hasil baca == sha256 file asal (c414cd0e...)	bytes match	TIDAK TERUJI di lokal: GET → ConnectTimeoutError (identik M2b); stub tidak menyipkan bytes → round-trip mustahil oleh desain driver	N/A-adapted	raw 21; sha256 sumber dipakai utk uji deployed
GET objek private TANPA signed query	M3: akses private/... tanpa signed → 403 storage	lokal: perilaku driver dicatat setara/beda	BEDA (bukan setara): driver lokal tidak punya signature untuk di-strip — unsigned===signed TRUE (0 query param); GET raw path → ConnectTimeoutError; enforcement-gap lokal BERBASIS KODE, bukan asumsi	N/A-adapted (LAPOR GAP)	raw 21,22; F-T8-01; uji R2 mentah = T8d deployed
lessons.is_preview FALSIFI	Klaim README "free preview bypass"	faktakan kolom/branch	(1) PRAGMA table_info(lessons) TANPA is_preview; (2) scan getVideoUrl tanpa branch preview; (3) student GET /lessons/<id>/video-url → 404 "Video lesson tidak ditemukan." — BUKAN 200-signed	PASS (falsifikasi)	raw 23,24,25; DOC-DRIFT TERBUKTI (F-T8-05 → §10)
<code>PATCH /api/lessons/:id</code>	L1 admin ubah title	lesson CRUD admin	HTTP 200, title berubah, videoObjectKey:null, status draft	PASS	raw 26
<code>PATCH /api/lessons/:id</code>	L2 student → 403	user → 403	HTTP 403 permission message	PASS	raw 27
<code>PATCH /api/lessons/:id</code>	L3 injeksi {"videoObjectKey":"private/projects/evil.png"}	edge whitelist	HTTP 400 "property videoObjectKey should not exist" → key TIDAK bisa dipasang via PATCH	PASS	raw 28; proyeksi bukti F-T8-04
<code>PATCH /api/lessons/:id/reorder</code>	L4 {orderIndex:5}	reorder admin	HTTP 200 orderIndex 5; DB order_index=5	PASS	raw 29,34
<code>PATCH /api/lessons/:id/reorder</code>	L5 {orderIndex:-1} → 400	400	HTTP 400 "orderIndex minimal 0."	PASS	raw 30
<code>POST /api/lessons/:id/publish</code>	L6 publish	CRUD+publish admin	HTTP 200 status:"published"	PASS	raw 31
<code>POST /api/lessons/:id/publish</code>	L7 publish ulang	400	HTTP 400 "Lesson sudah berstatus published."	PASS	raw 32
<code>POST /api/lessons/:id/unpublish</code>	L8 unpublish (karantina akhir)	—	HTTP 200 status:"draft" — kembali draft	PASS	raw 33
lessons.video_object_key state	Rekam kondisi null	key null	Semua respons lesson: videoObjectKey:null; DB <NULL>; TIDAK ADA rute API yang dapat mengisi key	PASS	raw 06,07,26-35,51; F-T8-04 — blokir struktural E4 (→ §2.4 PROMOSI)
GET /api/courses/:slug (adjacent)	L10 course qa-t8 draft → 404 publik	regression	HTTP 404 "Course tidak ditemukan."	PASS	raw 35
<code>POST /api/lessons/:lessonId/resources</code>	X2a admin valid (type pdf, objectKey = key M2, title)	create valid	HTTP 201; DB row persist penuh, object_key = key server-generated	PASS	raw 36,40; detail → §8 X2
<code>POST /api/lessons/:lessonId/resources</code>	X2b student	non-admin 403	HTTP 403	PASS	raw 37
<code>POST /api/lessons/:lessonId/resources</code>	X2c type:"quiz"	invalid → 400	HTTP 400 verbatim: "type hanya boleh salah satu dari: pdf, resource, assignment."	PASS	raw 38
<code>POST /api/lessons/:lessonId/resources</code>	X2c2 body {}	invalid → 400	HTTP 400, 7 pesan per-field	PASS	raw 39
<code>DELETE /api/resources/:id</code>	X2d admin delete → 2xx + DB gone	delete	HTTP 200 (echo baris terhapus) + SELECT COUNT(*) WHERE lesson_id=ours → 0	PASS	raw 41,42
<code>DELETE /api/resources/:id</code>	X2e id sendiri lagi / id asing	asing → 404	Re-DELETE id sendiri → 404 "Resource tidak ditemukan."; random-UUID v4 → 404	PASS	raw 43,44,45
<code>POST /api/lessons/:lessonId/resources</code>	X2e4 lessonId UUID tak dikenal	edge	HTTP 404 "Lesson tidak ditemukan."	PASS	raw 46
GET /api/courses?limit=100	REG adjacent: katalog sehat pasca mutasi; qa-t8 absen	regression	HTTP 200 {items:[],page:1,limit:100,total:0,query:null} — 0 slug qa-t8	PASS	raw 47,49
GET /api/home	REG adjacent	regression	HTTP 200 empty-shape wajar	PASS	raw 48

Target: **DEPLOYED** — verdict verbatim: T8-DEPLOYED: 12/14 PASS, FAILs: [BUG-T8d-01 (X2a resource-create 502 deterministik deployed; memblokir leg happy-path X2d)]. Kelompok inti R2 (M1-M3): **10/10 PASS** — keempat N/A-adapted lokal naik ke bukti nyata.

Endpoint	Skenario	Expected	Actual	Status	Catatan
<code>POST /admin/media/upload-url</code>	M1.1 contentType text/html → 400 pra-presign	MIME/size ditolak sebelum presign; non-admin 403	HTTP 400, verbatim: "contentType must be one of the following values: image/jpeg, image/png, image/webp, application/pdf, video/mp4" — body TANPA key/uploadUrl	PASS	raw 06; pesan identik byte-per-byte dgn lokal
<code>POST /admin/media/upload-url</code>	M1.2 size > 524288000 → 400	sama	HTTP 400 "size must not be greater than 524288000", tanpa key/uploadUrl	PASS	raw 07
<code>POST /admin/media/upload-url</code>	M1.3 prefix disallowed → 400	sama	HTTP 400 "prefix must be one of the following values: public/site, public/projects, private/projects, private/courses"	PASS	raw 08
<code>POST /admin/media/upload-url</code>	M1.4 student → 403	non-admin 403	HTTP 403 "You do not have permission to access this resource."	PASS	raw 09
<code>POST /admin/media/upload-url</code>	M1.5 no-token → 401	AD3	HTTP 401 "Missing or invalid authorization token." (401 bukan 403 — AD3 terpenuhi)	PASS	raw 10

Endpoint	Skenario	Expected	Actual	Status	Catatan	
POST /admin/media/upload-url	M2a filename jahat → key uuid server	key = \${prefix}/\${uuid}.ext	HTTP 201 {key:"private/projects/52247ee0---.png", uploadUrl:"https://xolvon-course-storage-staging-r2.cloudflarestorage.com/~?X-Amz-Algorithm=AWS4-HMAC-SHA256...", expiresIn:3600}; key regex uuid v4 TRUE; EVIL / .. TIDAK ada di key	PASS	raw 11; 201-bukan-200 = F-T8-07 paritas	
PUT uploadUrl (R2)	M2b bytes nyata 200	upload PUT nyata	HTTP 200 + etag:"2cd8bde4...". server: cloudflare.cf-ray → 70 B PNG MENDARAT DI R2 NYATA. BUKTI UPLOAD PERTAMA yang mustahil di lokal; TTL presign PUT terverifikasi dari surface: X-Amz-Expires=3600	PASS	raw 12; gap F-T8-01 CLOSED	
POST /admin/media/confirm	M2c (key) → 2xx; edge traversal/foreign → 400	sama	{key} nyata → HTTP 201 {key, confirmed:true}; kontrol fabricated key (TIDAK PERNAH di-PUT) → juga 201 {confirmed:true} → confirm TETAP tidak memvalidasi eksistensi objek di R2 nyata; traversal & etc/passwd → 400 "Invalid media key."	PASS (kontrak)	raw 13-16; DB-metadata sub-baris = N/A-DBAccess; F-T8-02 persists lintas target	
POST /admin/media/read-url + GET	M2d readUrl + bytes sha256 match	GET it → bytes match sha256	read-url → 201 {readUrl, expiresIn:300} (SigV4, X-Amz-Expires=300 terlihat); GET signed → 200. content-type: image/png. 70 bytes, sha256 = c414cd0e... == sumber → PLAYABLE PROVEN di R2 nyata; key invalid → 400 "Invalid media key."	PASS	raw 17-20; gap F-T8-01/M2e CLOSED	
GET raw R2 TANPA signed query	M3 → 403/AccessDenied storage	akses private tanpa signed → ditolak storage	query di-strip → DITOLAK storage: HTTP 400 <Code>InvalidArgument</Code> <Message>Authorization</Message> (deviasi kode-status dari literal "403 AccessDenied" dicatat, DENIAL tercapai — nol bytes); kontrol tamper 1 char signature → HTTP 403 SignatureDoesNotMatch	PASS	raw 21,22; unsigned===signed FALSE (kontras lokal TRUE); bukti level-storage yang mustahil diberikan stub lokal	
PRAGMA/D1 + video-url student	is_preview falsify (kolom + branch + surface 404)	sama	student (tanpa enrollment, lesson draft) GET /lessons/<id>/video-url → 404 "Video lesson tidak ditemukan." → verbatim identik lokal (branch null-key sebelum enrollment). Sub-baris skema kolom = N/A-DBAccess	PASS (surface-only)	raw 23; falsifikasi kolom tetap dari bukti lokal	
PATCH /lessons/:id +reorder/publish/unpublish	L1-L8 matriks lesson CRUD	CRUD admin; user → 403	L1 title→200; L2 student→403; L3 injeksi videoObjectKey→400 "property videoObjectKey should not exist"; L4 reorder 5→200; L5-1→400 "orderIndex minimal 0."; L6 publish→200; L7 ulang→400 "Lesson sudah berstatus published."; L8 unpublish→200 draft (karantina akhir). Semua pesan verbatim identik lokal	PASS	raw 24-31; hanya lesson milik course qa-t8d sendiri	
POST /lessons/:id/resources	X2a/b/c 201/403/400	create non-admin 403, valid	X2a FAIL — BUG-T8d-01 (deployed-only): admin valid → HTTP 502 deterministik 5/5 (halaman error edge Cloudflare "canadev.my.id \	502: Bad gateway , BUKAN envelope JSON aplikasi). Rute HIDUP di lapisan lain: X2b student → 403; X2c type="quiz" → 400 verbatim; X2c2 'j' → 400 7 pesan. GET list resource: rute tidak ada by design	FAIL	raw 32-35,44-49; lokal = 201 → DEFECT KHUSUS DEPLOYED; kandidat akar (T16/owner): drift skema/worker di jalur INSERT course_resources deployed — verifikasi butuh akses D1/worker-log
DELETE /resources/:id	X2d/e 200+gone / 404	delete; id asing → 404	Leg happy-path tidak tercapai — diblokir BUG-T8d-01 (tidak ada baris yang bisa dihapus). Leg error TERBUKTI: DELETE id non-UUID → 400 "Validation failed (uuid v4 is expected)"; random UUID v4 → 404 "Resource tidak ditemukan."; create ke lesson UUID asing → 404	FAIL	raw 36-39; mumi dependensi baris yang gagal tercipta oleh bug di atas	

Catatan deployed-parity: FINDING F-T8-03 (confirm tanpa HEAD-check) KINI TERBUKTI DI ATAS R2 NYATA (kontrol dua sisi key nyata vs fabricated → sama-sama 201) → BUG-T8-01 tetap Open-Parked (→ \$9). Residual jurur: 1 objek R2 70 B penanda QA tak terhapus (tidak ada endpoint delete objek — by design).

\$2.4 Enrollments/Orders/Signed-URL (mapping E1-E10 — PALING KRITIKAL; bukti task-9-critical-local.md + task-9-local/raw 48 + task-9-deployed/raw 52)
Target: LOKAL — verdict verbatim: T9-L0CAL: 19/20 PASS + 1 N/A-adapted (E6 driver-caveat – satu-satunya kaveat diizinkan acceptance), FAILs: [], E8: EXPLOIT-TERBUKTI.
Fixture-method WAJIB diingat saat membaca bukti E4/E6: video_object_key mustahil diisi
via API (F-T8-04) → SATU UPDATE lessons SET video_object_key=_ WHERE id=<lesson qa-t9>
sqlite terlabel (receipt raw/s2-db-lesson.txt).

Endpoint	Skenario	Expected	Actual	Status	Catatan			
POST /api/orders	E1a (L1) order pertama course published (A)	201 pending, amount server-side	201; amount=150000 == SELECT price FROM courses =150000; DB: order `22deb73e...`)	150000)	pending)	A`	PASS	total dihitung dari DB; klien tak kirim amount
POST /api/orders	E1b (L2) order ULANG course yang sudah enrollment ACTIVE	amati perilaku nyata	201 order baru b57ce3a1... pending; DB pending A+course: 0→1	PASS (observasi)	[OPEN DECISION] — checkout TIDAK cek enrollment aktif/duplikat/status published; order dup dibatalkan di teardown; aturan tidak di-invent			
POST /api/orders/:id/payment-proof	E2a (L3) proof objectKey arbitrer (private/courses/whatever.png)	diterima (DTO hanya string)	201 {message:"Bukti pembayaran berhasil diunggah.",proofId}:row payment_proofs tercatat	PASS	persis prediksi DTO			
payment-proof	E8 (L4) traversal key .././etc/passwd	amati	201 diterima, row DB object_key .././etc/passwd	PASS (probe)	bagian verdict E8			
payment-proof	E8 (L5) traversal percent-encoded ../..%2f..%2fprivate%2fcourses%2fx.png	amati	201 diterima	PASS (probe)				
payment-proof	E8 (L6) PROBE kunci lintas-user: A submit objectKey milik B	amati; diterima = EXPLOIT	A: 201 diterima (proofId 370b3dac...); B: 201 milik sendiri. DB: kedua baris payment_proofs memakai object_key yang sama, uploaded_by user berbeda	PASS (probe)	E8: EXPLOIT-TERBUKTI (write-surface) → BUG-T9-01; raw dua sisi tersimpan			
payment-proof	E7-guard adjacent (L7) B submit proof ke ordernya A	403 (order-IDOR guard)	403 Anda tidak memiliki akses ke order ini.	PASS	guard ORDER-ownership ADA; yang TIDAK ada = validasi KEY (L6)			
PATCH /api/orders/:id/verify	E2b (L8) admin verify pending→paid	200 paid + audit	200 status=paid; DB verified_by=<admin>; audit row verify	PASS				

Endpoint	Skenario	Expected	Actual	Status	Catatan	
verify	E2c (L9) verify order CANCELLED	4xx state-machine	400 "Order yang sudah dibatalkan tidak dapat diverifikasi."	PASS		
POST /api/orders/:id/cancel	E9 (L10) cancel order PAID; cancel DUA KALI	4xx konsisten	cancel-paid: 400 "Hanya order berstatus pending yang dapat dibatalkan." ; double-cancel: 400 pesan sama; pending→cancelled: 200	PASS	state machine satu arah	
POST /api/orders/:id/activate	E3 (L11) activate lalu activate LAGI (sekuensial)	200; enrollment tetap 1	#1 200 {activatedCoursesCount:1}; #2 200 body identik; DB COUNT enrollments (A,C1) = 1; /enrollments/me: 1 baris active	PASS	UPsert ON CONFLICT.FINDING minor: aktivasi ke-2 tetap menulis audit row ke-2 & count tetap 1 padahal no-op (FIND-T9-05)	
GET /api/lessons/:id/video-url	E4 (L12) TANPA enrollment	403	403 "Enrollment aktif diperlukan." (A & B)	PASS		
video-url	E4 (L13) DENGAN enrollment active: mint + GET URL	200 {videoUrl, expiresIn:300}; URL playable	200 {"url":"http://local-storage.test/read/private/courses/qa-t9-...mp4","expiresAt":"-"}; GET url → gagal resolusi DNS	PASS dengan 2 catatan	(1) deviasi bentuk kontrak {url, expiresAt} BUKAN {videoUrl, expiresIn} — prompt salah, kode otoritatif (FIND-T9-03 → §10); (2) local-test driver → "playable" tak terdefinisi lokal (FIND-T9-04); padanan R2 = DEPLOYED/T8d	
video-url	E4c (L14) setelah enrollment REVOKED, minta ulang	403 (cek per request)	403 "Enrollment aktif diperlukan." ; DB: enrollment'a2ce9380...)	revoked	PASS	entitlement diverifikasi per request, bukan cache
video-url	E5 (L15) IDOR: B minta lesson course yang HANYA A ikuti	403/404	403 "Enrollment aktif diperlukan."	PASS		
GET signed URL expired	E6 (L16) mint → tunggu >305s → GET lagi	catat AKTUAL (driver lokal mungkin tak menegakkan expiry)	GET @+124s lewat expiry: http_status=000 , curl_exit=28 DNS-fail — TIDAK ada 403 storage; URL expired & hidup tak terbedakan di lokal	N/A-adapted (driver caveat)	pembuktian TTL = DEPLOYED (R2 presigned terbukti di T8d: X-Amz-Expires=300)	
POST /api/progress , GET /api/progress	E7 (L17) A tulis own; spoof user_id B di body; GET per user; B tulis lesson milik A	2xx; 400 whitelist; hanya baris sendiri	A POST 201; A POST + user_id:"<B-uuid>" → 400 ["property user_id should not exist"]; A GET → hanya 1 baris milik A; B POST → 403 "Enrollment aktif diperlukan." ; B GET → []; DB: 1 row, user_id=A	PASS	progress ikut entitlement gate, bukan sekadar whitelist; race duplikat = T14C	
PATCH /api/enrollments/:id/revoke	E10 (L18) student revoke sendiri; B revoke A; admin revoke; admin revoke LAGI; /me isolation	non-admin 403; admin 200; ulang → idempotent-atau-4xx (record); me terisolasi	A: 403; B: 403; admin #1: 200; admin #2: 200 + audit revoke nambah 1→2; B /me: []; A /me pasca-revoke: 1 baris status:"revoked"	PASS dengan FINDING	FIND-T9-02: revoke ulang SELALU 200 + audit duplikat (tanpa guard status) — direkam apa adanya	
admin_audit_logs	spot-check (L19) verify/activate/revoke/publish/cancel/create QA actor	baris ada, actor=admin	LIMIT 12 berisi 12 aksi, semuanya actor admin; SELECT mismatch actor = 0	PASS	completeness penuh = T13	
teardown karantina	(L20) course berakhir UNPUBLISHED; order nyasar dibatalkan	draft + konsisten	course draft ; lesson draft ; orders: 1 paid (fakta historis), 3 cancelled, enrollment revoked; progress 1 baris; proofs 5 baris	PASS	Catatan karantina: users tak bisa dihapus via API → akun qa + proofs + orders historis TINGGAL (jujur)	

Target: DEPLOYED — verdict verbatim: T9-DEPLOYED: 17/20 PASS (incl 3 N/A justified: D13/D16 noKeyAttachment + D19 DBAccess — tanpa satu pun skip diam-diam), FAILs: [], E8-deployed: EXPLOIT-TERBUKTI (paritas penuh lokal; BUG-T9-01 tetap High).

Endpoint	Skenario	Expected	Actual	Status	Catatan
POST /api/orders	E1a (D1) order pertama course published	201, amount server-side	201 9ed781e1... status pending, amount=150000 == price respons POST /courses (klien hanya kirim courseId)	PASS	padanan bukti DB lokal via respons ber-side-server
POST /api/orders	E1b (D2) re-order saat enrollment ACTIVE	amati nyata	201 order baru deed505e... pending	PASS (observasi)	PARITAS deployed [OPEN DECISION] — order dibatalkan di teardown (D20)
payment-proof	E2a (D3) key arbitrator	diterima	201 {message:"Bukti pembayaran berhasil diunggah.",proofId:"48f0d35a..."}	PASS	persis lokal
payment-proof	E8 (D4) traversal ../etc/passwd	amati	201 diterima, proofId 80f6782d...	PASS (probe)	verdict E8-deployed
payment-proof	E8 (D5) encoded traversal	amati	201 diterima, proofId 5e5d940f...	PASS (probe)	
payment-proof	E8 (D6) probe kunci lintas-user B→A	amati; two-sided	B (key sendiri): 201; A submit key MILIK B di order A: 201 3712b0ab...	PASS (probe)	E8-deployed: EXPLOIT-TERBUKTI — raw dua sisi, + key asing private/courses/foreign-alien-* 201
payment-proof	adjacent (D7) B→order A	403	403 "Anda tidak memiliki akses ke order ini."	PASS	guard kepemilikan order ADA; validasi KEY tidak ada (D6)
PATCH /orders/:id/verify	E2b (D8) pending→paid	200 + audit	200 status=paid, respons memuat verifiedAt	PASS (actor audit = N/A-DBAccess)	audit row → D19/T13
verify	E2c (D9) verify cancelled	4xx	400 "Order yang sudah dibatalkan tidak dapat diverifikasi."	PASS	verbatim sama lokal
cancel	E9 (D10) cancel-paid; double-cancel; pending→cancelled	400/400/200	cancel-paid: 400 "Hanya order berstatus pending yang dapat dibatalkan." ; double-cancel: 400 sama; pending→cancelled: 200	PASS	paritas lokal
activate	E3 (D11) activate *2 sekuensial → 1 enrollment	200 + COUNT=1	#1 200 {activatedCoursesCount:1}; #2 200 identik; GET /enrollments/me: tepat 1 baris active	PASS	COUNT via /me (DB deployed non-akses); paralel = T14d
video-url	E4 (D12) tanpa enrollment	403	404 "Video lesson tidak ditemukan." (A pra-aktivasi, B, A pasca-aktivasi — ketiganya identik)	PASS sebagian; cabang 200/403 = N/A-noKeyAttachment	urutan kode: key-null 404 SEBELUM entitlement 403 — gerbang entitlement TIDAK terverifikasi di E4-deployed (jujur)
video-url	E4 (D13) mint + GET url playable	200 {url,expiresAt}	N/A-noKeyAttachment — tak ada jalur API melampirkan key (bukti hidup 400 kedua varian nama field; confirm tanpa bind lesson; tak ada sqlite)	N/A	PROMOSI temuan sistemik (blok di bawah)
video-url	E4c (D14) pasca-revoke	403	404 "Video lesson tidak ditemukan." (identik pra-entitlement)	PASS surface; cabang 403 tak terverifikasi	revoke tak bisa dibedakan dari enrolled pada lesson tanpa key

Endpoint	Skenario	Expected	Actual	Status	Catatan
video-url	E5 (D15) IDOR lintas-user	403/404	404 "Video lesson tidak ditemukan."	PASS (404 aktual; 403-branch tak terverifikasi)	
GET signed URL	E6 (D16) expired >305s	storage menolak (403)	N/A-noKeyAttachment — URL tak pernah bisa diminta (D13) → penegakan TTL 300s R2 tak teruji di baris ini	N/A	justifikasi = temuan terpromosi, BUKAN skip diami-diam; TTL R2 terbukti di T8d read-url
progress	E7 (D17) own + spoof user_id + isolation	2xx/400/own-only	APOST → 201; spoof user_id → 400 ["property user_id should not exist"]; A GET hanya milik A; B POST → 403 "Enrollment aktif diperlukan."; B GET []	PASS penuh	TIDAK bergantung key — teruji penuh deployed (entitlement progress nyata)
revoke	E10 (D18) matrix + /me isolation	403/200/actual[]	A: 403; B: 403; admin #1: 200 status=revoked; admin #2: 200 identik; B /me : [] ; A /me : 1 baris revoked	PASS dengan FINDING	FIND-T9-02 paritas deployed; audit-dup tak terverifikasi → D19
audit	(D19) spot-check actor	actor benar	N/A-DBAccess — tanpa akses SELECT deployed (D1) → pointer T13 (wangler)	N/A-DBAccess	jujur per aturan; bukan FAIL
teardown	(D20) course qa deployed UNPUBLISHED akhir	draft	unpublish 200; publik GET /courses/<slug> → 404; order tersisa cancelled 200*3, 1 tetap paid (historis) — receipt via GET /admin/orders	PASS	Catatan residu: 2+2 akun student qa-t9d, 6 proofs, 4 orders, 1 enrollment revoked TINGGAL (tak terhapuskan via API — jujur)
POST /api/orders	D-EX FIND-T9-06 deployed: order course DRAFT	record actual	201 order d49bca94... pending amount=75000 — order atas course DRAFT DAPAT dibuat (SELECT tanpa filter status)	PASS (observasi)	FIND-T9-06 paritas deployed; dibatalkan di teardown

PROMOSI temuan sistemik (T9d → \$9, High, Open-Parked product-decision): alur video premium TIDAK TERJANGKAU via API pada build ini — tidak ada satu pun rute yang menautkan video_object_key ke lesson (DTO create/update menolak field; confirm tidak bind lesson; deployed tanpa jalur DB tulis) ⇒ GET /lessons/:id/video-url SELALU 404 sebelum cabang entitlement. README "Secure Media & Video Streaming Flow" bersifat FIKSIONAL pada build ini. Perbaikan = surface baru (rute attach key / field DTO) → keputusan pemilik, BUKAN dibuat QA.

\$2.5 Projects/Marketplace/Collective (mapping P1–P6; bukti task-10-portfolio-local.md + task-10-local/raw 63 + task-10-deployed/raw 77)
Target: LOKAL — verdict verbatim: T10-LOCAL: 22/22 PASS, FAILS: []

Endpoint	Skenario	Expected	Actual	Status	Catatan
POST /api/projects	P1-a project DRAFT dibuat (semua field narasi terkirim)	draft → tak muncul publik	HTTP 201, status:"draft", id UUID v4 86bb81fb-...; body = ProjectResponseDto lengkap (problem/solution/techStack/result terisi)	PASS	raw 10
GET /api/projects?q=	P1-b draft qa ABSEN dari katalog publik	draft absent	HTTP 200; items = 0 entri berslug qa-t10-proj-1789279022	PASS	raw 11
GET /api/projects/:slug	P1-c slug DRAFT → 404, bukan datanya	404	HTTP 404 "Project not found." — identik slug tak dikenal (draft#missing)	PASS	raw 12
GET /api/projects/:slug	P1-d slug random → 404 envelope BUKAN 500	404 envelope, no 500	HTTP 404 (bukan 500), envelope keys tepat [statusCode,message,error,timestamp,path]	PASS	raw 13
POST /api/projects/:id/publish → list	P1-e publish → 200 → muncul di list	publish 200 → appears	publish HTTP 200 status:"published" → list ?q= kini total:1, slug hadir	PASS	raw 14,15
GET /api/projects	P1-f list keys	items/page/limit/total/query	keys ACTUAL = [items,page,limit,total]: page:1 limit:100 total:1	PASS	FINDING F-T10-01: query TIDAK di-echo — kontrak repo api-contract.md:98 memang mem-pin 4 kunci TANPA query → teks-tasks ter-generalisasi dari /courses; bukan bug produk
GET /api/projects/:slug	P2-a detail narasi: problem/solution/techStack/result non-null	field ada + non-null	HTTP 200; summary.problem.solution.result non-null; techStack di-parse jadi ARRAY ["Next.js","NestJS","D1","R2"]; key order respons: ... problem,solution,techStack,result,media,members...	PASS	raw 16,18; urutan RENDER visual = urusan frontend, di luar surface API
POST /api/projects/:id/media → detail	P2-b media attach → array media muncul	media array present	attach HTTP 201; detail: media [{id,mediaType:"image",sortOrder:0}] — objectKey TIDAK terekspos publik (scan 0 hit)	PASS	raw 17,18
POST /api/projects/:id/members	P3-a assign member collective NYATA dengan role	201, built-by role per member	HTTP 201 {projectId,memberId,role:"QA Lead Engineer"}; detail: members: [{memberId,name,role:"QA Lead Engineer"}] — role PER MEMBER terbukti	PASS	raw 20–23,30,31
DELETE /api/projects/:id/members:memberId	P3-b unassign member asli	2xx	HTTP 200 {"message":"Project member removed."}; follow-up detail: members:[]	PASS	raw 35,36; audit 'delete'
POST /api/projects/:id/members (UUID asing)	P3-c assign member ASING → 404	404	HTTP 404 "Collective member not found." — existence check SEBELUM insert (BUKAN FK-500); nol baris terbentuk	PASS	raw 33 (F-T10-04 positif)
DELETE /api/projects/:id/members:asing	P3-d unassign id ASING → 404	404	HTTP 404 "Member assignment not found for this project."	PASS	raw 34
GET /api/collective + /:slug	P4-a member DRAFT: absen list + detail 404	draft absent; 404	create 201 draft; list ?q= 0 entri; detail → HTTP 404 "Collective member tidak ditemukan."	PASS	raw 20–22
POST /api/collective/:id/publish → list	P4-b publish → muncul ONLY after publish	appears only after publish	publish 200 → list ?q= slug hadir (keys [id,name,slug,photo,role,skills,bio,socialLinks,status])	PASS	raw 23,24
GET /api/collective/:slug	P4-c detail TANPA email/phone (recursive scan)	no PII keys	recursive keys VERBATIM: bio,id,name,photo,platform,relatedProjects,role,skills,slug,socialLinks,status; forbidden-key scan [email,phone,password,...] → HITS: []; scan nilai 0 hit	PASS	raw 25; stripping berlapis: DTO + SELECT + mapper
POST/GET /api/marketplace	P5-a create+publish item external_url → detail	externalUrl present	create 201 draft; publish 200; detail keys [id,title,slug,description,capabilities,externalUrl,status,media], externalUrl:"https://qa-t10.example.test" verbatim	PASS	raw 40–42; field API = camelCase externalUrl (literal external_url = nama DB)
POST /api/marketplace/:id/pay	P5-b TIDAK ada endpoint pembayaran → 404	404	HTTP 404 — body = HTML fallback Express Cannot POST /api/marketplace/<id>/pay (rute tak terdaftar tak lewat AllExceptionsFilter)	PASS	raw 43; FINDING F-T10-02 (Low) dua kelas 404
POST /api/marketplace/:id/checkout	P5-c endpoint checkout → 404	404	HTTP 404, HTML fallback serupa	PASS	raw 44
POST/DELETE /api/marketplace/:id/media	P5-d media attach/delete per DTO	works per DTO	attach image → 201; mediaType:"screenshot" → 400 ["mediaType hanya menerima nilai image, video, atau deck."]; delete → 200 → media:[]; media item keys [id,objectKey,mediaType,sortOrder] — objectKey EKSPOS publik di marketplace (kontras projects)	PASS	raw 45–48; F-T10-03 asimetri by-design tercatat

Endpoint	Skenario	Expected	Actual	Status	Catatan
Xoligon Backend Documentation - xoligon-backend-full-documentation.pdf SEMUA 18 mutation (projects×8, collective×4, marketplace×6) sebagai STUDENT	P6-a student → 403 semua mutation	403	18/18 HTTP 403 "You do not have permission to access this resource."; verifikasi anti-sisi: GET detail setelahnya → tidak ada mutasi boor	PASS	raw 70–87
3 subset mutation TANPA token	P6-b no-token → 401 (distingsi 401≠403)	401	3× HTTP 401 "Missing or invalid authorization token," — berlainan pesan+status dengan 403 student	PASS	raw 90–92: AuthGuard vs RolesGuard terbukti
project detail + collective detail	Relasi dua arah project→member dari KEDUA sisi	both sides	Sisi A: members: [{,role:"QA Lead Engineer"}] . Sisi B: relatedProjects: [{id,title,slug,type,summary,status:"published"}] . TWO-WAY CONFIRMED; negative control pre-assign relatedProjects: []	PASS	raw 31,32 (F-T10-04)
POST :id/unpublish ×3 + GET ×3 + SQL RO	QUARANTINE EXIT: semua qa-t10 DRAFT	unpublish → 404 publik	3×200 status:"draft!"; GET slug → 3×404; DB RO: ketiganya draft, project_members=0, market_media=0, proj_media=1 (row stub pada project DRAFT, tak terekspos publik)	PASS	raw a1–a6

Target: **DEPLOYED** — verdict verbatim: T10–DEPLOYED: 4/8 PASS, FAILS: [P1, P2, P3, RELASI] – keempatnya blocked-by-BUG-T10d-01 (product defect deployed, ber-BUG-id per rubrik)

Endpoint	Skenario	Expected	Actual	Status	Catatan
POST /api/projects (+list/detail)	P1 draft lifecycle (create→absent→404→publish→appears; list keys)	sama	create GAGAL: HTTP 502 cPanel "Bad gateway" HTML, persisten 1×(run-1)×3×(resume @20s backoff)+1×(diag body minimal), latency ~325–400ms → tanpa PROJ_ID, publish/appears tak ter-exercise. Proksi-surface lain terbukti: draft/unknown slug → 404 "Project not found." envelope identik; list keys [items,page,limit,total] TANPA query — parity persis F-T10-01	FAIL (BUG-T10d-01)	raw 04,69–71,001; kandidat orphan draft invisible terdokumentasi (502 mendahului validasi; 409 tak pernah muncul)
GET /api/projects/:slug	P2 narasi 5 field + media array	sama	Tak tercapai: tidak ada proyek milik sendiri (create 502); attach media ke id placeholder → 400 uuid. Guardrail melarang memakai baris milik orang lain	FAIL (blocked-by-BUG-T10d-01)	raw 10,11,12; bukan deviasi perilaku — belum tersentuh
POST/DELETE /api/projects/:id/members	P3 built-by roles + alien 404 ×2	sama	Sisi-proyek tak tercapai (id placeholder → 400 ParseUUIDPipe). Sisi-member TERBUKTI sehat di P4. Alien-assign 404 service-level tak teruji deployed	FAIL (blocked-by-BUG-T10d-01)	raw 19–25; repro siap-jalan begitu create project hidup
GET /api/collective(/:slug)	P4 published-only + PII recursive scan	sama	PARITY PENUH: create 201 draft; list 0 saat draft; detail draft 404; publish → hadir; recursive keys VERBATIM sama lokal; forbidden scan → 0/0 hits	PASS	raw 13–18
POST/GET /api/marketplace(/:slug) + /pay /checkout	P5 externalUri + no-payment 404×2 + media enum	sama	PARITY PENUH: create/publish/detail verbatim; pay & checkout → 404 HTML fallback Express — F-T10-02 TERKONFIRMASI SAMA di deployed; enum 400 verbatim; delete 200 verbatim; objectKey EKSPLOS (F-T10-03 parity)	PASS	raw 26–34; bukti response-surface (no D1)
SEMUA mutation ×student + no-token	P6 403-grouped + 401×≥2	sama	18/18 HTTP 403 verbatim + 3/3 no-token HTTP 401 verbatim; fakta baru: RolesGuard MENDAHULUI ParseUUIDPipe (403 murni guard bahkan dgn id invalid)	PASS	raw 35–55
project detail + collective detail	Relasi dua arah dari kedua sisi	sama	Sisi-B tereksekusi sbg NEGATIVE CONTROL valid: relatedProjects: [] (200). Sisi-A TIDAK teruji — create 502	FAIL (blocked-by-BUG-T10d-01)	raw 20,21; dua-aras TERBUKTI di LOCAL → gap = infra/build deployed, bukan logika
unpublish ×3	QUARANTINE EXIT deployed	sama	Receipts response-surface: member & item unpublish 200 draft → 404 → list absent; proyek: tak pernah published (create 502). Residu terdokumentasi (draft tak publik): 3 collective draft, 1 marketplace draft, 2 students, ±1 project draft orphan — butuh owner D1-cleanup opsional; nol eksposur publik	PASS (adapted-caveat)	raw a1–a9, b-series

BUG-T10d-01 (detail §9): isolasi permukaan — rute lain modul projects SEHAT (publish id asing → 404 JSON service-level), tulis modul lain SEHAT pada detik yang sama (POST /collective → 201), baca GET /projects → 200 ⇒ kegagalan spesifik-rute pada permukaan tulis projects deployed — **satu keluarga dengan BUG-T8d-01** (502-CF family; hipotesis akar utk owner: worker crash mid-route ATAU rule edge/WAF — BUKAN keputusan QA).

§2.6 Admin aggregation (mapping AD1–AD4; bukti task-11-admin-local.md + task-11-local/raw + task-11-deployed/raw 26)

Target: **LOKAL** — verdict verbatim: T11–LOCAL: 23/23 PASS, 1 N/A-adapted (AD1 proof-ref = DL–015 locked), 1 DELEGATED-T13 (completeness), FAILS: []

(hitungan kontrak 23 = 4 AD1 + 3 AD2 + 14 AD3 [diagregasi per-rute 7×2] + 1 AD4 titik + 1 overview; baris AD3 diagregasi di bawah demi format kolom seragam).

Endpoint	Skenario	Expected	Actual	Status	Catatan
GET /admin/orders	AD1 top-level keys	{items,page,limit,total}	exactly that	PASS	
GET /admin/orders	AD1 10 mapped kolom (PRD §45 — bukan "Handbook §7")	semua ada per row	10/10 × 6 rows, jq has()=true — [id,user,courseTitles,amount,status,createdAt,updatedAt,activationStatus,grantedBy,grantedAt] ; user = {id,name,email,phone}	PASS	mapping: User/Email/ Course→ c Amount; Or Date→ c r Payment St Activation/C — lihat tabe evidence
GET /admin/orders	WhatsApp/Proof-Reference + "Actions" kolom	dedicated field	ABSENT — 'jq map(select(test("whatsapp\	proof\	action")))
GET /admin/orders?	invalid enum → status=bogus	400	400 ["status hanya menerima nilai pending, paid, atau cancelled."]	PASS	verbatim
GET /admin/users?	AD2 recursive scan	0×password_hash, 0×argon2	jq [paths] → 0 hits: grep argon2 → 0; grep password_hash → 0	PASS	SELECT co (admin.serv
GET /admin/users/id	AD2 same scan (student sendiri)	0/0	0 hits; keys= [createdAt,email,enrollments,id,name,orders,phone,role,status]	PASS	
/admin/users?	AD2 DL-011 bounds	400/400/200-empty	400 "limit maksimal 100." / 400 "page minimal 1." / 200 items=[] page=9999 total=9	PASS	total echo s range page
7 rute /api/admin/*	AD3 guard matrix 7 × (no-token, student) = 14 check	tanpa token → 401 (bukan 403); token user → 403	no-token → 401 verbatim Missing or invalid authorization token. pada SEMUA 7; student → 403 verbatim You do not have permission to access this resource. pada SEMUA 7 — 14/14, tidak ada rute yang collapse	PASS	class-level; admin.contr media.contr per pasang.
PATCH /api/courses/id	AD4 audited mutation (SELECT titik)	+1 row, actor=admin uuid	Δ+1 (87→88), row 3f0ee33a, actor= 85cf38c6—6974 == admin uuid; semua 88 rows actor=admin	PASS	completene (delegasi pl sini)

Endpoint	Skenario	Expected	Actual	Status	Catatan
POST /admin/media/{upload-url,confirm,read-url}	AD4 audit on media mutations	—	ZERO rows + zero call sites (grep audit.record(= 10 call-site map: media = nol)	PASS (re-proof)	→ BUG-T1 (alias BUG-parked, not
(completeness penuh)	AD4 semua aksi QA × actor	—	TIDAK dihitung mid-wave (paralel → non-deterministik); delegated	DELEGATED→T13	hasil T13: 8 admin — ltr
GET /admin/overview	parity vs SELECTs	{userCount,pendingOrders,activeEnrollments}	{ <code>"userCount":10,"pendingOrders":1,"activeEnrollments":1</code> } vs SELECT `10` user = {email,id,name,phone}	1\	1' — exact

Target: DEPLOYED — verdict verbatim: T11-DEPLOYED: 21/21 PASS, 1 N/A-adapted (AD1 proof-ref = DL-015, sama lokal), 1 N/A-DBAccess (AD4 → pointer T13; CF token names-only tersedia), FAILs: []

Endpoint	Skenario	Expected	Actual	Status	Catatan
GET /admin/orders	AD1 top-level keys	{items,page,limit,total}	exactly that; rows=4, total=4	PASS	DB deployed berisi data lintas pass — STRUKTUR kolom yang diuji, bukan count
GET /admin/orders	AD1 10 mapped kolom	semua ada per row	10/10 × 4 rows, 1 keyset unik [activationStatus,amount,courseTitles,createdAt,grantedAt,grantedBy,id,status,updatedAt,user] ; user = {email,id,name,phone}	PASS	mapping identik lokal
GET /admin/orders	WhatsApp/Proof/Actions kolom	dedicated field	ABSENT — scan keys = []	N/A-adapted	excluded by DL-015 (locked) — sama persis loka
GET /admin/orders?status=bogus	invalid enum	400	400 ["status hanya menerima nilai pending, paid, atau cancelled."]	PASS	verbatim identik lokal
GET /admin/users?limit=100	AD2 recursive scan + pagination	0×password_hash, 0×argon2	paths=0, argon2=0, suspicious keys= [] ; top-keys exact; total=14	PASS	
GET /admin/users/{id (student sendiri)}	AD2 same scan + shape	0/0	200; keys identik lokal; 0 hash/argon2/suspicious	PASS	
/admin/users bounds (101/0/9999)	AD2 DL-011	400/400/200-empty	400 "limit maksimal 100." / 400 "page minimal 1." / 200 items=[] page=9999 total-echo=14	PASS	parity lokal
7 rute admin × (no-token, student)	AD3 14 check	401 / 403	14/14 — 401 & 403 pesan verbatim identik lokal di semua 7 rute (termasuk 3 media); curl independen spot-check /admin/overview no-token → 401	PASS	tidak ada inkonsistensi
(D1 admin_audit_logs)	AD4 completeness deployed	rows per aksi	TANPA akses D1 dari pass T11d → tidak ada query dieksekusi di sini; token CF names-only HADIR → pointer T13 (wrangler read-only)	N/A-DBAccess	T13 kemudian MENGEKSEKUS 12 SELECT read only (lihat §3)
API-surface parity AD4	mutation oleh worker ini sendiri	audit per mutasi	Nol aksi-mutasi di pass T11d (read-only; satu-satunya create = akun student karantina via /auth/register, bukan aksi admin) → tidak ada objek audit surface-side	N/A (alasan)	BUG-audit-media tidak di-re-proof deployed di pass ini — milik T13
GET /admin/overview	parity shape	{userCount,pendingOrders,activeEnrollments} sorted-identik	200 keys= [activeEnrollments,pendingOrders,userCount] : values= {userCount:13,pendingOrders:0,activeEnrollments:0}	PASS	values = realitas deployed, tidak dinilai

QUARANTINE EXIT deployed: tidak ada konten dibuat/diubah (satu-satunya artefak = akun student karantina tanpa data); nol mutasi admin dieksekusi.

§2.7 Home aggregation (mapping H1; bukti task-12-home-local.md + task-12-local/raw + task-12-deployed/raw + artefak t12e-* ; sapuan X1 → §8)

Target: LOKAL — verdict verbatim: T12-LOCAL: 12/12 PASS, FAILs: [] (H1-b ditutup PASS via T12e Wave-4; hitungan mencakup baris §8 X1 + teardown receipt)

Endpoint	Skenario	Expected	Actual	Status	Catatan
GET /api/home	H1-a No-draft-leak: tokenless saat 4 entitas DRAFT + 1 course PUBLISHED ada	200; slug/nama draft 0 kemunculan di SEMUA array: course published hadir atau dijelaskan cap	200. Scan rekursif: 4/4 slug draft = 0 hits (courses/projects/marketplace/collective). qa-t12-pub-... HADIR di courses[] . Body hanya memuat 2 course (milik sendiri + sibling worker; provenans dicek via detail publik 2/2 = 200, draft akan 404)	PASS	Cap COURSE_LIMIT=4 tidak tercapai → kehadiran published ter-assert langsung, tanpa cabang "cap hides it"
GET /api/home	H1-b Empty-state (DB kosong)	200 bentuk kosong wajar, bukan 500	200 {"courses":[],"projects":[],"marketplace":[],"collective":[]} — keempat array kosong; adjacent GET /api/courses = 200 {"items":[],"page":1,"limit":20,"total":0,"query":null}	PASS (via T12e Wave-4)	Kaveat metode terdokumentasi: premis plan "boot fresh → auto-migrasi" TERBUKTI SALAH — DB tanpa schema membalas 500 sanitised (artefak t12e-literal-*noSchema.txt , DIARSIP bukan di-pass-kan); empty-state sah = schema valid (migrate via runMigrations bawaan repo, 8/8 file) + nol content. Temuan metode: F-T12e-01 (→ §9)

Target: DEPLOYED — verdict verbatim: T12-DEPLOYED: 7/7 PASS (H1-b N/A-adapted), FAILs: [], FINDING: BUG-T12d-01 (POST /projects 502 deterministik deployed – infra, CF-class sekelas BUG-T8d-01)

Endpoint	Skenario	Expected	Actual	Status	Catatan
GET /api/home	H1-a(d) No-draft-leak via fixture qa-t12d deployed	idem lokal	200. 3 slug draft (course/mkt/col) = 0 hits di SEMUA array: publish → courses[] berisi qa-t12d-pub-... (cap tidak tercapai → terassert langsung); UNPUBLISH → /home **0 string qa-t12d*"; draft publik detail 404	PASS	project-draft TIDAK bisa dibuat deployed: POST /projects → 502 deterministik (CF Bad Gateway HTML, 3× repro) → cakupan draft-project tak ter-exercise langsung; tetap aman by-code (filter status='published' identik home.service.ts:89-91) + array projects tak memuat slug-ku di 3 snapshot. FINDING = BUG-T12d-01 (dedup BUG-T10d-01)
GET /api/home	H1-b(d) Empty-state deployed	—	TIDAK DIUJI — akan butuh menghapus konten published milik orang (course-1 dst.)	N/A-adapted	sesuai plan T12; empty-state = T12e lokal

Temuan pendukung §2.7 lokal (F-T12-02: collective tidak terindeks FTS; F-T12-03: item /home tanpa field status → provenance via semantic 404) → §9/§10.

§3 Data integrity (T13 — SELECT langsung READ-ONLY; bukti task-13-integrity.md + task-13-raw/)

DB baseline lokal bersih-by-construction (reset T1) → angka global = murni efek Wave 1-4.

Semua sqlite3 memakai file:local.db?mode=ro ; semua wrangler --command "SELECT _" — rows_written:0 / changes:0 di SETIAP envelope D1 (bukti read-only dari respons D1 sendiri).

Adaptasi kolom task-text→aktual (semantik sama): orders amount (bukan total_amount) ;

enrollments granted_at / revoked_at (bukan activated_at).

Basis: LOKAL — T13-LOCAL: PASS (bugs baru: none)

Poriksa	Metode (aktual)	Actual (run sungguhan)	Status	Catatan		
Xolvon Backend Documentation - xolvon-backend-full-documentation.pdf (a) dupel enrollment aktif (user_id,course_id)	GROUP BY 1,2 HAVING c>1	0 rows; PRAGMA index_list → UNIQUE autoindex pada (user_id,course_id) TERKONFIRMASI; total enrollments=2 = persis 2 aktivasi ber-evidence (T9 + newman folder-06)	PASS	raw a-enrollments.txt		
(b) order paid ter-activate ↔ enrollment	LEFT JOIN order_id	kedua order paid (22deb73e T9, 5a14f2aa newman) punya enrollment tertaut — 0 yatim; arah invers juga bersih; 1 pending sah (T11)	PASS	raw b-orders.txt ; paid-but-not-activated memang LEGAL per state machine		
(c) admin_audit_logs COMPLETENESS (delegasi T11/AD4)	build expected dari SEMUA capture 2xx admin-mutation wave (classifier call-site) vs GROUP BY action	total 89 baris; AKTOR 89/89 = admin uuid tunggal, non-admin/NULL = 0; expected c actual di SEMUA kategori — tidak ada kategori kurang; delta semua over-capture teratribusi (newman window / siklus same-worker); media = 0 baris by-design (BUG-audit-media parked terkonfirmasi, bukan completeness-miss); double-audit activate*2/revoke*2 = persis prediksi FIND-T9-05/02	PASS	tidak ada BUG-T13 (nol under-capture); raw c-audit.txt , c-attribution.txt		
(d) tidak ada secret di log	grep 'password_hash\	\$argon2\	access token' server.log *3 = 0 hits; sample-hash 16-char (metode tercatat) grep ke SELURUH evidence tree = 0 hits; regex JWT penuh = 0 file	0 violations	PASS	hit kata di transcripts = field-name/label assertion (whitelisted, dikutip berkonteks raw d-secrets.txt)
(e) progress duplikat (baseline PRE-RACE)	GROUP BY user_id,lesson_id HAVING c>1	0 rows dup; total 1 baris → baseline diserahkan ke T14-C	PASS	progress TANPA UNIQUE (0003)		
(f) inventaris residu qa-*	SELECT per tabel	users 8 qa (total 11), courses qa 8 (semua draft final) + 1 contoh newman, orders 5 qa + 1 newman, proofs 6, enrollments 2 (1 active newman, 1 revoked T9), progress 1, sessions 20, projects/collective/marketplace 2/2/2 semua draft, audit 89 immutable	PASS (inventaris)	karantina-conformant; raw f-residue.txt		

Basis: DEPLOYED (wrangler D1 xolvon-staging --remote , SELECT-only — EKSEKUSI, bukan BLOCKED) — T13-DEPLOYED: PASS

Periksa	Metode (aktual)	Actual (run sungguhan)	Status	Catatan
(a) dupel active	SELECT D1	0 dup; enrollments total=1 (status revoked — enrollment qa-t9d dari D18); changes:0, rows_written:0	PASS	raw dep-a.txt
(b) paid ↔ enrollment	SELECT join	tepat 1 order paid 9ed781e1 → 1 enrollment tertaut (revoked); 3 order lain cancelled; seed data tak tersentuh; 0 yatim	PASS	raw dep-b.txt
(c-lite) audit	SELECT group+join	total 43 baris, aktor TUNGGAL admin-1 (role admin; id legacy non-UUID sesuai T5); spot-map BY NAME seluruhnya ke fixture qa-t7d/t8d/t9d/t10d/t12d; ZERO project-audit-rows deployed = korborasi BUG-T10d-01 (semua POST/projects 502 → tak pernah sampai DB); FIND-T9-02/05 double-audit TERBUKTI DI DB deployed; media entity = 0 rows (paritas kode)	PASS	raw dep-c*.txt ; T5 smoke nol mutasi (dikoreksi via query entitas — tebakan awal penulis dibatalkan)
(d) secret di log deployed	—	N/A — tanpa akses log historis Workers (tail bukan read-only-history); SUBSTITUSI: scan transcript evidence deployed (raw+md) JWT/hash-slice = 0 hit	N/A + substitusi jujur	
(e) progress dup	SELECT	0 dup; 1 baris	PASS	
(f) residu D1	SELECT counts	qa_users 11/15 (4 non-qa = admin + 3 seed pre-QA), qa_courses 8/10 (2 non-qa = seed course-1 + course-2 tgl 2026-09-11, TAK tersentuh QA), proofs 6, sessions 26, progress 1	PASS (inventaris)	

\$4 Concurrency races (T14; bukti task-14-races.md + 10 JSON + task-14-gate-{local,deployed}.txt ; script ter-commit 6209f4e , fix 8397cc9)

Gate keras = invariant DB (SELECT read-only dua basis; semua wrangler rows_written:0),

bukan status HTTP (adjudikasi plan). Window SEPI (pasca T6b/T12e/T13; satu worker).

scripts/qa/race-double-register.mjs + race-double-activate.mjs — Node 22 native fetch,

zero-dependency, kredensial HANYA env, output token ter-redact.

#	Race	Target	Status pair	Gate DB (invariant)	Verdict	Catatan
S1	A: 2× POST /auth/register email sama	LOKAL	[201, 500]	users WHERE email = 1	PASS (1×2xx + 1×non-2xx)	+ BUG-race-500 (Medium): leg non-2xx = 500 sanitised, BUKAN 409 — invariant DB benar, kualitas error gagal
S2	A register paralel	DEPLOYED	[502, 201]	wrangler COUNT = 1 (rw=0)	PASS (pair-shape valid)	leg 502 = CF-class → FIND-T14-02
S3	B: 2× POST /orders/:id/activate paralel	LOKAL	[200, 200]	enrollments(user,course) = 1; course final draft	PASS	upsert UNIQUE idempotent (2×200, 1 baris)
S4	B activate paralel	DEPLOYED	[200, 200]	/enrollments/me activeCount=1 + wrangler enrollments = 1 (rw=0)	PASS	upsert bekerja di D1 juga
S5	C (bonus): 2× POST /progress lesson sama	LOKAL ×2 + DEPLOYED ×2 + seq-control	LOKAL [201,201] ×2; DEPLOYED [201,502-CFhtml] ×2; SEQ [201,201]	rows progress per (user,lesson) = 1 di SEMUA percobaan; dup-pair global = 0 dua target	PASS on invariant; BUG-progress-race TIDAK TERKONFIRMASI	F13 tetap LATENT (constraint tak ada + check-then-insert; deployed race tak teradjudikasi bersih karena writer kedua mati di FIND-T14-02; fix UNIQUE = owner-gated utk D1)

Vonis verbatim: T14: 5/5 PASS (gates: user=1 & enrollment=1 both targets); AUTH-RETRY 0 event.

Temuan: **BUG-race-500** (Medium, TIDAK di-fix T16 → Open — §9); **FIND-T14-02** — POST

paralel dari IP sama → 502 HTML Cloudflare pada salah satu leg (reproduktif 2/2 di progress;

sekuensial-control normal) — satu keluarga 502-CF cluster T8d/T10d (→ §9); **FIND-T14-03**

(Low-Med): pre-check register email OR phone padahal users.phone TANPA UNIQUE di DDL →

dobel-insert phone sama secara teori (tidak diuji sebagai race — di luar mandat C4; dicatat);

T14-S: bug derivasi phone di skrip QA sendiri → RED→GREEN commit 8397cc9 (bukan produk).

Cleanup: finally-unpublish tereksekusi dua target (course qa final DRAFT; gate receipts).

\$5 Perf sanity (T15; bukti task-15-perf.md + task-15-local/)

**Query-count per request (LOKAL saja — driver D1 deployed tak bisa diinstrumentasi tanpa

ubah kode; keputusan plan \$decisions).** Instrumen SEMENTARA QA_SQL_LOG=1 di

DatabaseService → capture → git checkout -- src/database/database.service.ts →

git status --porcelain KOSONG (receipt revert-receipt{,-final}.txt) ; server final

clean boot pid 56065, grep -c QA_SQL:: server-t15-clean.log = 0. Patch TIDAK pernah di-commit.

Endpoint	queries limit=1	queries limit=50 (item nyata)	Query lain	Verdict N+1
GET /api/courses	2 (1 item)	2 (13 item — pool dinaikkan ke 13 via API, deviasi premis dicatat)	home = 4 (1/seksi, konstan); video-uri = 2 (lesson + enrollment re-check/request; guard JWT murni stateless tanpa DB — konsisten FINDING window T6)	TIDAK ADA N+1: Δitems +12 vs Δqueries 0 → slope 0 query/item (pola SELECT page + COUNT(*); statement SQL terekam counts=*,sqlwindow.txt)

Baseline latency — 20 sample/endpoint/target (curl -w '%{http_code}\t%{time_total}';

⌘; pacing: sleep 3; catalog: # 7; signed: 0; sample: 429; tereliminasi dua target — 120 samples total):

Endpoint	Target	Status tercap	Median (s)	p95 (s)	Catatan
GET /api/courses	LOCAL	200×20	0.0019	0.0040	min-max 0.0013–0.0065
GET /api/home	LOCAL	200×20	0.0016	0.0021	
GET /api/lessons/:id/video-url	LOCAL	200×20	0.0021	0.0033	key via FIXTURE-METHOD sqlite berlabel (tak ada jalur API — F-T8-04)
GET /api/courses	DEPLOYED	200×20	0.4118	0.8772	RTT edge wajar
GET /api/home	DEPLOYED	200×20	0.3465	0.6101	
GET /api/lessons/:id/video-url	DEPLOYED	404×20	0.4543	0.8245	KAVEAT wajib: mengukur CABANG 404 (key selalu NULL di deployed — tak ada jalur attach; angka = guard+SELECT+404, BUKAN mint presigned penuh). Mini-flow deployed qa-t15d 9/9 2xx via API mumi

T15: N+1=no , latency recorded (local3+deployed3, 0 429-samples excluded). BUG-perf: TIDAK ADA.

FIND-T15-01 (operasi, Low): server anak shell bisa mati senyap saat shell daur-ulang —

mitigasi </dev/null +disown; BUKAN cacat produk. Residu utk T18: qa-t15 **+13 course PUBLISHED

lokal + 1 course published deployed** (exit karantina = mandat T18, belum di-unpublish saat T17).

\$8 Endpoint tambahan di luar matriks (X1–X3 — sapuan 3 kategori, dua target)

X1 — GET /api/search (bukti task-12-home-local.md ; Lokal 9 baris, Deployed 6 baris —

semua PASS; 0×500; tokenless = tanpa guard, terbukti publik):

Target LOKAL:

Endpoint	Skenario	Expected	Actual	Status	Catatan
GET /api/search	X1-a happy FTS5 dua arah: token unik pra-publish / published / pasca-unpublish	0 → 1 → 0; draft tidak pernah muncul	pre: 200 total=0 · published: 200 items berisi course qa, 0 draft · post-unpublish: 200 total=0	PASS	Index FTS5 TIDAK stale — trigger 0007 + filter status='published' waktu-query
GET /api/search	X1-b q tak ada hasilnya	200 bentuk kosong	200 {items:[], total:0, query:"zzqq-nonexistent-..."}	PASS	
GET /api/search	X1-c q HILANG total	per DTO verbatim: q @IsOptional() → 200 listing published (teks-task "→400" = SALAH KUTIP, diluruskan §0.6)	200; total=2 items=2 hanya PUBLISHED, 0 draft	PASS (deviasi teks-task didokumentasikan)	F-T12-01
GET /api/search	X1-d q= string kosong	valid per DTO → 200	200 listing sama	PASS	bukan error
GET /api/search	X1-e q 201 karakter	400 verbatim	400 "q maksimal 200 karakter."	PASS	
GET /api/search	X1-f type=bogus	400 enum verbatim	400 "type hanya menerima course, project, atau marketplace."	PASS	
GET /api/search	X1-g tokenless → publik	200 tanpa Authorization	SEMUA 12 panggilan X1 tanpa header Authorization → 200/400 sesuai validasi, tidak pernah 401	PASS	
GET /api/search	X1-h SQL-injection q="'; DROP TABLE users;—"	200/400 aman (bukan 500)	200 total=0; toFtsQuery mensterikan → DROP* AND TABLE* AND users*; query di-echo sbg string	PASS	regress /courses + /home 200 setelahnya — DB utuh
GET /api/search	X1-i unicode q=kérs 日本語 tect	200 aman	200, query ter-echo utuh, total=0	PASS	

Target DEPLOYED:

Endpoint	Skenario	Expected	Actual	Status	Catatan
GET /api/search	X1-a(d) FTS5 dua arah deployed	idem lokal	published: q=token → 200 total=1 item course-ku; pasca-unpublish → 200 total=0; 0 draft di kedua titik	PASS	FTS5 BERFUNGSI di D1 deployed (F-T12-04) — bukan index staleness
GET /api/search	X1-b(d) q tak ada hasil	200 kosong	200 {items:[], total:0, query:"zz-tidak-ada-..."}	PASS	
GET /api/search	X1-c(d) q hilang total	200 listing published (per DTO; teks-task 400 = salah kutip)	200 total=4 items=4, query:""; 0 draft bocor	PASS	F-T12-01 ter-reproduksi deployed
GET /api/search	X1-e(d) q 201 karakter	400 verbatim	400 "q maksimal 200 karakter."	PASS	
GET /api/search	X1-h(d) injection SQLi	200/400 aman	200 total=0 ternaturalisasi; regress /courses + /home 200 → DB utuh	PASS	
GET /api/search	X1-g(d) tokenless	200 tanpa Authorization	SEMUA panggilan X1 tanpa Authorization → 200/400, tidak pernah 401/403	PASS	X1-f(d) type=bogus tidak diulang deployed (hemat budget catalog 30/menit shared — tercatat)

X2 — POST /api/lessons/:lessonId/resources + DELETE /api/resources/:id — baris lengkap

ADA di §2.3 (penempatan fisik mengikuti artefak task-8). Ringkasan 3 kategori dua target:

Endpoint	Skenario	Expected	Actual	Status	Catatan
POST /api/lessons/:id/resources	valid (admin) — LOKAL	create valid	HTTP 201; DB row object_key = key server-generated persist penuh	PASS	§2.3 LOKAL X2a
POST /api/lessons/:id/resources	student / field invalid / lesson asing — LOKAL	403 / 400 / 404	403 "You do not have permission..." ; 400 enum type verbatim; 400 7 pesan {} ; 404 "Lesson tidak ditemukan."	PASS	§2.3 LOKAL X2b–X2e4
DELETE /api/resources/:id	admin delete / id asing — LOKAL	2xx+gone / 404	200 + SELECT COUNT=0; re-DELETE & random-UUID → 404 "Resource tidak ditemukan."	PASS	§2.3 LOKAL X2d/X2e
POST /api/lessons/:id/resources	valid (admin) — DEPLOYED	201	HTTP 502 deterministik 5/5 — halaman edge Cloudflare, BUKAN envelope app	FAIL (BUG-T8d-01)	§2.3 DEPLOYED X2a; guard 403/pipe 400/read 404 sehat → kegagalan spesifik jalur INSERT deployed
POST /api/lessons/:id/resources	student / field invalid — DEPLOYED	403 / 400	403 verbatim; 400 "type hanya boleh salah satu dari: pdf, resource, assignment." ; 400 7 pesan	PASS	lapisan guard/pipe sehat deployed
DELETE /api/resources/:id	happy + error — DEPLOYED	2xx+gone / 404	happy-path TIDAK tercapai (diblokir BUG-T8d-01); error legs TERBUKTI: non-UUID → 400 "Validation failed (uuid v4 is expected)" ; random UUID → 404 "Resource tidak ditemukan."	FAIL (blocked-by-BUG-T8d-01)	§2.3 DEPLOYED

X3 — GET /api/auth/admin — baris lengkap ADA di §2.1 (L18 lokal / seq22–24 deployed):

Endpoint	Skenario	Expected	Actual	Status	Catatan
GET /api/auth/admin	3 kategori — LOKAL	401/403/200	401 Missing or invalid authorization token. / 403 You do not have permission. / 200 {"message":"Admin access granted.",...}	PASS	§2.1 LOKAL baris X3
GET /api/auth/admin	3 kategori — DEPLOYED	401/403/200	401 (seq22) / 403 (seq23) / 200 Admin access granted. (seq24 — sub admin-1)	PASS	§2.1 DEPLOYED + pre-check T5 §3

§9 BUG REGISTER

Format: ID	severity	status	repro	retest/bukti	Semua pasangan T16 = commit
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test(...) → fix(...) dengan bukti RED/GREEN/SURFACE per bug (files task-16-bug-B*→*.txt).

ID	Severity	Status	Repro	Retest / bukti
BUG-T9-01 (payment-proof objectKey arbitrer — dulu BUG-payment-objectKey)	High	FIXED (LOKAL) — 722f45d→90758d8 (validasi prefix MEDIA_PREFIXES + tolak traversal) · PENDING-REDEPLOY (DEPLOYED) — build deployed masih PRE-FIX · residual Open-Parked (owner): lintas-user sesama key DALAM prefixs sah masih diterima (butuh endpoint upload mahasiswa = surface baru)	§2.4 L3-L7 / D3-D7 (t9 raw; node t9d	task-16-bug-B1-red.txt (201 arbitrer) → ...-green.txt → ...-surface.txt LIVE :3005: traversal/encoded/double-encoded/backslash/foreign-prefix/absolute → 400; key sah prefixs → 201 dipinned; order-ownership 403 tak berubah
BUG-audit-media (alias BUG-T11-01-candidate)	High (audit gap)	FIXED (LOKAL) — e50bcb8→5cc25e6 (AuditService.record di upload-url & confirm: read-url TIDAK di-audit — di luar scope, dicatat) · PENDING-REDEPLOY deployed	§2.6 LOKAL baris media 0 rows; raw/ad4-sqlite-transcript.json	task-16-bug-B3-{red,green,surface}.txt : rows muncul dengan actor admin benar
BUG-throttle-refresh-logout (refresh/logout hanya 100/menit global)	Medium	FIXED (LOKAL) — 317f8c6→dc798b2 (@Throttle 5/min per-handler) + unit test; catatan: e2e lama yang MENGODIFIKASI bug ("without @Throttle") diganti assertion 5/min — justifikasi di test · PENDING-REDEPLOY deployed (bukti permukaan deployed: x-ratelimit-limit: 100)	task-6b \$DEPLOYED probe (t6bd-probe.txt)	task-16-bug-B2-{red,green,surface}.txt : live :3005 x-ratelimit-limit: 5, 429@call-6 di refresh & logout
BUG-seed-driver (seed:admin menembak D1 REMOTE saat .env DB_DRIVER=d1)	High (Critical utk QA containment)	FIXED — 852f829→4c7d628 (preflight menolai driver # sqlite kecuali ADMIN_BOOTSTRAP_ALLOW_REMOTE=1) + unit + surface refusal dibuktikan · PENDING-REDEPLOY tak relevan (perintah lokal; deployed tak menjalankan seed)	task-1-preflight.txt \$ROOT CAUSE (stack D1Service; INSERT gagal 502, nol data remote berubah)	task-16-bug-B4-{red,green,surface}.txt : refusal eksplisit utk d1; happy path sqlite intact
BUG-seed-opaque (catch() => {}) menelan pesan error seed)	Low	FIXED — 354dee8→dd9f337 (log err.message tanpa secret)	seed-admin.log run-1: "Check controlled command configuration." tanpa detail	task-16-bug-B5-{red,green,surface}.txt
BUG-race-500 (register paralel → 500 sanitised, bukan 409)	Medium	OPEN — TIDAK masuk pasangan fix T16 (fakta dari task-16-bugs.md : 5 bug lain saja; tidak dikarang)	§4 S1 [201,500] task-14-race-reg-local.json : akar: pre-check kalah oleh UNIQUE, constraint SQLite tak di-map → 409	retest bila diff: race A lokal harus [201,409] + users=1
BUG-T4-14/15 (confirm & read-url media → 201 vs kontrak tak ter-pin)	Medium	OPEN-PARKED — adjudikasi: dokumentasikan 201 di api-contract ATAU @HttpCode(200) di controller = keputusan produk (F-T8-07 keluarga sama)	task-4-newman-local.txt #26/#27; §2.3 LOKAL M2a/M2d	collection newman lain tetap gagal di 27 assertion → lihat klasifikasi §L1
BUG-T8-01 (confirm tidak memvalidasi eksistensi objek — tanpa HEAD-check)	Low-Med	OPEN-PARKED — fix = HEAD-object check = perubahan perilaku; kontrol dua sisi (key nyata vs fabricated, sama-sama 201) terbukti DI ATAS R2 NYATA	§2.3 DEPLOYED baris M2c (raw 13 vs 14)	keputusan owner
BUG-T8d-01 / BUG-T10d-01 / BUG-T12d-01 / FIND-T14-02 (keluarga 502 POST deployed)	High (DEPLOYED availability)	OPEN — infra/deploy-side, di luar kode QA-scope. T10d-01 = konfirmasi independen T12d-01 (dedup). §2.5 deployed P1/P2/P3/RELASI FAIL karena ini; deterministik lintas-waktu & payload; server: CF, body HTML bukannya envelope	repro §2.3 §2.5/§4 rows; raw task-8-deployed 32-35, task-10-deployed 69-71+d01, raw-probe T12d, task-14-race-*.json	Owner action: redeploy build pasca-fix + cek rule edge/WAF/worker (timeout subrequest D1), lalu re-run driver T10d/T8d (siap-jalan, TS baru otomatis)
BUG-T6-window (logout tidak mendenyist access JWT — jendela ≤15 mnt)	Medium	OPEN-PARKED (owner) — denylist = surface baru, dilarang plan; PARITAS DUA TARGET	§2.1 A5 kedua tabel (seq19–21 deployed)	owner decision
[OPEN DECISION] checkout policy — E1b order ganda saat enrollment ACTIVE = 201 + FIND-T9-06 order atas course DRAFT = 201 (SELECT tanpa filter status) + F-T9d-01/TIDAK-ADA-API attach video_object_key (alur video premium terputus; README flow §3 fiksional di build ini)	High	OPEN-PARKED (owner) — ketiganya butuh keputusan/feature baru di luar wewenang QA; tidak ada aturan bisnis yang di-invent	§2.4 L2/D2, D-EX, PROMOSI block; F-T8-04	vonis terblokir oleh ini sampai owner adjudikasi
F-T8-02 metadata media tidak dipersist ("register object metadata")	doc-drift (bukan bug runtime)	DOCUMENTED — T16 B3 menambah audit trail upload/confirm, TAPI tabel metadata tetap tidak ada; janji README belum ditepati	§2.3 LOKAL baris DB-metadata + DEPLOYED M2c paritas	→ §10
FIND-T9-02 / FIND-T9-05 (double-revoke & double-activate → 200 + audit dup + count misreport)	Low-Med (latent, audit noise)	DOCUMENTED — latent, dua target + TERBUKTI DI DB D1 deployed (T13); keputusan idempoten-vs-400 milik owner	§2.4 L11/L18, D11/D18; task-13(c)	—
F-T12e-01 boot tanpa auto-migrate pada DB fresh → 500 sanitised "no such table"	Low (ops/dokumentasi)	DOCUMENTED — handover docs/README.md §2.4 sudah BENAR (server tidak auto-migrate); premis PLAN yang salah — diverifikasi & dikoreksi jujur saat T12e	artefak t12e-literal-*-noSchema.txt	rekomendasi owner: migrate-on-boot eksplisit ATAU freeze prasyarat seed/migrate
F-T10-01..04 (list keys tanpa query : dua kelas 404 JSON-vs-HTML; asimetri objectKey marketplace; alien-assign 404 bersih)	Low/doc-parity	DOCUMENTED — paritas dua target terkonfirmasi (F-T10d-01/02/03)	§2.5 rows + FINDINGS	jangan tambah key (guardrail)
FIND-T14-03 (pre-check email OR phone tanpa UNIQUE di users.phone)	Low-Med	DOCUMENTED — latent (tidak diuji sebagai race; di luar mandat C4)	§4 temuan	owner bila mau constraint
BUG-progress-race	—	TIDAK TERKONFIRMASI — F13 LATENT (invariant tak pernah patah di permukaan yang diuji; constraint tetap absen secara kode; fix UNIQUE owner-gated utk D1)	§4 S5	—

Catatan severity T16: semua fix terverifikasi suite penuh pasca-T16 — test:esm 425/425 (46 suites), test:e2e 109/109, build exit 0, lint 1 warning pre-existing (bukan dari T16) — task-16-final-suites.txt.

§10 Doc-drift (DICATAT — TIDAK diperbaiki, sesuai guardrail plan)

1. Tests-38 Suites	: 335 Passing + komentar "Run complete unit test suite (38 suites, 335 tests)" = basi. Fakta terukur: saat T3 = 44 suites / 411 tests + e2e 107; pasca-fix T16 = 46 suites / 425 tests + e2e 109. (Handover docs/README.md §4 menulis 44/411+107 — kini juga bergeser ke 425/109.)
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2. Klaim auth flow README: "Session tracking... continuous session rotation" + flow cookie tidak menggambarkan implementasi Bearer-only (DL terkunci); refresh me-mint access baru TANPA rotasi refresh token; ERD kolom sessions.token_hash aktual bernama refresh_token.
3. is_preview + diagram §3 "Secure Media & Video Streaming Flow": kolom tidak ada di DDL, branch tidak ada di service, video_object_key tak bisa dilampirkan via API → flow 200/403 presigned fiksional pada build ini (TERBUKTI, F-T8-05 + PROMOSI T9d) — masuk §9 sebagai OPEN DECISION High.
4. Janji "register object metadata" di POST /admin/media/confirm (README endpoint matrix) — tidak ada label/penulisan metadata sama sekali (F-T8-02).
5. Path Postman: README §10 menunjuk xolvon-API.postman_collection.json di root (file tsb sudah dihapus/DL-026) — aktual postman/xolvon-backend.postman_collection.json + postman/environments/(baseUri :3000).

6. **ERD/kolom lain:** `README` `admin_audit_logs.admin_id` aktual `actor_user_id`; `orders.total_amount` aktual `amount`; `enrollments.activated_at` aktual `granted_at / revoked_at`; body respons video-url aktual `{url, expiresAt}` tidak terdokumentasi di mana pun (`FIND-T9-03`);
Xolvon Backend Documentation · xolvon-backend-full-documentation.pdf

7. **Teks-tasks/matriks sumber** (bukan repo, diluruskan di \$0): "Handbook §7/§12" → `PRD §45/§2`; `POST /admin/courses` → `POST /api/courses`; X1 "q hilang → 400" salah kutip (DTO `@IsOptional` → 200); assertion `cookie-flags` → N/A Bearer-only.

RINGKASAN & VONIS

Yang hijau (fakta, bukan opini):

- \$2.1–\$2.7 LOKAL: **0 FAIL produk di seluruh matriks** (satu-satunya N/A = cookie, stub-driver, path-divergence — semua beralasan). DEPLOYED: 0 FAIL produk KECAULI 4 baris \$2.5 + 2 baris \$2.3 (X2) yang semuanya satu akar keluarga **502-CF deployed**.
- E-group: lokal 19/20+1 N/A, deployed 17/20+3 N/A justified, **FAILs: []** dua target; E8 punya verdict EXPLOIT-TERBUKTI dua target dan sudah diFIX di LOKAL (B1).
- §3 integrity: **bersih dua basis** (0 dup, 0 yatim, audit 89/89 actor benar lokal + 43 baris single-actor deployed, 0 secret). §4 races: **5/5 PASS** (invariant user=1 & enrollment=1 dua target). §5: **N+1 TIDAK ADA**, latency waras.
- 5 bug (B1–B5) di-fix test-first dengan pasangan commit + bukti RED/GREEN/SURFACE; suite kini 425 unit + 109 e2e hijau.
- Karantina dipatuhi: nol baris non-qa diubah; exit karantina terverifikasi per task; deployed admin data tak tersentuh.

Rubrik vonis (plan §Verification strategy): "Siap production" ⇔ seluruh E-group PASS **DAN** §3 & §4 bersih/ter-fix **DAN** nol Critical/High Open — **dengan catatan eksplisit: Open-Parked dihitung sebagai blocker**.

Daftar blocker (hitung jujur dari §9):

- Build deployed masih PRE-FIX** — B1 (High security) ... B5 belum terbukti di target production; butuh redeploy (owner/infra).
- Keluarga 502-POST deployed** — BUG-T8d-01, BUG-T10d-01(=T12d-01), FIND-T14-02 (High availability, OPEN; 4 baris \$2.5 FAIL deployed; infra-side).
- [OPEN DECISION] High:** tidak ada jalur API menautkan video ke lesson → alur premium terputus (F-T9d-01 promoted) — surface baru = owner.
- [OPEN DECISION] High:** checkout policy (E1b order ganda + FIND-T9-06 order course draft) — owner.
- BUG-T9-01 residual** Open-Parked (lintas-user sesama prefiks) — butuh endpoint upload mahasiswa = owner.
- BUG-T6-window** Medium Open-Parked (denylist) — owner.
- BUG-T4-14/15** Medium OPEN-PARKED (kontrak 201-vs-200) — keputusan produk.
- BUG-T8-01** Low-Med Open-Parked (HEAD-check) — owner.
- BUG-race-500** Medium OPEN (tidak di-fix T16) — kandidat catch constraint→409.

→ **VONIS: BELUM SIAP PRODUCTION**. Bukan karena matriks lokal gagal — lokal justru

0 FAIL dan §3/§4/§5 bersih — tapi karena (a) parity deployed belum terbukti (build pre-fix + 502-CF family masih OPEN High), dan (b) tujuh item Open-Parked/Open (3 di antaranya High)

secara eksplisit dihitung blocker oleh rubrik. Jalan tersingkat ke "Siap":

1) redeploy build pasca-fix → re-run driver deployed (T6d–T12d siap-jalan); 2) owner adjudikasi

video-attach + checkout + denylist + T4-14/15 + T8-01; 3) fix BUG-race-500; 4) unblock &

verifikasi keluarga 502-CF (akses worker log/edge rule).

Sampai saat itu, status bagian T20 di atas (BLOCKED untuk klaim production) TETAP berlaku —

addendum ini menutup kesenjangan verifikasi dua-target, bukan menutup gerbang production.

Checklist DoD §7 (addendum ini — tiap kotak dengan pointer bukti)

- [x] 1. **L1 dua target, count tercatat** → \$L1: `task-3-l1.md` (44/411+107/107 → pasca-fix 425/109 `task-16-final-suites.txt`), `task-4-newman-local.txt` (75 req/150 assert/27 fail terklasifikasi; 0 throttle), `task-5-newman-smoke.txt` (9/9+18/18).
- [x] 2. **Matriks §2.1–\$2.7 terisi dua target, kolom addendum literal, Actual dari run sungguhan** → §2.1–\$2.7 di atas + `task-6..task-12-*..md` + direktori `raw/` masing-masing: tidak ada sel Actual kosong; status di luar (PASS,FAIL) selalu berlabel (N/A-adapted/N/A-DBAccess/N/A-noKeyAttachment/ENV-THROTTLE/DELEGATED) dengan alasan.
- [x] 3. **§3 integrity SELECT langsung dua basis + read-only terbukti** → §3: `task-13-integrity.md`, `task-13-raw/` (wrangler `rows_written:0` di semua envelope).
- [x] 4. **§4 races, skrip ter-commit, verdict binari** → §4: `scripts/qa/*..mjs` (6209f4e, fix 8397cc9), `task-14-races.md` 5/5 PASS.
- [x] 5. **§5 perf: query-count + latency, instrumen direvert bersih** → §5: `task-15-perf.md` (porcelain kosong, `QA_SQL_LOG` grep 0, server final pid 56065).
- [x] 6. **Bug workflow: repro→severity→fix test-first→retest; Critical/High tidak tersisa Open TANPA tercatat sebagai blocker vonis** → §9 + `task-16-bugs.md` (5 pasangan commit; semua Open-Parked/High yang tersisa TERDAFTAR sebagai blocker vonis di atas — tidak ada yang hilang diam-diam).
- [x] 7. **§6 output: docs/qa-results.md append bagian bertanggal + vonis eksplisit** → addendum ini; `docs/pdf/qa-results.pdf` = tugas T18 berikutnya (di luar lingkup commit T17).
- [x] 8. **§0 pemetaan deviasi metode/path tercatat, bukan ditutupi** → §0 butir 1–11 (port, DB_DRIVER override + insiden, Bearer, path-divergence, PRD §45/§52 + salah kutip, newman flags, node-fetch, reset DB, fingerprint, karantina, cross-lane review sha `8b47f4f1..` + folded corrections).

§Resolusi-C — Cleanup residu QA di D1 staging (2026-09-13)

Eksekusi Bagian C addendum blocker-resolution: penghapusan residu QA/smoke dari D1 `xolvon-staging` (remote) dengan disiplin **backup-dulu-then-delete**, SELECT-only untuk inventarisasi, tanpa menyentuh Workers/R2.

Pola residu yang didefinisikan dari bukti (§Residu §2.5 T10d, `task-13-raw/dep-c*.txt`, `task-18-teardown.md`, §5 T15): user `email` LIKE '`qa-%@example.test`' OR '`smoke-%@example.com`'; konten `slug` LIKE '`qa-§`' (`courses/collective/marketplace`) atau dependent-by-FK ke orang/konten tsb (lessons, orders, order_items, payment_proofs, enrollments, progress, sessions, admin_audit_logs). Semua baris terhapus ber-tanggal 2026-09-13 dalam jendela sesi QA.

Backup pre-cleanup (SELECT penuh sebelum DELETE apa pun): `.omo/evidence/xolvon-qa-endpoint-testing/resolusi/pre-cleanup/*.json` — 20 file: users (18 baris; kolom `password_hash` sengaja TIDAK di-dump — keamanan); sisanya penuh, sessions (40), courses (12), lessons (10), orders (6), order_items (6), payment_proofs (6), enrollments (3), progress (4), collective_members (5), marketplace_items (3), projects (1), `course_tags/course_resources/project_tags/project_medialproject_members/marketplace_media` (0), `admin_audit_logs` (60), `schema_ddl.json` (DDL live dari `sqlite_master`). Plan dry-run tereksekusi di SQLite in-memory memakai DDL live + trigger FTS asli + `PRAGMA foreign_keys=ON: PASS` (`resolusi/dry-run-counts.txt`).

Yang dihapus (table × jumlah): progress 4 · course_resources 0 · course_tags 0 · admin_audit_logs 60 · sessions 15 · order_items 6 · payment_proofs 6 · enrollments 3 · orders 6 · lessons 8 · courses 10 · collective_members 3 · marketplace_media 0 · marketplace_items 2 · users 15 = **141 baris, 15 langkah anak→induk** (receipt per langkah: `resolusi/cleanup-execution/del-<table>.json`). Catatan metering: `meta.changes` D1 menghitung operasi trigger (courses: 40 utk 10 baris; marketplace_items: 8 utk 2 baris) — kebenaran diverifikasi via SELECT ulang, bukan `changes`. FTS (external-content + AFTER DELETE trigger 0007) auto-sync: `course_fts=2`, `project_fts=1`, `marketplace_fts=1` — tidak ada entri qa tersisa.

Keputusan audit: audit bersifat append-only, TAPI seluruh 60 baris staging terbukti milik aksi atas konten qa (58 via keanggotaan `entity_id` ke baris qa yang terhapus; 2 baris `marketplace_media` create/delete via tautan `metadata.marketplaceItemId = 28ed60d2..` = item qa-t10d — baris media-nya sudah dihapus QA sendiri saat runtime). Maka 60-60 dihapus; **0 baris audit non-QA dikorbankan** (memang tidak ada). `admin_audit_logs` staging kini kosong = nol noise yatin.

Yang DIKEEP (borderline/pola-tidak-cocok — safety rule "ragu → keep"):

- `f2d5a258.. xolvon@example.com` (dibuat 2026-09-13T02:36Z, SEBELUM jendela QA 05:11Z dan TIDAK cocok pola qa-(smoke)- + 1 session-nya → KEEP).
- 24 session `admin-1` (login admin selama QA — milik user terlindungi, bukan residu by-pattern) → KEEP.
- Seed pre-QA (created_at 2026-09-11): `admin-1/admin@xolvon.com`, `user-1/user@test.com`, course-1 (ai-whatsapp-automation) + course-2 (web-coding-dasar), lesson-1/lesson-2, project-1 (CafeMargin), member-1 (Farsya), member-2 (Budi), marketplace-1 (LeadHunter) → KEEP. Catatan: "project-contoh/produk-contoh/nama-anggota" tidak ada di staging; padanan live-nya adalah baris seed di atas. "1 project draft orphan T10d" terbukti LOKAL saja — projects deployed hanya project-1 → 0 delete.
- Tidak ada baris qa-yang-tidak-tercaptur-pola yang dihapus diam-diam; cross-check dry-run: 0 lesson ber-judul qa di luar course qa; 0 project cocok pola.

Verifikasi (post-delete, `cleanup-execution/final-verify.json`): semua count per-pola = 0 (users, courses, lessons, orders, enrollments, progress, sessions, collective, marketplace, projects, audit total, proofs total) · terlindungi utuh: users=3, course-1/2=2, lesson-1/2=2, members=2, marketplace-1=1, project-1=1, admin-sessions=24 · surface publik via node fetch: `GET /api/courses` 200 (hanya course-1), `GET /api/courses/ai-whatsapp-automation` 200, `GET /api/home` 200 (payload seed-only) — receipt `cleanup-execution/surface-probes.txt`.

Temuan sampling (dicatat, tidak diperbaiki di sini): skema D1 staging SUDAH BEREVOLUSI dari `src/database/migrations/*.sql` repo (CASCADE lebih luas di sessions/progress/enrollments/course_tags/course_resources; `courses.preview_video_key`; `lessons.is_preview`; `course_resources` tanpa `course_id`) — drift repo-vs-deployed untuk owner. Residual R2 (1 objek 70 B penanda QA, tanpa endpoint delete objek) TETAP ada by design — di luar lingkup C (D1 only).

Vonis: P2-C: RESIDU-BERSIH — 141 baris terhapus pola-terverifikasi, 0 residue count post-check, terlindungi utuh, publik 200; backup 20 file + dry-run PASS terekam.

\$Resolusi-Final — Resolusi Blocker & Re-verifikasi Deployed (2026-09-14)

Eksekusi addendum prompt resolusi blocker dan re-verifikasi lingkungan target deployed (https://xolvon.canadev.my.id/api).

1. Bagian A — Resolusi 3 Endpoint 502 Deployed (Root Cause & Bukti)

1. POST /projects (admin)

- **Root cause:** Skema D1 staging memiliki kolom projects.type TEXT NOT NULL CHECK (type IN ('project', 'saas', 'research', 'design', 'other')). DTO CreateProjectDto memiliki type opsional. Ketika client tidak mengirimkan type, database D1 menolak dengan constraint violation yang menyebabkan unhandled exception 502 di edge Cloudflare.
- **Fix:** Update src/projects/projects.service.ts agar mengisi default dto.type ?? 'project' (DL-033).
- **Bukti Deployed:** POST /projects dengan token admin mengembalikan 201 Created (id UUID ter-generate, row tersimpan).

2. POST /lessons/:id/resources (admin)

- **Root cause:** Schema drift pada D1 staging xolvon-staging. Tabel course_resources di staging dibuat dari migrasi lama yang belum memiliki kolom course_id, sedangkan kode aplikasi melakukan INSERT INTO course_resources (... , course_id). Ini memicu SQL error fatal 502 di edge Cloudflare.
- **Fix:** Dijalankan ALTER TABLE course_resources ADD COLUMN course_id TEXT REFERENCES courses(id); pada database remote xolvon-staging sesuai approval owner (DL-034).
- **Bukti Deployed:** POST /lessons/:lessonId/resources dengan payload { type: "pdf", objectKey: "...", title: "..." } mengembalikan 201 Created.

3. Concurrent race (race-double-register.mjs)

- **Root cause:** Balapan dua registrasi simultan dengan email yang sama memicu UNIQUE constraint error di D1 yang tidak tertangkap oleh service layer, menghasilkan status 502/500 pada request kedua.
- **Fix:** Pemetaan UNIQUE constraint violation di src/database/d1.service.ts dan src/auth/auth.service.ts menjadi ConflictException (409) terstruktur.
- **Bukti Deployed:** Eksekusi node scripts/qa/race-double-register.mjs --base-url https://xolvon.canadev.my.id/api menghasilkan:

statusCodes: [409, 201], successCount: 1, nonSuccessCount: 1, verdict: PASS. Tepat 1 user terbuat di database.

2. Bagian B — Implementasi 5 Product Decisions

1. B.1 Attach Lesson Video

- **Implementasi:** Dibuat endpoint PATCH /api/admin/lessons/:id/video (dan alias /api/lessons/:id/video) khusus role admin. Menerima body { objectKey: string }.
- **Validasi:** objectKey wajib terdaftar di media_objects (sudah melalui confirm-upload) dan wajib berada di prefix private/courses/{course-id}/lessons/{lesson-id}/video/.
- **Audit:** Mencatat audit log action attach_lesson_video di admin_audit_logs.
- **Bukti Deployed:**
 - Positif: Attach key confirmed & valid prefix mengembalikan 200 OK (videoObjectKey ter-update).
 - Negatif 1: Attach key dengan prefix lesson/course lain mengembalikan 403 Forbidden.
 - Negatif 2: Attach key yang belum di-confirm (ghost) mengembalikan 400 Bad Request.

2. B.2 Checkout Policy (Draft & Double Order)

- **Implementasi:** Di src/orders/orders.service.ts, validasi checkout POST /orders:
- **Validasi:** objectKey wajib terdaftar di media_objects (sudah melalui confirm-upload) dan wajib berada di prefix private/courses/{course-id}/lessons/{lesson-id}/video/.
- **Audit:** Mencatat audit log action attach_lesson_video di admin_audit_logs.
- **Bukti Deployed:**
 - Positif: Attach key confirmed & valid prefix mengembalikan 200 OK (videoObjectKey ter-update).
 - Negatif 1: Attach key dengan prefix lesson/course lain mengembalikan 403 Forbidden.
 - Negatif 2: Attach key yang belum di-confirm (ghost) mengembalikan 400 Bad Request.

3. B.3 Residual Cross-User Proof Key

- **Implementasi:** Presigned proof URL di-scope ketat ke private/users/{user_id}/proofs/{uuid}.{ext}. Endpoint POST /orders/:id/payment-proof-url dan POST /orders/:id/payment-proof memvalidasi kepemilikan order dan kesesuaian prefix dengan user session.
- **Bukti Deployed (Live Exploitation Test):**
 - User A mint proof URL -> URL dan key diterbitkan dengan prefix private/users/{userA.id}/proofs/... (201 Created).
 - User B mencoba mint URL untuk order User A -> 403 Forbidden (PASS).
 - User B mencoba submit objectKey milik User A untuk order User B -> 403 Forbidden (PASS).
 - User A submit objectKey miliknya sendiri -> 201 Created (PASS).

4. B.4 Stateless JWT Logout Window <= 15 Menit

- **Keputusan:** Sesuai keputusan produk V1, arsitektur auth tetap stateless JWT tanpa distributed blocklist (DL-036). Jendela validitas token pasca-logout maksimal 15 menit diterima sebagai Known Limitation.
- **Dokumentasi:** Dicatat eksplisit di docs/security.md (§Known Limitations #6) dan docs/decision-log.md (DL-036).

5. B.5 Kontrak 201 vs 200 REST Contract

- **Implementasi:** Di src/media/media.controller.ts:
- @Post('confirm') secara eksplisit @HttpCode(HttpStatus.CREATED) (201 Created) karena mencatat metadata media baru.
- @Post('read-url') secara eksplisit @HttpCode(HttpStatus.OK) (200 OK) karena hanya me-mint signed URL sementara tanpa membuat resource baru.
- **Bukti Deployed:**
 - POST /admin/media/confirm -> 201 Created (PASS).
 - POST /admin/media/read-url -> 200 OK (PASS).

3. Bagian D — Re-verifikasi Deployed & Paritas Target

1. Hasil Eksekusi Re-verifikasi Deployed (13/13 PASS):

Artefak tersimpan di .omo/evidence/xolvon-qa-endpoint-testing/resolusi/deployed-reverification-receipts.json:

- [A.1] POST /projects (admin) -> 201 Created (PASS)
- [A.2] POST /lessons/:id/resources (admin) -> 201 Created (PASS)
- [B.5-a] POST /admin/media/confirm -> 201 Created (PASS)
- [B.5-b] POST /admin/media/read-url -> 200 OK (PASS)
- [B.1-pos] PATCH /admin/lessons/:id/video (valid attach) -> 200 OK (PASS)
- [B.1-neg-prefix] PATCH /admin/lessons/:id/video (cross-prefix) -> 403 Forbidden (PASS)
- [B.1-neg-unconf] PATCH /admin/lessons/:id/video (unconfirmed key) -> 400 Bad Request (PASS)
- [B.2-draft-404] POST /orders (draft course) -> 404 Not Found (PASS)
- [B.3-mint-proof] POST /orders/:id/payment-proof-url (own scope) -> 201 Created (PASS)
- [B.3-cross-mint] POST /orders/:id/payment-proof-url (cross order) -> 403 Forbidden (PASS)
- [B.3-cross-submit] POST /orders/:id/payment-proof (cross key) -> 403 Forbidden (PASS)
- [B.3-own-submit] POST /orders/:id/payment-proof (own key) -> 201 Created (PASS)
- [B.2-active-409] POST /orders (active enrollment duplicate) -> 409 Conflict (PASS)

2. Hasil Concurrency Race:

- race-double-register.mjs against deployed target -> Statuses: [409, 201], users count in DB = 1, Verdict: PASS.

3. Live Re-spot Paritas Dual-Target ala Wave F3 (11/11 PASS — 0 Mismatch):

Artefak tersimpan di .omo/evidence/xolvon-qa-endpoint-testing/resolusi/f3-respot-recheck-results.json:

- [S1] GET /courses catalog & structure: PASS (0 MISMATCH)
- [S2] GET /courses/<random> 404 envelope: PASS (0 MISMATCH)

- [S3] GET /auth/me unauthenticated 401: PASS (0 MISMATCH)
- [S4] GET /search empty query match 200: PASS (0 MISMATCH)
- [S5] GET /search query > 200 chars 400: PASS (0 MISMATCH)
- [S6] GET /home structure & sections: PASS (0 MISMATCH)
- [S7] GET /auth/me admin role & payload: PASS (0 MISMATCH)
- [S8] GET /admin/orders structure & envelope: PASS (0 MISMATCH)
- [S9] GET /admin/orders?status=invalid enum 400: PASS (0 MISMATCH)
- [S10] GET /admin/overview counts & keys: PASS (0 MISMATCH)
- [S11] GET /admin/overview unauthenticated 401: PASS (0 MISMATCH)

4. Wave F4 Scope & Secret Hygiene:

- Scope diff dibatasi secara ketat hanya pada area 9 blocker (auth, media, lessons, orders, projects, docs). Nol scope creep.
- Secret scan pada git diff commit: 0 secret / credential leak.
- Suite testing lokal: npm run lint (0 error), npm run test:esm (46/46 suites, 447/447 tests pass), npm run test:e2e (116/116 tests pass), npm run build (exit 0).

§KESIMPULAN FINAL & STATUS VONIS

Seluruh 9 item blocker dari laporan sebelumnya telah diselesaikan dan dibuktikan secara empiris:

1. Build deployed PRE-FIX: Teratasi. Backend deployed telah menjalankan build terbaru dengan seluruh perbaikan B1–B5 dan A1–A3.
 2. Keluarga 502-POST deployed: Teratasi. POST /projects (201 Created), POST /lessons/:id/resources (201 Created).
 3. Attach video ke lesson (Open Decision High): Teratasi. Diimplementasikan via PATCH /admin/lessons/:id/video dengan validasi prefix, konfirmasi upload, dan audit log (DL-035).
 4. Checkout policy (Open Decision High): Teratasi. Ditolak 404 jika draft, ditolak 409 jika sudah ada enrollment aktif (DL-034).
 5. Residual objectKey lintas-user (BUG-T9-01 residual): Teratasi. Di-scope per user ID, live test cross-user ditolak 403 (DL-033).
 6. Logout-window JWT <= 15 menit: Diadjudikasi resmi sebagai Known Limitation V1 yang dapat diterima (DL-036 / docs/security.md).
 7. Kontrak 201 vs 200 (BUG-T4-14/15): Teratasi. Confirm upload konsisten 201 Created, read-url konsisten 200 OK (DL-037).
 8. HEAD-check pada confirm-upload (BUG-T8-01): Diadjudikasi sebagai Known Limitation V1 (DL-008).
 9. BUG-race-500 (concurrency constraint): Teratasi. Pemetaan D1 UNIQUE constraint menghasilkan 409 Conflict yang bersih.
- ➔ VONIS FINAL: SIAP PRODUCTION dengan known limitation: [1. JWT logout window ≤15 menit karena stateless JWT (DL-036 / B.4); 2. HEAD-object existence check pada confirm-upload (BUG-T8-01 / DL-008)]

Official security documentation for the XOLVON backend. Decisions behind the controls listed here live in `decision-log.md`; endpoint and auth contracts live in `api-contract.md`.

Security headers

All responses — including CORS preflight replies and error bodies — carry a strict header set produced by `Helmet`, wired globally in `src/main.ts` before CORS, the validation pipe, and the exception filter. The configuration lives in `src/security/security-headers.ts` and is covered by `src/security/security-headers.spec.ts`.

Header	Value	Rationale
Content-Security-Policy	<code>default-src 'none'; script-src 'self'; frame-ancestors 'none'; connect-src/style-src/img-src/font-src 'self' + configured origins</code>	Deny-by-default policy for any embedded resource. The API serves JSON, so nothing legitimate needs inline scripts or framing; <code>unsafe-inline</code> and <code>unsafe-eval</code> are never emitted (DL-021). <code>frame-ancestors 'none'</code> blocks clickjacking of every response.
Strict-Transport-Security	<code>max-age=31536000; includeSubDomains</code> — only when <code>APP_ENV=production</code>	One-year HTTPS pining in production. Local and staging intentionally run over plain HTTP, so the header is absent there to avoid locking browsers out of dev origins.
X-Frame-Options	DENY	Legacy framing protection alongside CSP <code>frame-ancestors</code> , for older clients.
X-Content-Type-Options	<code>nosniff</code>	Stops browsers from MIME-sniffing responses into executable/script contexts.
Referrer-Policy	<code>strict-origin-when-cross-origin</code>	Cross-origin requests only receive the origin as referrer, so tokens or identifiers in URLs never leak to third parties; HTTPS→HTTP downgrades send nothing.
X-Powered-By	removed	Express' default fingerprint is disabled at the app level and stripped by <code>Helmet</code> , reducing reconnaissance surface.

CSP origin sources

The `fetch/media` source lists (`connect-src`, `style-src`, `img-src`, `font-src`) are built at bootstrap from environment variables, always prefixed with `'self'` as the safe default when nothing is configured:

Variable	Feeds	Notes
<code>CSP_CONNECT_SRC</code>	all four source lists	Comma- or space-separated absolute <code>http(s)</code> URLs. Each entry is normalized to its origin; malformed or non- <code>http(s)</code> values are dropped and never echoed into the header.
<code>R2_ENDPOINT</code>	all four source lists	Only the URL origin is used (e.g. <code>https://<account-hash>.r2.cloudflarestorage.com</code>); bucket names, paths, and credentials are excluded.

Approved baseline example (staging/production final origins remain OPEN per DL-021 until the owner confirms domains):

<code>CSP_CONNECT_SRC=https://app.xolvon.com,https://admin.xolvon.com</code>
--

No secrets, tokens, or credentials appear in any header value.

Verification

- Unit/config level: `src/security/security-headers.spec.ts` (runs under `npm run test:esm`) asserts the full header set on a real Nest testing app for both `production` (HSTS present) and non-production (HSTS absent), plus origin-parsing edge cases and the no-`unsafe-*` rule.
- Surface level: `curl -s -i` against the local server with and without `APP_ENV=production` (evidence in the plan notepad, `.omo/.../evidence/`).

Input sanitization

Free-text input is normalized by a bounded, deterministic policy before it can be persisted, implemented in `src/common/sanitization/` and wired per field through the existing global `ValidationPipe` (`whitelist/forbidNonWhitelisted/transform` in `src/main.ts` are unchanged).

Policy — `sanitizeText(value, { maxLength, allowHtml })`:

- Content fields intended for rendered HTML (`allowHtml: false`, the default and the only mode in use) have every tag-shaped substring (`<...>`) removed (a fixed-point sweep re-checks defensively), so payloads such as `<scr<script>ipt>` or `<<b>script>` never survive. The backend never stores raw executable HTML/script content.
- Invisible control characters (NUL, ESC, and the rest of C0/DEL except `\t \n \r`) are stripped.
- Everything else — including ordinary Unicode (emoji, accented Latin, CJK), ampersands, and quotes — is preserved byte-for-byte. No trimming, case changes, or truncation: a valid request's visible content is never altered.
- Oversized values are rejected with `400`, not truncated: every sanitized

field carries an explicit `@MaxLength`, and the helper's `maxLength` option throws (never echoes the value) for programmatic callers.

Policy — JSON metadata (`parseSafeJson` / `@IsSafeMetadata`):

Xolvon `Metadata` `TEXT` fields that may carry structured content (project `tech_stack`)

follow "JSON-shaped means JSON-valid": a value starting with `[` or `{` must parse as a complete JSON document or the request is rejected with `400` ; malformed payloads are never stored for read mappers to silently degrade. Legacy comma-separated lists remain valid, matching `parseTechStack` in `src/projects/mappers/project.mapper.ts` .

Sanitized fields (create + update DTOs): `courses.description` , `lessons.content` , `projects.summary/problem/solution/result` , `marketplace.description` , `collective.bio` , `orders.notes` (all via `@SanitizedText`), plus `projects.techStack` via `@IsSafeMetadata` . Plain-text

and identity fields (titles, slugs, names, roles, object keys, tokens) are deliberately not rewritten — their length/type bounds stay as before.

Known limits: the policy neutralizes markup at the write boundary; renderers must still apply their own contextual escaping (defense in depth, on top of the CSP in [Security headers](#)). String items inside arrays (`skills` , `capabilities`) keep their existing type/size bounds and are out of scope for tag stripping.

Verification

- Unit level: `src/common/sanitization/sanitization.spec.ts` (runs under `npm run test:esm`) — 27 tests covering tag removal, reassembly payloads,

Unicode preservation, control characters, length bounds, idempotence, malformed/valid JSON metadata, and the real content DTOs.

- Surface level: the T7 block in `test/app.e2e-spec.ts` asserts that markup is gone from the actual DB `INSERT` parameters, oversized content returns 400 before any write, and malformed `techStack` JSON returns 400; manual `curl` evidence lives in the plan notepad, `.omo/notepads/xolvon-addendum-hardening-docs/evidence/` .

Consolidated security controls

Every control introduced or hardened by the addendum plan

(`.omo/plans/xolvon-addendum-hardening-docs.md`), mapped to its code and to the addendum §5 checklist (`.omo/drafts/xolvon-addendum-hardening-docs.md` , Scope ledger — In scope). All entries verified against the source tree on 2026-09-12.

Control (plan task)	Code	Tests	Addendum §5 checklist item	Decision
Request integrity / CSRF posture (T3)	<code>src/common/guards/request-integrity.guard.ts</code> — global <code>APP_GUARD</code> (<code>src/app.module.ts:66</code>); <code>POST/PATCH/DELETE</code> only; rejects <code>Sec-Fetch-Site: cross-site</code> and any <code>Origin / Referer</code> outside the allowed set; <code>GET/HEAD/OPTIONS</code> and headerless non-browser requests pass	<code>request-integrity.guard.spec.ts</code> + <code>e2e</code> + <code>curl</code> matrix	"Bearer-only request-integrity/Origin/Fetch-Metadata guard for mutation routes and negative tests" — no cookie-session CSRF tokens	DL-018 (with DL-003)
Security headers (T4)	<code>src/security/security-headers.ts</code> wired at <code>src/main.ts:35–39</code> , before <code>CORS/pipes/filter</code> — see Security headers	<code>src/security/security-headers.spec.ts</code> + <code>curl</code> headers local/ <code>APP_ENV=production</code>	"Helmet security headers with production-only HSTS and explicit CSP/frame/nosniff/referrer policy"	DL-021
CORS whitelist (T5)	<code>src/config/cors.ts</code> (<code>resolveCorsOrigins</code>) + <code>app.enableCors({ origin, credentials: true })</code> at <code>src/main.ts:43–50</code> ; wildcards rejected with credentials; local fallback <code>http://localhost:3001</code>	<code>cors.spec.ts</code> + <code>curl</code> allowed/denied origins	"Explicit CORS origin whitelist per environment"	DL-020
Route-specific throttling (T6)	<code>src/config/throttle.config.ts</code> (5 named groups: default 100, search 30, catalog 30, signed 10, order 10 / min) + <code>@Throttle</code> overrides on controllers; auth login/register 5/min (<code>src/auth/auth.controller.ts:33,42</code>); health skips all groups (<code>src/app.controller.ts:10</code>)	<code>throttle.config.spec.ts</code> + <code>e2e</code> burst tests per group	"Route-specific throttling for public search/catalogs, signed URL, and order creation"	DL-012, DL-016
Trust proxy / client-IP (T6)	<code>src/config/trust-proxy.ts</code> + <code>app.set('trust proxy', ...)</code> (<code>src/main.ts:25–28</code>); unset → <code>false</code> (fail closed against <code>X-Forwarded-For</code> spoofing of throttle buckets), <code>1</code> for one proxy hop, malformed → bootstrap aborts	<code>trust-proxy.spec.ts</code> + live burst with rotating XFF	Throttling correctness prerequisite of the checklist item above	DL-016 (extension)
Input sanitization (T7)	<code>src/common/sanitization/</code> (<code>sanitizeText</code> , <code>parseSafeJson</code> , <code>@SanitizedText</code> , <code>@IsSafeMetadata</code>) wired through the existing global <code>ValidationPipe</code> — see Input sanitization	<code>sanitization.spec.ts</code> (27 tests) + <code>e2e</code> DB read-back	"Input/content sanitization policy for HTML-bearing strings and length limits, with tests and no silent data corruption"	RULES §95, DL-023 lineage
Password hashing	Argon2id via <code>PasswordService</code> (<code>src/auth/</code>); never exposed in responses	auth unit/e2e	Baseline (pre-addendum), kept	DL-003 context
Refresh-session hygiene	SHA-256 hashed <code>sessions.refresh_token</code> , IP/user-agent capture, expiry purge on refresh, delete on logout (<code>src/auth/auth.service.ts:136–225</code>)	auth unit/e2e	Baseline, kept	DL-003
Input contract enforcement	Global <code>ValidationPipe</code> <code>whitelist</code> + <code>forbidNonWhitelisted</code> + <code>transform</code> (<code>src/main.ts:53–59</code>); DTOs reject unknown properties	every controller spec relies on it	Baseline, kept	—
Safe error surface	<code>AllExceptionsFilter</code> (<code>src/common/filters/all-exceptions.filter.ts</code>); generic 500, no stack/upstream-body/secrets	filter + <code>e2e</code> assertions	Baseline, kept	—
Immutable admin audit	<code>AuditService.record()</code> → <code>admin_audit_logs</code> for verify/activate/cancel/revoke/publish/create/update/delete; no public endpoint, metadata never leaves the API	audit unit + admin e2e	Baseline, kept (HANDBOOK §7 obligation)	—

Dependency audit status

`npm audit —json` was captured during plan T2 on 2026-09-12

(evidence: `.omo/notepads/xolvon-addendum-hardening-docs/evidence/t2-audit.json` , exit code 1 as expected with nonzero findings):

Xolvon Backend Documentation - xolvon-backend-full-documentation.pdf

Severity	Count
Critical	0
High	4
Moderate	1
Low	2
Total	7

Remediation of the high/moderate residuals and the Dependabot-vs-scanner policy are plan T13 work; no paid scanner has been adopted (DL-023 — OPEN).
No claim of a clean audit is made here.

Known limitations (honest state)

1. **Production D1/R2 live path is BLOCKED, not deployed.** There is no verified staging/production deployment target, secret store, or HTTPS domain in this workspace, and no live evidence is fabricated — see `qa-results.md`. All security verification above is local (unit/e2e/curl against locally booted instances with non-secret config).
2. **CSP and CORS final origins are OPEN DECISIONS.** Staging/production frontend origins and `CSP_CONNECT_SRC` values await owner confirmation (DL-020, DL-021, DL-027); until then the lists default to `'self'` plus the configured `R2_ENDPOINT` origin only.
3. **Monitoring/alerting is interface-only.** `MonitoringPort` (`src/observability/monitoring.port.ts`) has no implementation or provider; vendor and recipients remain OPEN (DL-022).
4. **The request-integrity guard is a browser-CSRF control, not an auth layer.** A `POST/PATCH/DELETE` with no `Origin`, no `Referer`, and no `Sec-Fetch-Site` header (curl, Postman, server-to-service clients) passes the guard by design; such clients still need valid Bearer credentials on every protected route. This is the explicit non-browser bypass policy recorded under DL-018, not an oversight.
5. **Sanitization neutralizes markup at the write boundary only;** renderers must still apply contextual escaping (defense in depth over CSP).
6. **JWT logout window ≤15 menit (stateless JWT limitation, B.4).** Setelah logout, JWT access token yang sudah diterbitkan tetap valid secara kriptografis hingga kedaluwarsa (maksimal 15 menit) karena arsitektur token bersifat stateless tanpa denylist/Redis terpusat. Ini merupakan acceptable risk yang disetujui untuk V1 guna mempertahankan skalabilitas edge zero-state Cloudflare Workers; revokasi server-side instan dijadwalkan sebagai pertimbangan arsitektur token blacklist di V2. Refresh token di sisi lain di-revoke secara instan di database saat logout.